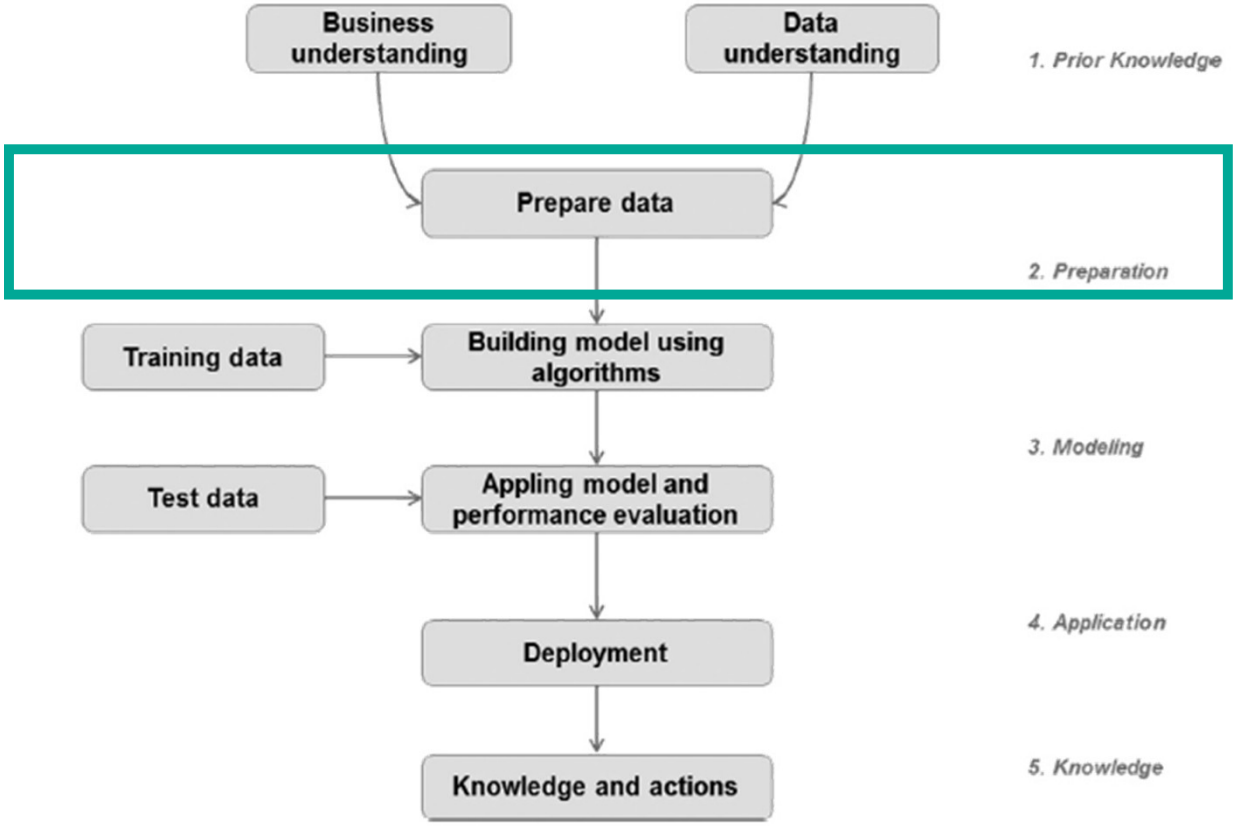
DATA EXPLORATION USING VISUALIZATION & DESCRIPTIVE STATISTICS

From the Latin “dare” to Modern Data Science — Understanding Data Before You Model It

What Is Data Exploration?





Data Exploration

Also called Exploratory Data Analysis (EDA) — a set of tools for gaining a fundamental understanding of a dataset's structure, distributions, extreme values, and the relationships between its attributes.

Types of Data Exploration

DESCRIPTIVE STATISTICS

Condensing key characteristics of a dataset into simple numeric metrics.

- Mean — the typical value
- Standard deviation — the spread
- Correlation — how attributes move together

VISUALIZATION

Projecting data into multi-dimensional space or abstract images.

- Line charts — trends over a sequence
- Bar charts — comparisons across groups
- Scatter plots — relationships & clusters

Objectives of Data Exploration

Where EDA shows up across the data science process.

01

Data Understanding

Typical values, spread, and extremes for each attribute.

02

Data Preparation

Handle outliers, missing values & correlated attributes.

03

Data Science Tasks

Sometimes exploration alone can substitute full modeling.

04

Interpreting Results

Histograms & charts explain predictions and error rates.

Objective 1: Data Understanding

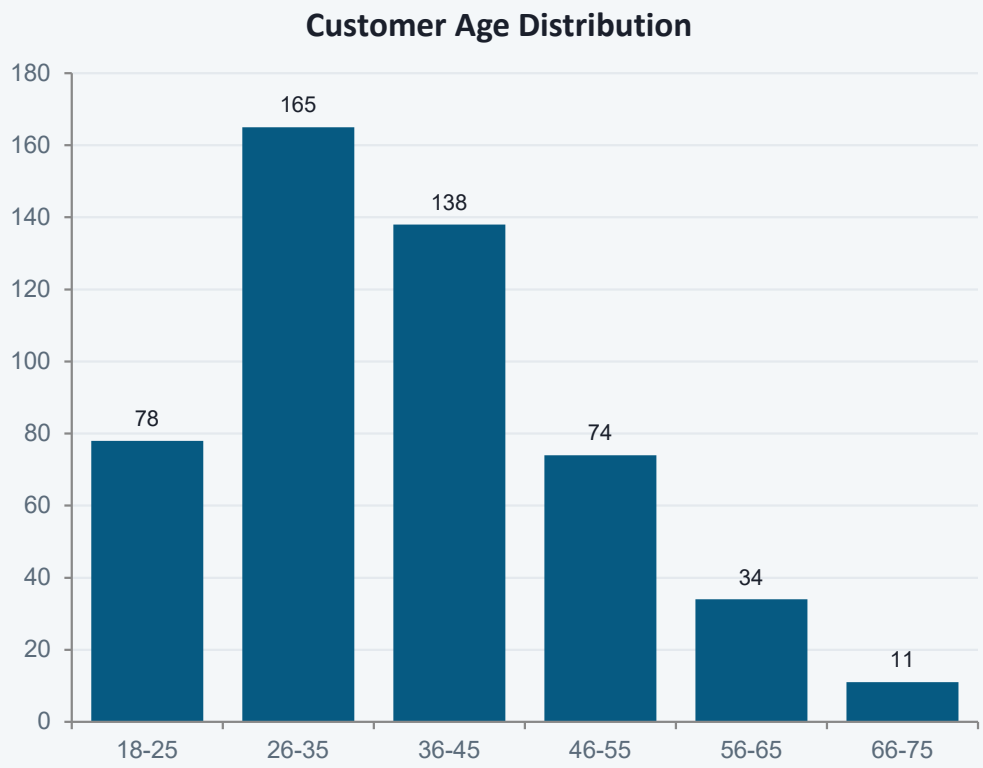
DATASET: Retail Customers (n = 500)

Attribute explored: Customer Age

Mean	**35.4 yrs**
Std. Dev.	**±8.2 yrs**
Min / Max	**18 / 75**

MEANING

- ✓ The typical customer fall between 26–45.
- ✓ This tells the business who its “core” customer before building any model
- ✓ And flags that ages above ~60 are comparatively rare (potential outliers to watch).



Objective 2: Data Preparation

Finding outliers, missing values, and correlated attributes before modeling.

DATASET: Residential House Listings (n = 1,200)

Outlier Detected

One house cost way too much.

One home was priced around ₹43 crore. Most homes cost between ₹1.2 crore and ₹3.3 crore.
This one price is so different, it would throw off the average and confuse the prediction. Look at it separately.

Missing Values

Some information is missing

144 out of 1,200 homes (12%) don't have balcony size listed.

Computers can't learn from blanks. Fill in a typical number, or drop this if too much is missing..

High Correlation

Two numbers say the same thing.

House size and room count move together almost every time (match: 0.89 out of 1).

When two things always change together, using both just confuses the model. Keep only one..

Some models get confused when a piece of information repeats itself. Catching this early saves a lot of trouble later..

Objective 3: Data Science Tasks

Sometimes a simple visual can substitute for the entire modeling process.

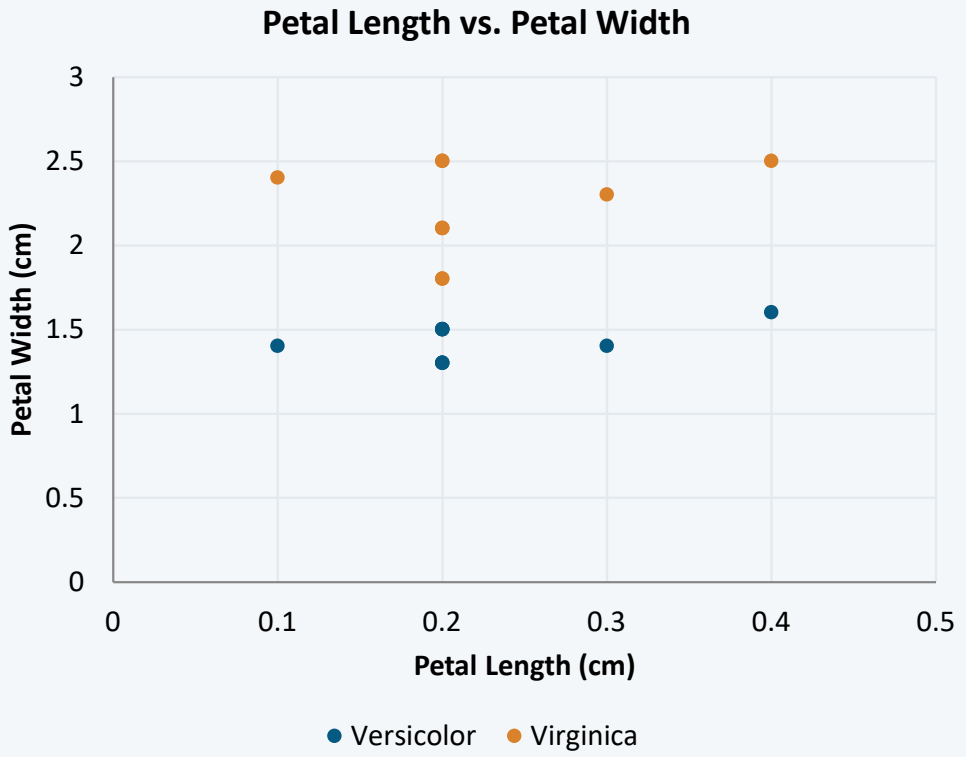
DATASET: Iris Flowers (n = 150)

Attributes plotted: Petal Length vs. Petal Width

A scatterplot of just two attributes visually separates the two flower species into distinct clusters with no classification algorithm run at all.

MEANING

A basic scatterplot can identify clusters in low-dimensional data or suggest simple visual rules for classification replacing the need for a full modeling pipeline in straightforward cases.



Objective 4: Interpreting the Results

Histograms help evaluate predictions, classifications, and error rates.

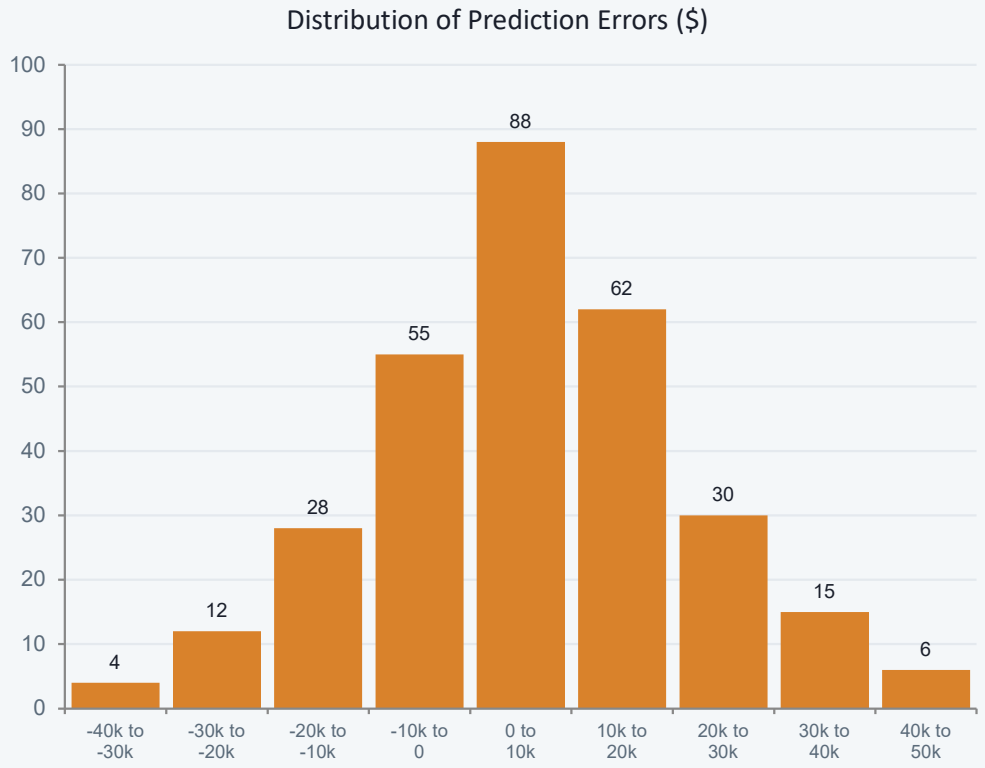
DATASET: House Price Model (n = 300)

Attribute: Prediction Error = Actual − Predicted Price

The histogram of residuals is centered near zero the model is broadly unbiased but has a longer right tail, showing the model under-predicts price for a subset of high-end homes.

MEANING

This single chart tells us the model's accuracy AND where it fails guiding the next iteration (e.g., add a luxury-home feature) without any further statistical testing.



Syllabus Content

Unit 2 · Data Exploration using Descriptive Statistics

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

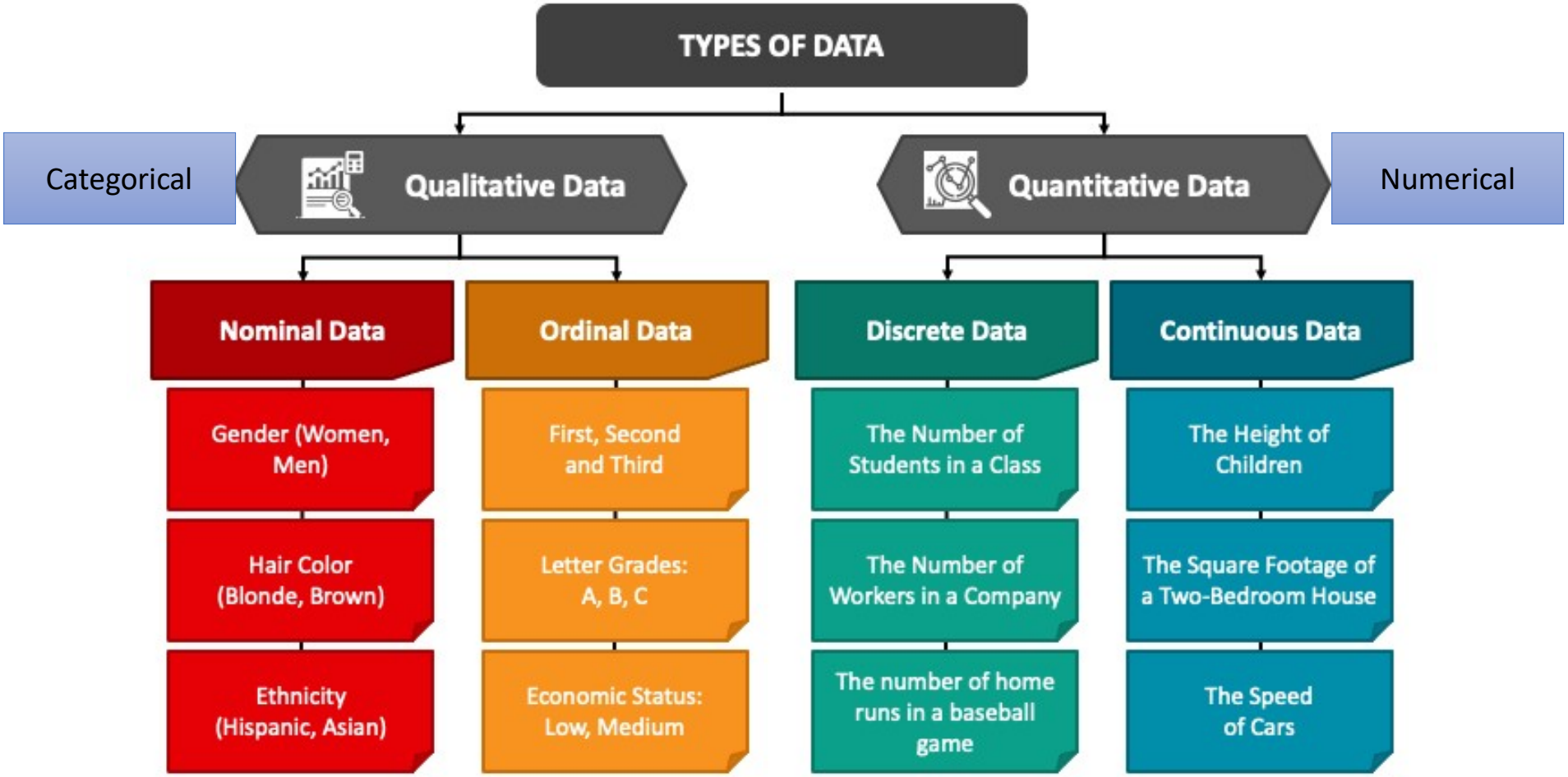
Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

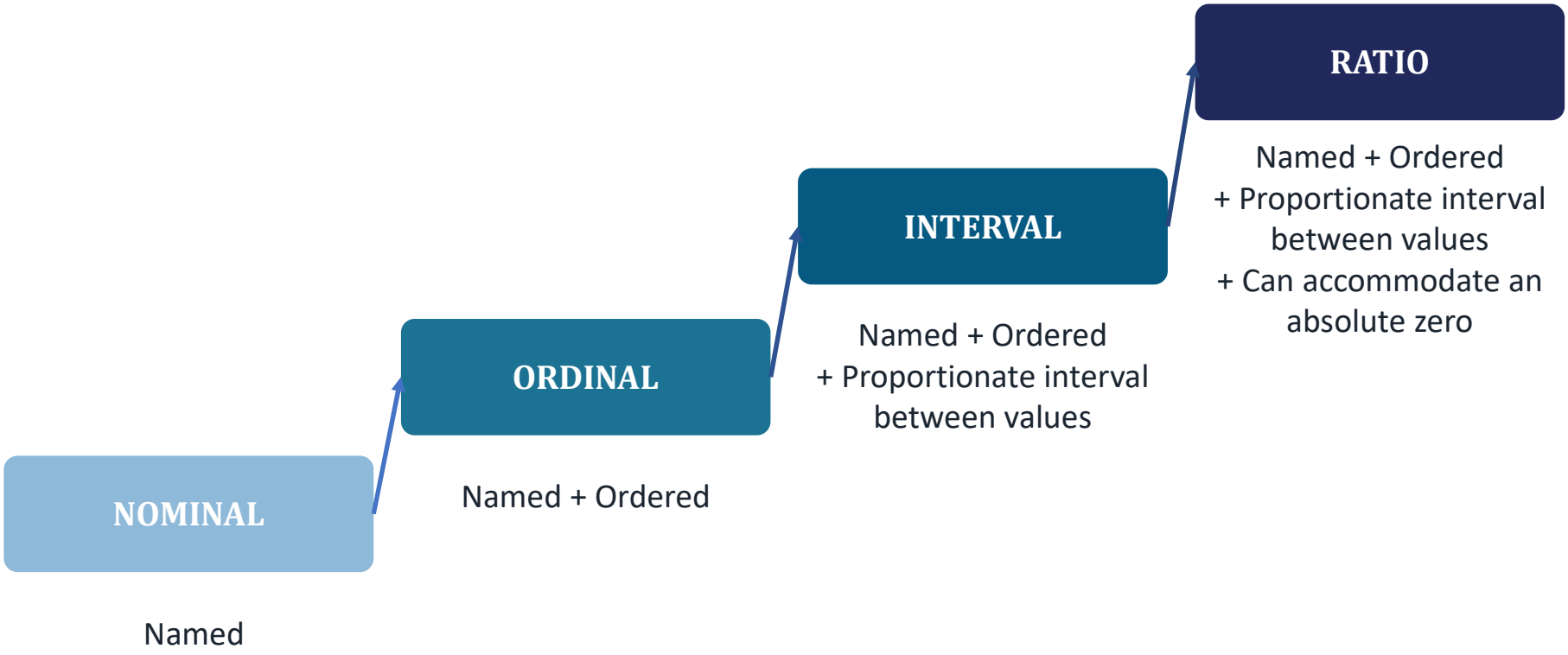
TYPES OF DATA

- *How the format of an attribute decides what operations you can perform on it*

Dataset consist of two types of data

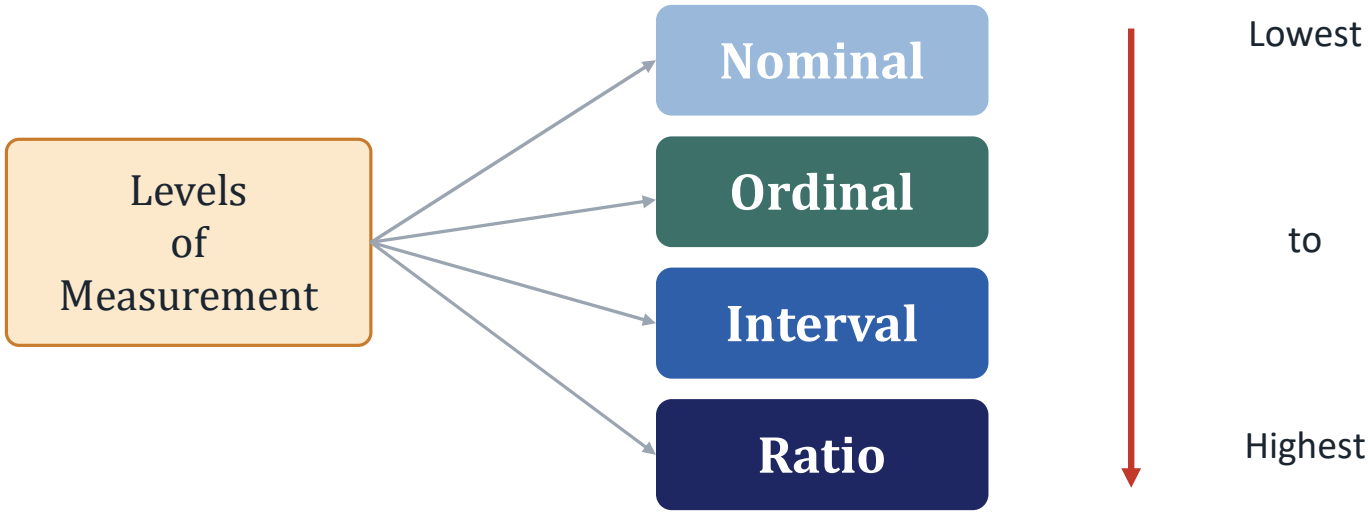


Levels of Measurement



Level of Measurement

Every dataset column sits at one of four levels. The level determines which statistical calculations is important.



Each level builds on the one before it — Ratio data can do everything Nominal, Ordinal, and Interval data can do, plus more.

Campus Student Survey: Dataset

Name	Favorite Cuisine	Class Standing	Year Enrolled	GPA
Emma R.	Italian	Senior	2021	3.8
Marcus T.	Mexican	Junior	2022	3.4
Priya K.	Thai	Sophomore	2023	3.6
Diego M.	Italian	Senior	2021	3.2
Aisha N.	Indian	Freshman	2024	3.9
Liam O.	Mexican	Junior	2022	3.1

Campus Student Survey: Nominal level

Name	Favorite Cuisine	Class Standing	Year Enrolled	GPA
Emma R.	Italian	Senior	2021	3.8
Marcus T.	Mexican	Junior	2022	3.4
Priya K.	Thai	Sophomore	2023	3.6
Diego M.	Italian	Senior	2021	3.2
Aisha N.	Indian	Freshman	2024	3.9
Liam O.	Mexican	Junior	2022	3.1

Favorite Cuisine

Italian • Mexican • Thai

Data at the nominal level of measurement are qualitative only.

Calculated using names, labels, or qualities. No mathematical computations can be made at this level: values can only be checked for equality.

Campus Student Survey: Ordinal level

Name	Favorite Cuisine	Class Standing	Year Enrolled	GPA
Emma R.	Italian	Senior	2021	3.8
Marcus T.	Mexican	Junior	2022	3.4
Priya K.	Thai	Sophomore	2023	3.6
Diego M.	Italian	Senior	2021	3.2
Aisha N.	Indian	Freshman	2024	3.9
Liam O.	Mexican	Junior	2022	3.1

Class Standing

Senior • Junior • Sophomore

Data at the ordinal level of measurement are qualitative or quantitative.

Freshman → Sophomore → Junior → Senior. Ordinal data gives *order* or *rank*, but the *gap* between consecutive values isn't a fixed, measurable quantity

Campus Student Survey: Interval level

Name	Favorite Cuisine	Class Standing	Year Enrolled	GPA
Emma R.	Italian	Senior	2021	3.8
Marcus T.	Mexican	Junior	2022	3.4
Priya K.	Thai	Sophomore	2023	3.6
Diego M.	Italian	Senior	2021	3.2
Aisha N.	Indian	Freshman	2024	3.9
Liam O.	Mexican	Junior	2022	3.1

Year Enrolled

2021 • 2022 • 2023

Data at the interval level of measurement are quantitative. A zero entry simply represents a position on a scale; it is not an inherent zero

Arranged in order, and the differences between data entries can be calculated. The gap between 2021 and 2022 equals the gap between 2022 and 2023.

Campus Student Survey: Ratio level

Name	Favorite Cuisine	Class Standing	Year Enrolled	GPA
Emma R.	Italian	Senior	2021	3.8
Marcus T.	Mexican	Junior	2022	3.4
Priya K.	Thai	Sophomore	2023	3.6
Diego M.	Italian	Senior	2021	3.2
Aisha N.	Indian	Freshman	2024	3.9
Liam O.	Mexican	Junior	2022	3.1

GPA

3.8 • 3.6 • 3.4 • 3.2 • 3.9 • 3.1

Data at the ratio level of measurement are similar to the interval level, but a zero entry is meaningful.

✓ It's a number (quantitative)
✓Differences between values are meaningful — 3.8 minus 3.4 is a real, calculable difference

Summary

Level of Measurement	Put Data in Categories	Arrange Data in Order	Subtract Data Values	Determine if One Value Is a Multiple of Another
Nominal	Yes	No	No	No
Ordinal	Yes	Yes	No	No
Interval	Yes	Yes	Yes	No
Ratio	Yes	Yes	Yes	Yes

Properties of Data

Characteristic	How It's Measured
Accuracy	Is the information correct in every detail?
Completeness	How comprehensive is the information?
Reliability	Does it contradict other trusted sources?
Relevance	Do you really need this information?
Timeliness	How up-to-date is it, usable for real-time reporting?

Syllabus Content

Unit 2 · Data Exploration using Descriptive Statistics

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

STATISTICAL METHODS · CHAPTER 2

Classification of Data

From raw responses to organised, analysable facts

Classification

Classification is the grouping of related facts into classes sharing a common characteristic — like sorting post-office letters by destination city.

Objectives of Classification

- Condense masses of data into manageable classes
- Facilitate comparison
- Highlight the most significant features
- Enable further statistical treatment

Types of Classification

Geographical

By area — cities, states, zones

Chronological

By time — years, periods

Qualitative

By attribute — sex, literacy

Quantitative

By magnitude — height, income

Why Classify?

RAW VS. CLASSIFIED

A factory's maintenance log records every breakdown as it happens. Raw, it's just a timestamped list; classified by fault type, the picture snaps into focus instantly.

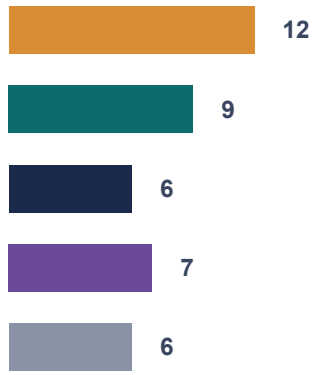
RAW · 40 unsorted maintenance tickets

```
T101 Overheating   T102 Bearing     T103 Electrical
T104 Belt slip     T105 Overheating T106 Software
T107 Bearing       T108 Electrical  T109 Overheating
T110 Software      T111 Overheating T112 Belt slip
T113 Bearing       T114 Overheating T115 Electrical
T116 Overheating   T117 Bearing     T118 Software
T119 Belt slip     T120 Overheating T121 Bearing
T122 Electrical    T123 Overheating T124 Belt slip
T125 Software      T126 Overheating T127 Bearing
T128 Electrical    T129 Overheating T130 Bearing
T131 Belt slip     T132 Overheating T133 Software
T134 Bearing       T135 Overheating T136 Electrical
T137 Overheating   T138 Belt slip   T139 Bearing
T140 Overheating
```

Quick — which fault happens most? You'd have to read all 40 lines and keep a tally.

CLASSIFIED · Same 40 tickets, grouped by fault type

Fault Type	Tickets
Overheating	12
Bearing	9
Electrical	6
Belt slip	7
Software	6
Total	**40**



One glance at the tallest bar answers it: Overheating (12 tickets) — the top failure mode to fix first.

Geographical Classification

AREA-WISE

Data grouped by geographical or vocational area — country, state, city, zone. Reference tables list areas alphabetically; summary tables rank them by size.

Milk production & per-capita availability in India, 2005-06 to 2016-17

Year	Production (Million Tonnes)	Per-capita Availability (g/day)
2005-06	97.1	241
2006-07	102.6	251
2007-08	107.9	260
2008-09	112.2	266
2009-10	116.4	273
2010-11	121.8	281
2011-12	127.9	290
2012-13	132.4	299
2013-14	137.7	307
2014-15	146.3	322
2015-16	155.5	337
2016-17	165.4	356

Chronological Classification

TIME-WISE

Data observed over a period of time — population, production, sales. Usually listed earliest → latest; a reverse order is used when recent events matter most.

Population of India, 1951 – 2011 (in crore)

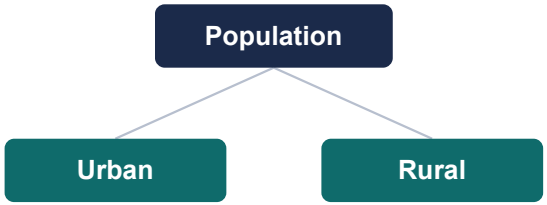
Year	Population	Year	Population
1951	36.11	1991	84.64
1961	43.92	2001	102.87
1971	54.82	2011	121.00
1981	68.33		

Qualitative Classification

BY ATTRIBUTE

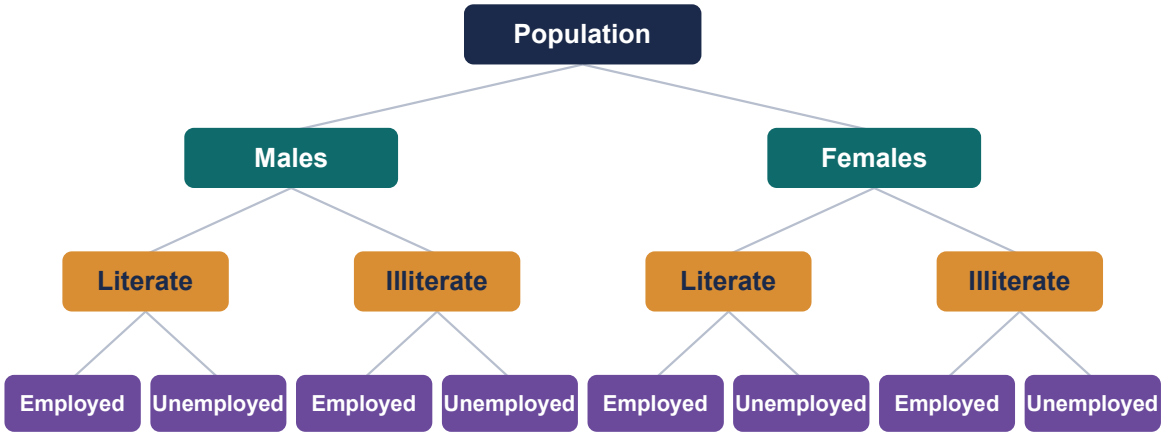
Data grouped by an attribute that can only be present or absent — gender, literacy, religion - never measured, only counted.

SIMPLE (TWO-FOLD) · Population by residence



One attribute → exactly two classes

MANIFOLD · Population by gender, literacy & employment



Gender → Literacy → Employment: each split subdivides further, forming many small classes.

Quantitative Classification

BY MAGNITUDE

Data grouped by a characteristic that can be **measured** — height, weight, income, sales, production.

Students of a college classified by weight

Weight (lbs)	No. of Students
90 - 100	50
100 - 110	200
110 - 120	260
120 - 130	360
130 - 140	90
140 - 150	40
Total	**1,000**

This kind of table a variable plus its frequency is called an empirical (or simple) frequency distribution.

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

What Is a Frequency Distribution?

A frequency distribution (or frequency table) groups raw data into classes and records the number of cases — the frequency — falling in each class. When items are expressed as a proportion of the total, the table becomes a relative frequency, or percentage, distribution.

Two Types of Frequency Distribution

- Discrete Frequency Distribution — for variables with exact, countable values (e.g. number of children per family).
- Continuous Frequency Distribution — for variables grouped into class intervals (e.g. income, marks, time spent).

WHY IT MATTERS

- Condenses large, unwieldy datasets into a compact, readable form.
- Reveals patterns, concentration, and spread in the data at a glance.
- Forms the base for further statistical work — averages, dispersion, graphs.

Frequency Distributions

DISCRETE VS CONTINUOUS

A frequency distribution groups data into classes and records how many cases fall in each — built from a variable that is either **discrete** (whole-number jumps, e.g. children) or **continuous** (any fractional value, e.g. weight).

DISCRETE · Families by number of children

Children	Families
0	10
1	40
2	80
3	100
4	250
5	150
6	50
Total	**680**

CONTINUOUS · Persons by weight

Weight (lbs)	Persons
100 - 110	10
110 - 120	15
120 - 130	40
130 - 140	45
140 - 150	20
150 - 160	4
Total	**134**

Formation of Discrete Frequency Distributions

Survey Data: Children per Family (35 Families)

In a survey of 35 families in a village, the number of children per family was recorded as follows:

1	0	2	3	4	5	6
7	2	3	4	0	2	5
8	4	5	12	6	3	2
7	6	5	3	3	7	8
9	7	9	4	5	4	3

Task: represent this raw data as a discrete frequency distribution.

No. of Children	Talley Bars	Frequency
0	||	2
1	|	1
2	||||	4
3	𝍸 ||	6
4	𝍸	5
5	𝍸	5
6	|||	3
7	||||	4
8	||	2
9	||	2
10		0
11		0
12	|	1
Total		**35**

Formation of Discrete Frequency Distributions

Number of letters in each word of a passage

> *Today, to a very striking degree, our culture has become a statistical culture. Even a person who may never have heard of an index number is attached, in an intimate fashion, by the gyrations of those index numbers which describe the cost of living.*

Letters	Talley Bars	Frequency
1	|||	3
2	𝍸 ||||	9
3	𝍸 |	6
4	||||	4
5	𝍸 ||	7
6	𝍸	5
7	||||	4
8	||||	4
9	|	1
~~10~~		—
11	|	1

Formation of Continuous Frequency Distributions

This classification method is the most common in practice. Four technical terms matter whenever data are grouped into class intervals:

Class Limits

- Lowest and highest values a class can include.
- Lower limit: no item in the class falls below it. Upper limit: none falls above it.
- E.g. class 20–40 → lower limit 20, upper limit 40.

Class Intervals

- The difference between a class's upper and lower limits.
- E.g. class 100–200 → interval = 200 − 100 = 100.
- Width (10, 20, 50, 100 …) depends on the data's range and the detail required.

Class Frequency

- The number of observations falling in a particular class.
- E.g. if ₹10,000–₹15,000 has frequency 50, then 50 persons earn in that range.
- Summing all class frequencies gives the total number of observations.

Class Mid-point (Class Mark)

- The value halfway between a class's lower and upper limits.
- Mid-point = (Upper limit + Lower limit) / 2.
- Used to represent the whole class in further statistical work.

Formation of Continuous Frequency Distributions

Class Limits

- The lowest and highest values that can be included in a class.
- Lower limit: the value below which no item in that class can fall.
- Upper limit: the value above which no item in that class can fall.
- Example: for the class 20–40, the lower limit is 20 and the upper limit is 40.

Class Interval

- The difference between the upper and lower limits of a class.
- Example: for the class 100–200, the class interval is 100 (i.e. 200 minus 100).
- Choosing the width (10, 20, 50, 100, 500 …) depends on the range of the data, the details required, and the number of classes desired.

Formation of Continuous Frequency Distributions

Class Intervals: Estimating Class Interval & Number of Classes

Class Interval Formula

$$i = (L - S) / k$$

- L = Largest item, S = Smallest item, k = Number of classes.

Example: salaries of 100 employees range from ₹10,000 to ₹30,000; 10 classes are wanted.

$$i = (30{,}000 - 10{,}000) / 10 = ₹2{,}000$$

Starting class: 10,000–12,000, then 12,000–14,000, and so on.

Number of classes: Sturges' Rule (for k)

$$k = 1 + 3.322 \log N$$

N = total number of observations

N	k = 1+3.322logN
10	4.322 ≈ 4
100	7.644 ≈ 8

k is conventionally kept between 4 and 20 classes.

Formation of Continuous Frequency Distributions

Class Frequency

- The number of observations corresponding to a particular class.
- Example: if the frequency of the class ₹10,000–₹15,000 is 50, then 50 persons have incomes in that range.
- Adding the frequencies of all classes gives the total frequency — the full number of observations studied.

Class Mid-point (Class Mark)

The value lying halfway between the lower and upper limits of a class interval.

Mid-point = (Upper limit + Lower limit) / 2

The mid-point of each class is taken to represent that entire class for the purpose of further statistical calculations.

Classification Methods based on Class Intervals

Exclusive Method vs. Inclusive Method

Exclusive Method = upper limit of previous class - lower limit of the next class

Ensures continuity of data; the item equal to the upper limit is excluded and placed in the next class.

Income (₹)	No. of Persons
10,000–15,000	50
15,000–20,000	100
20,000–25,000	200
25,000–30,000	150
30,000–35,000	40
35,000–40,000	10
Total	**550**

Inclusive Method — upper limit is included in the class itself

Avoids ambiguity for discrete data; a value exactly at 14,999 vs 15,000 is unambiguous.

Income (₹)	No. of Persons
10,000–14,999	50
15,000–19,999	100
20,000–24,999	200
25,000–29,999	150
30,000–34,999	40
35,000–39,999	10
Total	**550**

Rule of thumb: use the upper-limit-exclusive method for continuous variables; use the inclusive method for discrete variables.

Correction Factor — Adjusting the Inclusive Method

Correction factor = (Lower limit of 2nd class − Upper limit of 1st class) / 2

Subtract this value from every lower limit, and add it to every upper limit, to make the classes continuous.

Before adjustment (Weekly Wages, ₹)

Wages	Workers
2800–2899	5
2900–2999	10
3000–3099	15
3100–3199	8
3200–3299	2

After adjustment (correction factor = 0.5)

Wages	Workers
2799.5–2899.5	5
2899.5–2999.5	10
2999.5–3099.5	15
3099.5–3199.5	8
3199.5–3299.5	2

Correction factor = (2900 − 2899) / 2 = 0.5. The class interval was 99 before adjustment and becomes exactly 100 after adjustment.

Considerations in Constructing Continuous Frequency Distributions

- Number of classes: preferably between 5 and 20 (rarely fewer than 6 or more than 15 in practice) — depending on the number of figures, their magnitude, the detail required, and ease of calculation.
- Class width: avoid awkward intervals like 3, 7, 11, 26, 39. Prefer multiples of 5 (5, 10, 25, 100), since the human mind is accustomed to thinking in such units.
- Starting point: the lower limit of the first class should be zero or a multiple of 5 — e.g. if the lowest value is 63 with interval 10, start at 60, not 63.
- Equal class sizes are generally preferred for easy comparison, though unequal or open-end intervals may be used where data are concentrated in a narrow range with a few extreme values.
- Continuity: prefer the exclusive method to ensure continuity; if the inclusive method is used, apply a correction factor to restore continuity.

Continuous Frequency Distribution Example 1: Time Spent on Internet

Prepare a frequency distribution of the following data of time spent on Internet (in minutes) by 50 students in a day. Take the width of each class interval as 10 and use the exclusive method of classification.

57	44	80	75	00	18	45	14	04	64
72	51	69	34	22	83	70	20	57	28
96	56	50	47	10	34	61	66	80	46
22	10	84	50	47	73	42	33	48	65
10	34	66	53	75	90	58	46	39	69

Raw data — 50 observations, in minutes

- Class interval = 10
- Minimum Value=00, Maximum value=96
- So, first class will start at
  00 - 10, 10 - 20 up to 90-100

Frequency Distribution of Data Spent on Internet

Time Spent on Internet (minutes)	Tally Marks	Number of Students (Frequency)
0 – 10	||	2
10 – 20	𝍸	5
20 – 30	||||	4
30 – 40	𝍸	5
40 – 50	𝍸 |||	8
50 – 60	𝍸 |||	8
60 – 70	𝍸 ||	7
70 – 80	𝍸	5
80 – 90	||||	4
90 – 100	||	2
Total		**50**

Continuous Frequency Distribution Example 2: Marks Distribution

Prepare a frequency distribution of the marks obtained out of 100 from the following data

15	45	40	42	50	60	62	68	70	42
75	75	80	81	25	26	31	32	78	45
31	45	42	43	55	56	78	80	81	62
60	62	58	69	70	45	50	56	72	58
75	62	62	65	60	70	35	37	40	55

Raw data — marks of 50 students (out of 100)

- Total number of observations=50
- Minimum Value=15, Maximum value=81
- Number of classes

  $k = 1 + 3.322 \log N$

  $k = 1 + 3.322 \log 50 = 7$
- Class interval

  $i = (L - S) / k$

  $i=(81-15)/7 =9.4 =10$

Frequency Distribution of Marks

Marks	Tally Bars	No. of Students
15 – 25	|	1
25 – 35	𝍸	5
35– 45	𝍸 |||	8
45 – 55	𝍸 |	6
55 – 65	𝍸 𝍸 ||||	14
65 – 75	𝍸 ||	7
75 – 85	𝍸 ||||	9
Total		**50**

Continuous Frequency Distribution Example 3: Marks Distribition

Classify the following data by taking class intervals such that their mid-values are 17, 22, 27, 32, and so on.

30	42	30	54	40	48	15	17	51	42	25	41
30	27	42	36	28	26	37	54	44	31	36	40
36	22	30	31	19	48	16	42	32	21	22	46
33	41	21									

Raw data — marks of 39 students

Since mid-values must be 17, 22, 27, 32 ... ,

Marks	Tally Bars	Number of Students
Total		**39**

Finding the First-Class Interval from its Mid-value

Given: mid-value = 17,22,27.... → class width =22-17 = 5 → Let lower limit = L, upper limit = U

Equation 1 — from the mid-value

$$(L + U) / 2 = 17 \implies L + U = 34$$

Equation 2 — from the class width (Inclusive Method)

$$U - L = \text{width} - 1 = 5 - 1 = 4 \implies U - L = 4$$

Step 1 — Add equation 1 and equation 2	$(L + U) + (U - L) = 34 + 4 \implies 2U = 38$
Step 2 — Solve for U	$U = 38 / 2 = 19$
Step 3 — Substitute U = 19 into equation 1	$L + 19 = 34 \implies L = 34 - 19 = 15$
Step 4 — Check both equations	$L + U = 15 + 19 = 34$ $\quad$ $U - L = 19 - 15 = 4$

Result: $L = 15, U = 19 \rightarrow$ First class = 15 – 19 (mid-value = (15+19)/2 = 17)

Continuous Frequency Distribution Example 3: Marks Distribution

Classify the following data by taking class intervals such that their mid-values are 17, 22, 27, 32, and so on.

30	42	30	54	40	48	15	17	51	42	25	41
30	27	42	36	28	26	37	54	44	31	36	40
36	22	30	31	19	48	16	42	32	21	22	46
33	41	21									

Raw data — marks of 39 students

Since mid-values must be 17, 22, 27, 32 ... , the first class is 15–19
[mid = (15+19)/2 = 17], second class 20–24, and so on.

Frequency Distribution of Marks

Marks	Tally Bars	Number of Students
15 – 19	||||	4
20 – 24	||||	4
25 – 29	||||	4
30 – 34	𝍸 |||	8
35 – 39	||||	4
40 – 44	𝍸 ||||	9
45 – 49	|||	3
50 – 54	|||	3
Total		**39**

Finding the First-Class Interval from its Mid-value

Given: mid-value = 17,22,27.... → class width =22-17 = 5 → Let lower limit = L, upper limit = U

Equation 1 — from the mid-value

$(L + U) / 2 = 17 \implies L + U = 34$

Equation 2 — from the class width (Exclusive Method)

$U - L = \text{width} = 5 \implies U - L = 5$

Step 1 — Add equation 1 and equation 2	$(L + U) + (U - L) = 34 + 5 \implies 2U = 39$
Step 2 — Solve for U	$U = 38 / 2 = 19.5$
Step 3 — Substitute U = 19 into equation 1	$L + 19.5 = 34 \implies L = 34 - 19.5 = 14.5$
Step 4 — Check both equations	$L + U = 14.5 + 19.5 = 34$ $U - L = 19.5 - 14.5 = 5$

Result: $L = 14.5$, $U = 19.5$ → First class = 14.5 – 19.5 (mid-value = (14.5+19.5)/2 = 17)

Continuous Frequency Distribution Example 3: Marks Distribution

Classify the following data by taking class intervals such that their mid-values are 17, 22, 27, 32, and so on.

30	42	30	54	40	48	15	17	51	42	25	41
30	27	42	36	28	26	37	54	44	31	36	40
36	22	30	31	19	48	16	42	32	21	22	46
33	41	21									

Raw data — marks of 39 students

Since mid-values must be 17, 22, 27, 32 ... , the first class is 15–19
[mid = (15+19)/2 = 17], second class 20–24, and so on.

Frequency Distribution of Marks

Marks	Tally Bars	Number of Students
14.5 – 19.5	||||	4
19.5 – 24.5	||||	4
24.5 – 29.5	||||	4
30.5 – 34.5	𝍸 |||	8
34.5 – 39.5	||||	4
39.5 – 44.5	𝍸 ||||	9
44.5 – 49.5	|||	3
49.5– 54.5	|||	3
Total		**39**

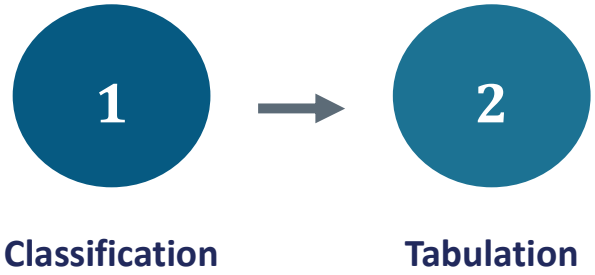
STATISTICAL METHODS

Tabulation of Data

What Is Tabulation?

Tabulation is the systematic arrangement of statistical data in rows and columns to simplify presentation and aid comparison.

Rows are horizontal arrangements; **columns** are vertical. Classification is the first step — grouping data by common characteristics — before it is displayed in a table.



Role of Tabulation

01

Simplifies complex data

Removes unnecessary detail; a clear picture emerges fast.

02

Facilitates comparison

Totals and sub-totals let parts of data be compared easily.

03

Gives identity

Numbered tables can be used as a source reference.

04

Reveals patterns

Shows relationships that narrative text would hide.

Parts of a Table

Seven elements combine to make a table self-explanatory

1 Table Number

Identifies the table for reference; usually top-centre or left.

2 Title

Clear, brief, self-explanatory — states what, where, when.

3 Caption

Column headings, placed at the middle of each column.

4 Stub

Row headings, at the extreme left; kept as narrow as possible.

5 Body

The numerical information itself — the most vital part.

6 Headnote

Explanatory statement below the title, e.g. “in thousands”.

7 Footnotes

Clarify exceptions, special circumstances, or the data source.

Body = numbers.
Stub = row labels.
Caption = column labels.

Tables May Be Classified Two Ways

Every statistical table falls into one of two broad classification schemes

BY CHARACTERISTICS

Simple & Higher Order Tables

Based on the number of characteristics studied at once — from a single variable (simple) to several shown together (complex).

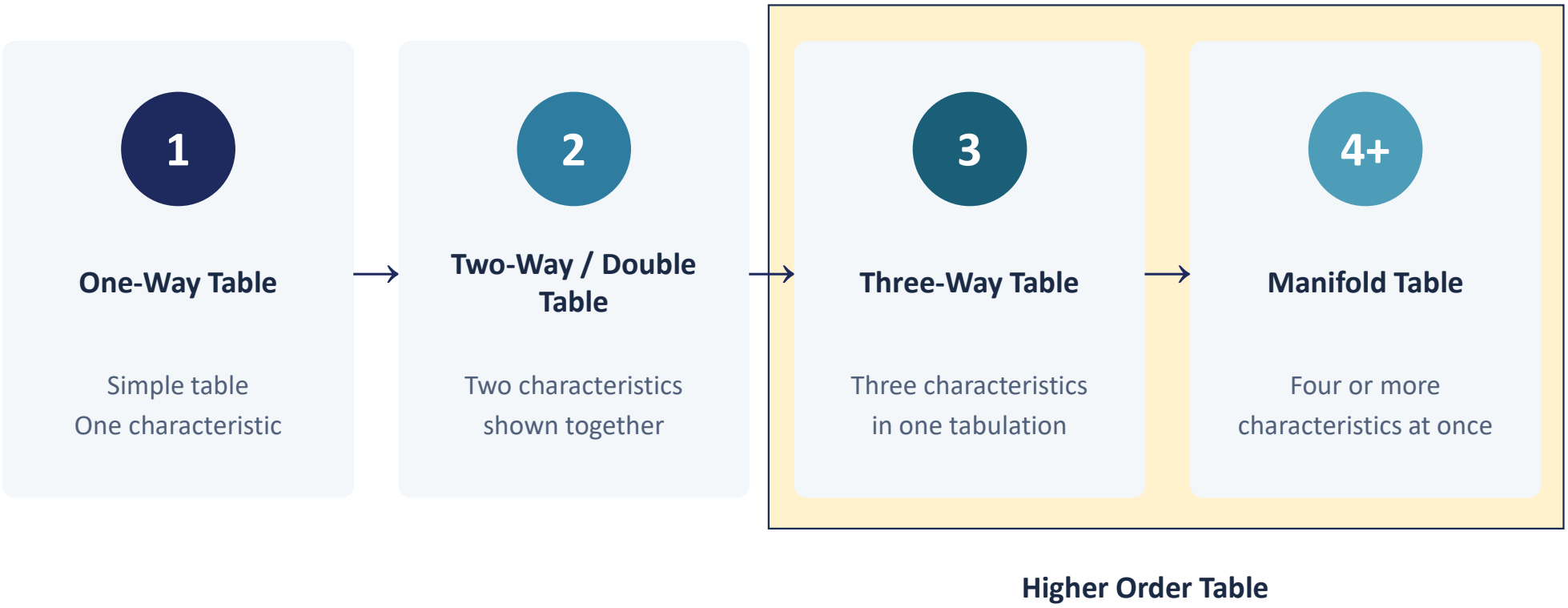
BY PURPOSE

General & Special Purpose Tables

Based on why the table was built — for general reference (general purpose) or to summarize/highlight specific findings (special purpose / summary).

From Simple to Manifold Tabulation: By Characteristics

The distinction between simple and complex tables rests on how many characteristics are shown together



Simple Table (One-Way Table)

Only one characteristic is shown — the most basic form of tabulation

In a simple table, only one characteristic is studied.

Example below:

Employees of a State Bank are classified by a single characteristic — their **age group.**

Why it matters

One-way tables are the building block for two-way and higher-order tables, which combine several such characteristics together.

NUMBER OF EMPLOYEES IN STATE BANK ACCORDING TO AGE GROUP

Age (in years)	Below 25	25–35	35–45	45–55	Above 55	Total
No. of Employees	...	...	...	...	...	**...**

Key features of a one-way table

- Stub (row heading) carries the single characteristic — here, age group.
- Caption (column heading) shows the frequency being counted — here, number of employees.
- A Total column or row closes the table, giving the grand total.

Two-Way Table

Two characteristics are shown, formed by dividing either the stub or the caption into two coordinate parts

Example below:

Employees are shown by **age group** AND **gender** together — two characteristics in one table.

Note

This is also called a Double Table — formed when either the stub or the caption is split into two coordinate parts.

NUMBER OF EMPLOYEES OF STATE BANK IN DIFFERENT AGE-GROUPS ACCORDING TO GENDER

Age (in years)	Males	Females	Total
Below 25	—	—	—
25–35	—	—	—
35–45	—	—	—
45–55	—	—	—
Above 55	—	—	—
Total	—	—	—

Worked Example: Two-Way Frequency Table

Height & weight of 20 persons, tallied into a 5 × 6 grid of cells (Illustration 7)

Height (X) \ Weight (Y)	115-125	125-135	135-145	145-155	155-165	165-175	Total
62-64	2	1	—	—	—	—	**3**
64-66	2	—	3	—	—	—	**5**
66-68	—	1	2	1	—	—	**4**
68-70	—	2	—	2	—	—	**4**
70-72	—	1	1	—	1	1	**4**
Total	**4**	**5**	**6**	**3**	**1**	**1**	**20**

Marginal distribution: the row totals give the frequency distribution of height (X); the column totals give the frequency distribution of weight (Y).

Reading a cell: e.g. 2 persons have height 62–64" and weight 115–125 lbs. Height is tallied first (row), then weight determines the column.

Higher Order Table

Three or more characteristics represented in the same table

1

Why it's used

Arises when several characteristics must be presented simultaneously in one table.

2

What it requires

An established order of precedence among the attributes or characteristics, based on their relative importance.

3

The trade-off

As the number of characteristics increases, the table becomes more and more confusing — so it should normally not go beyond a manageable number of characteristics.

Worked Example: Higher-Order Table

Three characteristics shown together — age, rank, and gender

Age (years)	Officers			Assistants			Clerks			Total		
	M	F	Total	M	F	Total	M	F	Total	M	F	Total
Below 25	—	—	—	—	—	—	—	—	—	—	—	—
25-35	—	—	—	—	—	—	—	—	—	—	—	—
35-45	—	—	—	—	—	—	—	—	—	—	—	—
45-55	—	—	—	—	—	—	—	—	—	—	—	—
55 & above	—	—	—	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—	—	—	—

Number of Employees of State Bank According to Age Group, Gender and Rank

Precedence matters: before building a higher-order table, rank the characteristics by importance — here, rank has the widest split, age forms the stub. Adding a 4th characteristic (e.g. religion) is possible, but tables rarely go beyond four — more becomes confusing to read.

Drafting a Blank Table: Higher-Order Table

Skeleton for personnel by gender, salary grade, age group, over two years

Year	Age Groups	Below ₹20,000			₹20,000-40,000			₹40,000 & above			Total		
		M	F	Total	M	F	Total	M	F	Total	M	F	Total
2020	Below 25	—	—	—	—	—	—	—	—	—	—	—	—
	25 & under 40	—	—	—	—	—	—	—	—	—	—	—	—
	40 & above	—	—	—	—	—	—	—	—	—	—	—	—
2021	Below 25	—	—	—	—	—	—	—	—	—	—	—	—
	25 & under 40	—	—	—	—	—	—	—	—	—	—	—	—
	40 & above	—	—	—	—	—	—	—	—	—	—	—	—
	Total	—	—	—	—	—	—	—	—	—	—	—	—

Four characteristics, one table: year, age group, salary grade, and gender — title and headnote make the scope explicit; the body is left blank until data is tallied in.

Examlpe1 — Spot the Mistakes

A poorly constructed table on population by Gender, age, and literacy

Gender

Gender	0 to 25	25 to 50	50 to 75	75 to 100
Males				
Females				

The guiding rule

Every characteristic mentioned in the problem — here, age, gender, AND literacy — must appear somewhere in the table: in the title, the headnote, or the headings themselves. If a table needs a reader to “assume” what a heading means, the table hasn't done its job.

What's wrong with this table?

1. **Missing characteristic**

   The title and headings is missing. Only age and gender appear.

2. **Ambiguous column headings**

   "0 to 25", "25 to 50"... are age groups, but without a heading that says “Age (in years)” the reader has to guess.

3. **Needs rearranging**

   Even once literacy is added, the table's layout has to be restructured to fit three characteristics clearly — see the corrected version next.

Example 1 — Corrected Table

Distribution of Population According to Age, Gender and Literacy

Age groups	Literates			Illiterates			Total		
	M	F	Total	M	F	Total	M	F	Total
0 to 25	—	—	—	—	—	—	—	—	—
25 to 50	—	—	—	—	—	—	—	—	—
50 to 75	—	—	—	—	—	—	—	—	—
75 to 100	—	—	—	—	—	—	—	—	—
Total	—	—	—	—	—	—	—	—	—

Notice: the row label is now Age groups only, while Gender (M/F) sits inside each literacy group — three characteristics, one table.

Example 2 — Draft a Blank Table

Distribution of personnel by

- *Gender*
- *The three grades of salary: below 20,000, 20,000-40,000 and above*
- *Three age groups: below 25, 25 and under 40, and above*
- *Two years: 2020 and 2021*

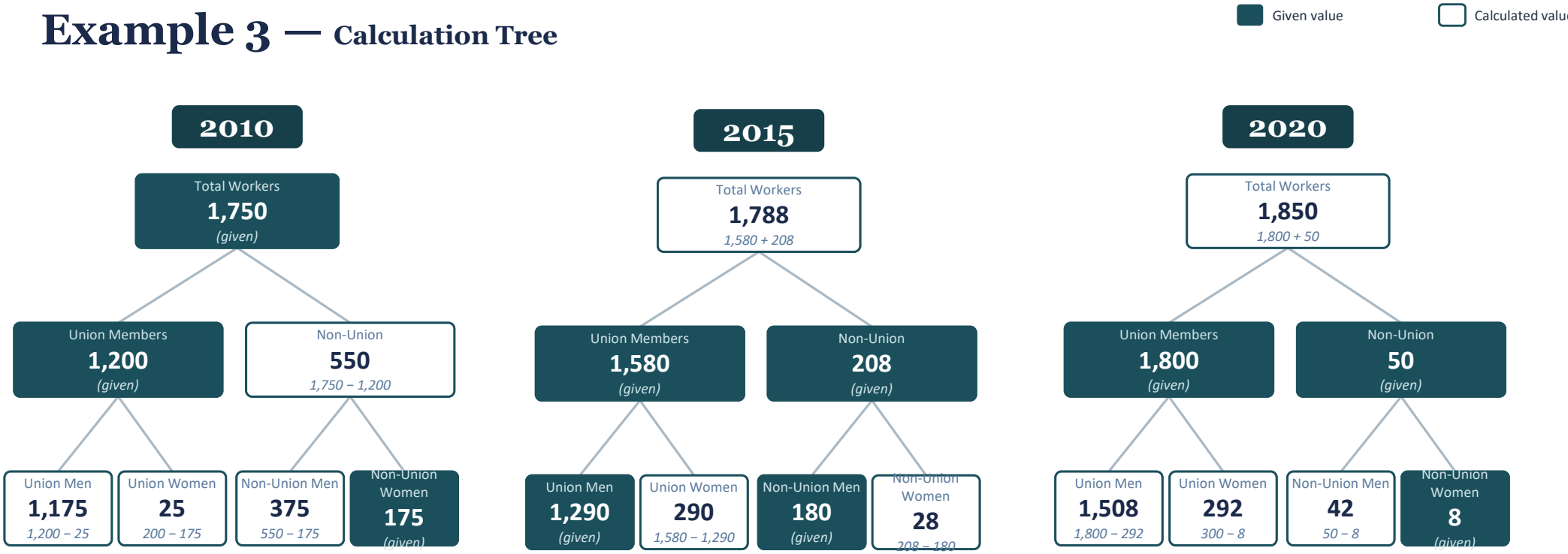
Distribution of personnel by gender, salary and age groups for two years

Year	Age Groups	Below ₹20,000			₹20,000 to 40,000			₹40,000 & above			Total		
		M	F	Total	M	F	Total	M	F	Total	M	F	Total
2020	Below 25	—	—	—	—	—	—	—	—	—	—	—	—
	25 and under 40	—	—	—	—	—	—	—	—	—	—	—	—
	40 and above	—	—	—	—	—	—	—	—	—	—	—	—
2021	Below 25	—	—	—	—	—	—	—	—	—	—	—	—
	25 and under 40	—	—	—	—	—	—	—	—	—	—	—	—
	40 and above	—	—	—	—	—	—	—	—	—	—	—	—
Total		—	—	—	—	—	—	—	—	—	—	—	—

Example 3 — Present the following information in a suitable tabular form:

In 2010, out of a total of 1,750 workers of a factory, 1,200 were members of a trade union. The number of women employees was 200, of which 175 did not belong to a trade union. In 2015, the number of union workers increased to 1,580, of which 1,290 were men. On the other hand, the number of non-union workers fell down to 208, of which 180 were men. In 2020, there were 1,800 employees who belonged to a trade union and 50 who did not belong to a trade union. Of all the employees in 2020, 300 were women, of whom only 8 did not belong to a trade union.

Example 3 — Calculation Tree



Given: Total = 1,750 | Union = 1,200 | Women = 200 | Non-union women = 175

Given: Union = 1,580 | Union men = 1,290 | Non-union = 208 | Non-union men = 180

Given: Union = 1,800 | Non-union = 50 | Women = 300 | Non-union women = 8

Example 3 — Present the following information in a suitable tabular form:

Distribution of Union and Non-Union Members for 2010, 2015 and 2020

Year	Category	Men	Women	Total
2010	Union Members	1,175	25	1,200
	Non-union Workers	375	175	550
	Total	**1,550**	**200**	**1,750**
2015	Union Members	1,290	290	1,580
	Non-union Workers	180	28	208
	Total	**1,470**	**318**	**1,788**
2020	Union Members	1,508	292	1,800
	Non-union Workers	42	8	50
	Total	**1,550**	**300**	**1,850**

Example 7— Problem Statement

In certain data, the following four Main Characteristics and Sub-Characteristics are present. *Prepare a suitable table showing the gender-wise distribution of people of different religions and age-groups, split by Urban and Rural locality.*

Main Characteristics

- Locality
- Religion
- Gender
- Age

Sub-Characteristics

Urban, Rural
Hindus, Non-Hindus
Males, Females
0–30, 30–60 and above

Gender-wise Distribution of the People of Different Religions and Age-groups

Age (years)	Hindus									Non-Hindus								
	Urban			Rural			Total			Urban			Rural			Total		
	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total	M	F	Total
Less than 30	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
30–60	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
60 and over	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Total	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Example 8— Problem Statement

A survey of 370 students from Commerce Faculty and 130 students from Science Faculty revealed that 180 students were studying for only CA Examination, 140 for only Costing Examination and 80 for both CA and Costing Examinations. The rest had offered part-time Management Courses. Of those studying for Costing, only 13 were girls and 90 boys belonged to Commerce Faculty. Out of 80 studying for both CA and Costing, 72 were from Commerce Faculty, amongst which 70 were boys. Amongst those who offered part-time Management Course, 50 boys were from Science Faculty and 30 boys and 10 girls from Commerce Faculty. In all, there were 110 boys in Science Faculty. Present the above information in a tabular form. Find the number of students from Science Faculty studying for part-time Management Courses.

Distribution of Students According to Professional Courses

Course/Faculty	Commerce			Science			Total		
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total
Part-time Management	30	10	40	50	10	60	80	20	100
CA	150	8	158	16	6	22	166	14	180
Costing	90	10	100	37	3	40	127	13	140
CA and Costing	70	2	72	7	1	8	77	3	80
Total	**340**	**30**	**370**	**110**	**20**	**130**	**450**	**50**	**500**

Answer: 60 students from the Science Faculty studied part-time Management courses (50 boys + 10 girls), highlighted above. Total Number of Students = 370 + 130 = 500

Example 10— Building a Two-Way Table from Raw Data

30 paired values of X and Y are given. Task: form a two-way table.

Class intervals — X: 10–20, 20–30, 30–40, 40–50 Y: 100–200, 200–300, …, 500–600

X	14	20	33	25	41	18	24	29	38	45
Y	148	242	296	312	518	196	214	340	492	568
X	32	37	19	28	34	38	29	44	40	23
Y	400	288	292	431	440	500	512	415	514	282
X	22	39	43	44	12	27	39	38	17	26
Y	282	481	516	598	122	200	451	387	245	413

Each of the 30 (X, Y) pairs is tallied into its X-class and Y-class cell.

Y \ X	10–20	20–30	30–40	40–50	Total
100–200	**3**	—	—	—	**3**
200–300	**2**	**5**	**2**	—	**9**
300–400	—	**2**	**1**	—	**3**
400–500	—	**2**	**5**	**1**	**8**
500–600	—	**1**	**1**	**5**	**7**
Total	**5**	**10**	**9**	**6**	**30**

Illustration 16 — Problem Statement

Marks obtained by 25 students in Statistics and Economics are given below, as (Statistics, Economics) pairs. ***Prepare a two-way table*** *taking the magnitude of each class interval as 4 marks, the first class being 0 and less than 4.*

Each pair is (Statistics, Economics):

(7,3)	(15,9)	(2,8)	(12,18)	(9,11)
(5,3)	(17,13)	(19,18)	(11,7)	(10,13)
(13,16)	(16,14)	(6,10)	(4,1)	(14,12)
(0,2)	(1,5)	(11,14)	(8,3)	(9,14)
(13,11)	(14,17)	(0,10)	(11,7)	(15,15)

Number of students by marks in Statistics (rows) and Economics (columns)

Statistics \ Economics	0–4	4–8	8–12	12–16	16–20	Total
0–4	1	1	2	0	0	**4**
4–8	3	0	1	0	0	**4**
8–12	1	2	1	3	0	**7**
12–16	0	0	2	2	3	**7**
16–20	0	0	0	2	1	**3**
Total	**5**	**3**	**6**	**7**	**4**	**25**

Class intervals for both subjects: 0–4, 4–8, 8–12, 12–16, 16–20

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

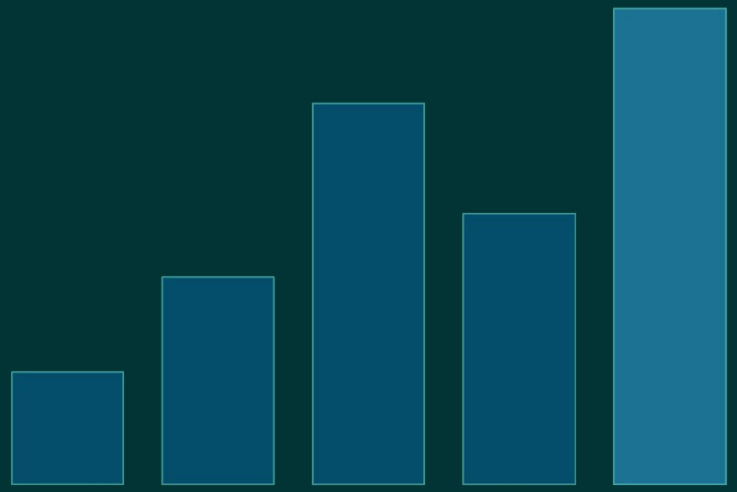
Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

STATISTICAL METHODS · CHAPTER 6.2

Diagrams

Presenting Statistical Data Through Visual Communication

A picture is worth 10,000 words.



Introduction

Classified and tabulated data does not always engage the common reader — figures alone can confuse rather than communicate.

Diagrams, graphs, and maps are among the most convincing ways to present results.

1. **Everyday visuals**

   Found widely in newspapers, magazines, journals, and advertisements.

2. **Many display forms**

   Numerous chart and graph types exist for pictorial data presentation.

3. **A creative challenge**

   Choosing the best method among many options takes artistic judgment.

Significance of Diagrams and Graphs

Why diagrams and graphs are extremely useful in statistical presentation

1

Bird's-eye view

Give an overall picture of the data; information is easily and quickly understood.

2

Attractive to the eye

Figures are dry but diagrams delight the eye and create greater interest.

3

Great memorising effect

The impressions created by diagrams last far longer than tabular figures.

4

Aid comparison

Facilitate comparison of data across different time periods or regions.

5

Stimulate analysis

Bring out facts and relationships, aiding analytical and investigative thinking.

6

Popular & versatile

Widely used in exhibitions, fairs, conferences, and public functions.

Tables vs. Diagrams

Both forms of presentation have their own usefulness — choose with care

Tables

- Contain precise, exact figures
- More information fits in one table than a single graph or diagram
- Usually require closer reading
- More difficult to interpret quickly
- Constructed on graph paper
- Represents a mathematical relationship between two variables
- Used heavily by statisticians and researchers for analysis
- Best for frequency distributions and time series

Diagrams & Graphs

- Give only an approximate idea of values
- Have strong visual appeal — more impressive to laymen
- Easier and quicker to grasp at a glance
- Better suited to publicity and general audiences
- Constructed on plain paper
- Does not represent a functional relationship
- More attractive to the eye; suited for publicity and propaganda
- Rarely used for frequency distributions

General Rules for Constructing Diagrams

1 Title

Brief, self-explanatory; usually placed at the top.

2 Proportion

Keep height:width near the “Root-two”

3 Scale

Use even numbers or multiples of 5/10; state the unit clearly.

4 Footnotes

Clarify special points about the diagram, placed at the bottom.

5 Index

Show what each line, shade or colour represents.

6 Neatness

Diagrams must be absolutely neat and clean.

7 Simplicity

Don't overload one diagram — several simple charts beat one complex one.

Types of Diagrams

Diagrams are grouped by the number of dimensions used to represent data.

1

One-Dimensional

(Bar Diagrams)

Only the length of the bar matters. 6 sub-types: simple, sub-divided, multiple, percentage, deviation, broken bar.

2

Two-Dimensional

(Area Diagrams)

Length and width both matter — area represents data. 3 sub-types: rectangles, squares, circles/pie.

3

Three-Dimensional

(Volume Diagrams)

Length, width and height all matter. 3 sub-types: cubes, cylinders, spheres.

4

Pictograms & Cartograms

(Pictorial / Map)

Data shown via picture symbols (pictograms) or shaded statistical maps (cartograms).

One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

Two-Dimensional Diagrams

Area Diagrams — both length and width represent the value

a

Rectangles

Area is proportional to the value; width and height can both encode data, useful for comparing two variables at once.

b

Squares

Side length is proportional to the square root of the value — keeps figures compact when values vary widely.

c

Circles / Pie Diagrams

Circle area (or its radius) represents the total; slices (sectors) show how it splits into components.

Three-Dimensional Diagrams

Volume Diagrams — length, width and height all represent the value

a

Cubes

Side length is proportional to the cube root of the value.

b

Cylinders

Volume of the cylinder is made proportional to the value.

c

Spheres

Volume of the sphere is made proportional to the value.

Used when the gap between the smallest and largest values is very large. Example: for values in the ratio 1 : 1,000, bars would need a 1 : 1,000 length ratio — but cubes only need a 1 : 10 side ratio.

Pictograms & Cartograms

Pictorial and map-based ways to represent data

1

Pictograms

Data is represented using small picture symbols or icons, where each symbol stands for a fixed unit of quantity (e.g., one icon of a person = 1 million people). Easy to understand but limited in precision.

2

Cartograms

Statistical maps where data is shown through shading, colouring, dots, or symbols placed over geographical regions — useful for showing regional distribution and density of a variable.

Types of Diagrams

Diagrams are grouped by the number of dimensions used to represent data.

1

One-Dimensional

(Bar Diagrams)

Only the length of the bar matters. 6 sub-types: simple, sub-divided, multiple, percentage, deviation, broken bar.

2

Two-Dimensional

(Area Diagrams)

Length and width both matter — area represents data. 3 sub-types: rectangles, squares, circles/pie.

3

~~Three-Dimensional~~

(Volume Diagrams)

Length, width and height all matter. 3 sub-types: cubes, cylinders, spheres.

4

~~Pictograms & Cartograms~~

(Pictorial / Map)

Data shown via picture symbols (pictograms) or shaded statistical maps (cartograms).

One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

What is a Simple Bar Diagram?

A simple bar diagram represents only one variable — for example, sales, production, or population figures across different years. Since all bars share the same width and only their length varies, the reader can compare values at a glance. Bars may be drawn vertically or horizontally; vertical bars are more common in practice.

Key Limitation

A simple bar diagram can show only one classification at a time. For instance, it can depict the total population over five decades — but not its sex-wise split.

1

Vertical Bars

Bars rise from a horizontal base. Most common form used in practice.

2

Horizontal Bars

Bars extend sideways from a vertical axis — useful for long category labels.

Illustration 1: Service Tax Collection

Represent the following data by a simple bar diagram

Years	2012–13	2013–14	2014–15	2015–16	2016–17	2017–18	2018–19
Service Tax	51,301	60,941	58,422	71,016	97,507	1,32,697	1,80,141

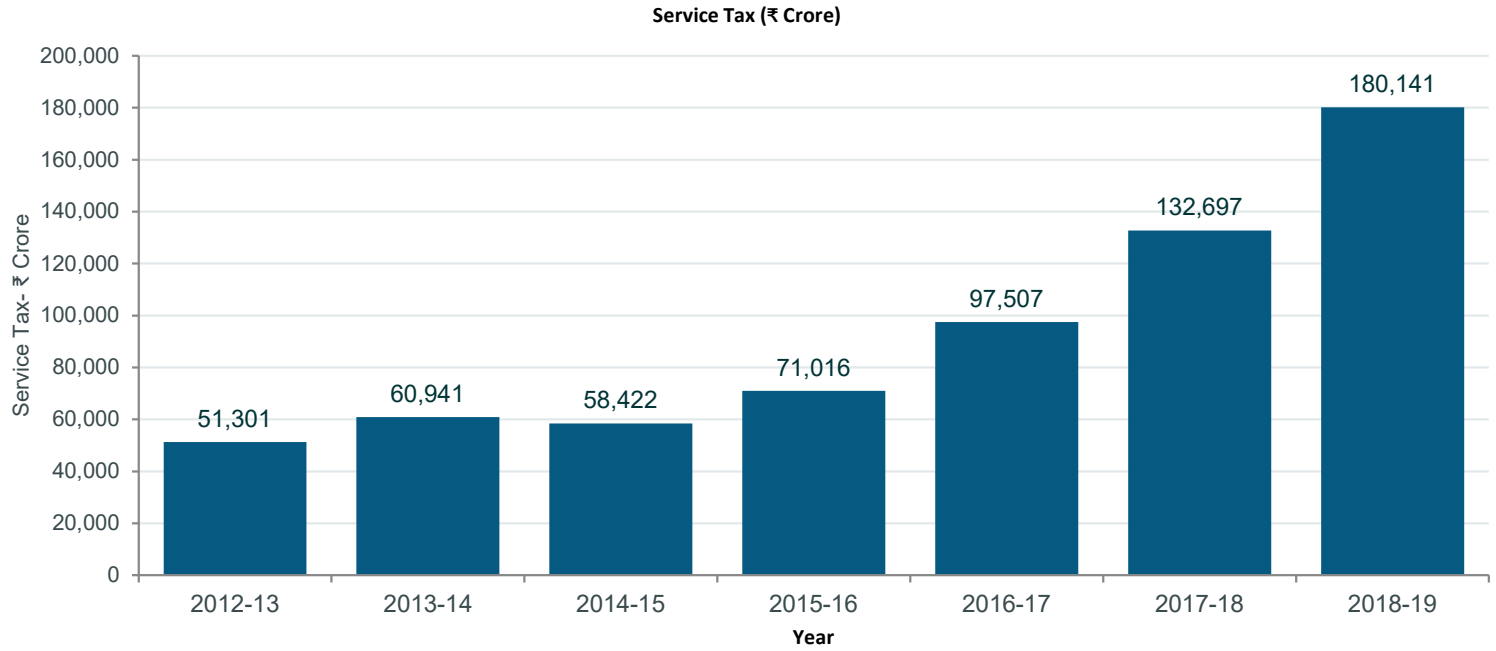
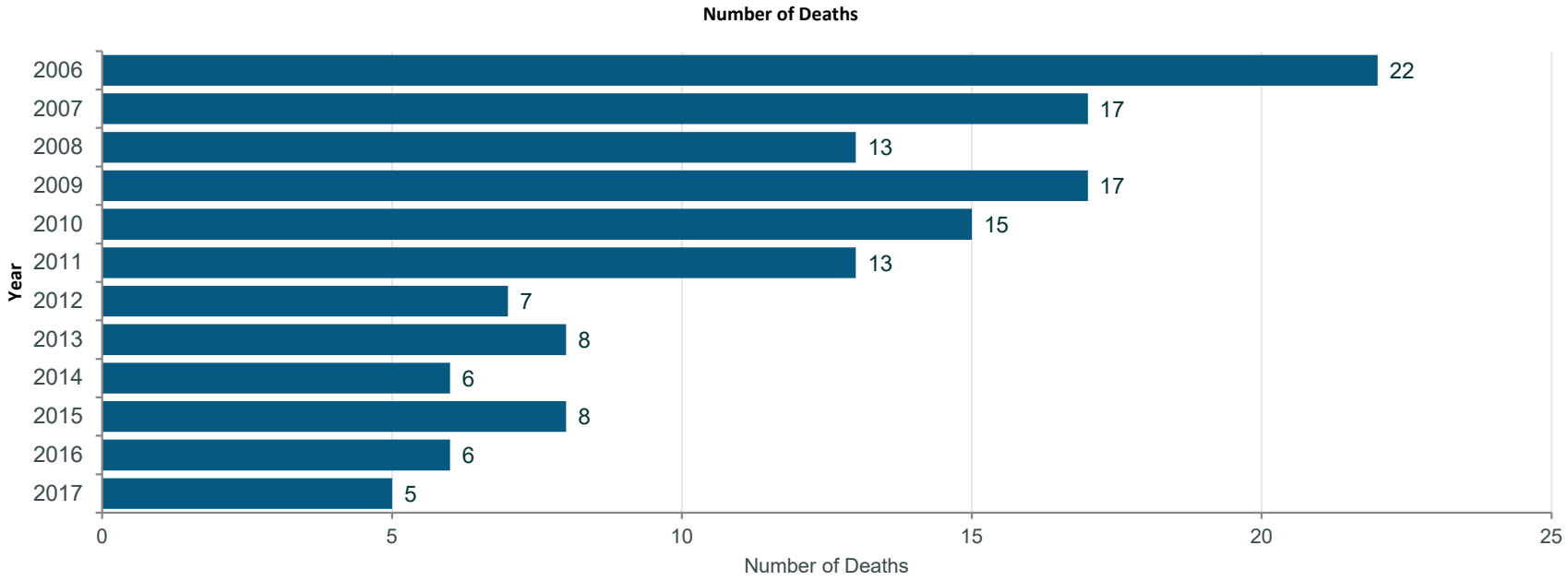


Illustration 3: Road Deaths in Delhi on Holi

The following data relate to road deaths in Delhi on the day of Holi from 2006 to 2017:

Years	2006	2007	2008	2009	2010	2011
Number of deaths	22	17	13	17	15	13
Years	2012	2013	2014	2015	2016	2017
Number of deaths	7	8	6	8	6	5



One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

What is a Sub-divided Bar Diagram?

Each bar representing a total is split into its components, with every segment sized in proportion to its share of that total.

1 Consistent ordering

Keep components in the same order in every bar — usually largest at the base, smallest at the top.

2 Shades, colours & a key

Use distinct shades or colours for each component, with an index/key explaining them.

3 A practical limit

Avoid using this diagram when there are more than 10–12 components — it becomes cluttered and hard to read.

4 Absolute or percentage data

Works well for either raw totals or distribution ratios (e.g., percentage shares).

5 Vertical or horizontal

Can be constructed on either a vertical or a horizontal base.

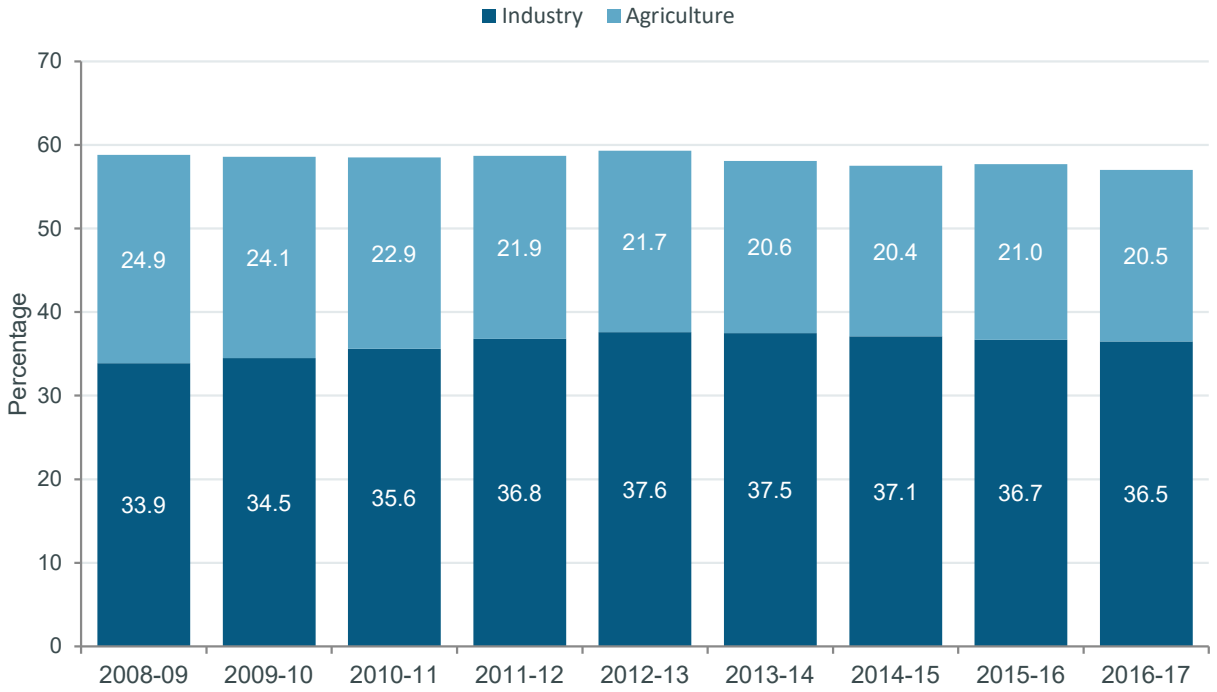
6 Typical use case

E.g., number of students across B.Com., M.Com., B.A. and M.A. in several colleges.

Illustration 4: Pattern of Electricity Consumption

The following table gives the pattern of electricity consumption. Represent the data by a suitable diagram.

Years	% Industry	% Agriculture	Total
2008–09	33.9	24.9	58.8
2009–10	34.5	24.1	58.6
2010–11	35.6	22.9	58.5
2011–12	36.8	21.9	58.7
2012–13	37.6	21.7	59.3
2013–14	37.5	20.6	58.1
2014–15	37.1	20.4	57.5
2015–16	36.7	21.0	57.7
2016–17	36.5	20.5	57.0



One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

Multiple Bar Diagrams

Two or more sets of interrelated data are represented side by side, using the same technique as a simple bar diagram.

1

Same technique as simple bars

Drawn just like a simple bar diagram — only now, more than one bar sits per category.

2

Distinguishing the bars

Different shades, colours, dots or crosses are used to tell the bars apart.

3

Best for comparison

Whenever two or more related variables need to be compared, this is the preferred diagram.

4

Typical use case

E.g., comparing imports and exports, or sales, gross profit and net profit, across years.

Example

Represent the following data by a suitable diagram:

Years	2013	2014	2015	2016	2017	2018	2019	2020
Imports ('000 tonnes)	400	450	560	620	560	450	500	540
Exports ('000 tonnes)	220	235	385	420	420	380	360	400

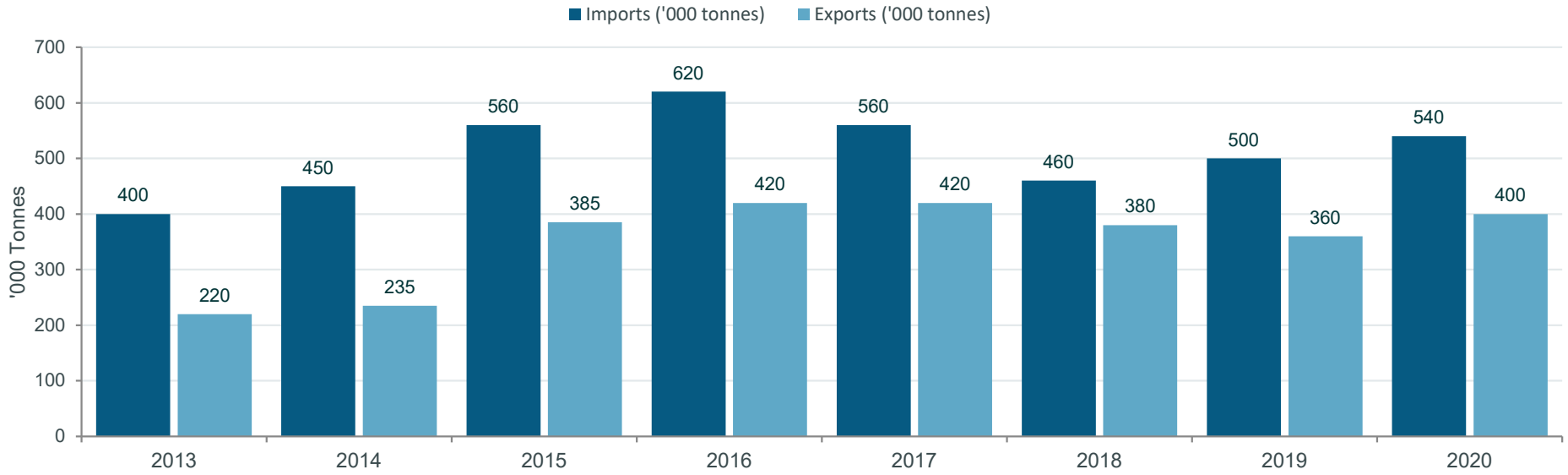
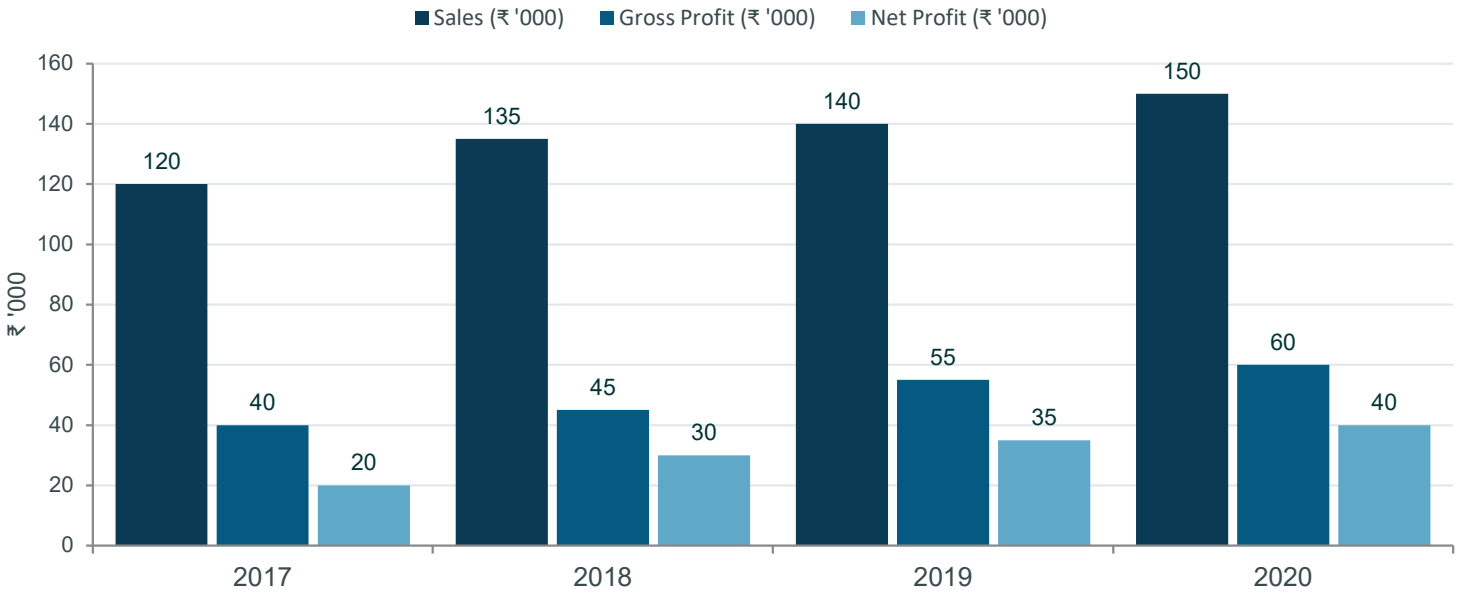


Illustration 7

Draw a multiple bar diagram from the following data:

Years	Sales (₹ '000)	Gross Profit (₹ '000)	Net Profit (₹ '000)
2017	120	40	20
2018	135	45	30
2019	140	55	35
2020	150	60	40



One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

Percentage Bar Diagrams

Useful when the goal is to compare the relative (not absolute) change in data across categories.

1

Every bar is equal in length

Each bar is scaled to represent 100%, regardless of the underlying total.

2

Segments show components

The bar is cut into segments, each representing a component's percentage share of the aggregate.

3

A special case

It is essentially a sub-divided bar diagram drawn on a percentage basis, rather than absolute values.

Percentage Bar Diagrams

Represent the following data by sub-divided bar diagram drawn on the percentage basis. ***Sale price per chair is taken as ₹100.***

Particulars	**2018**	**2019**	**2020**
1. Cost per chair			
(a) Wages	90	150	210
(b) Other costs	60	100	140
(c) Polishing	30	50	70
Total	180	300	420
2. Proceeds per chair	200	300	400
3. Profit (+) Loss (-)	+20	—	–20

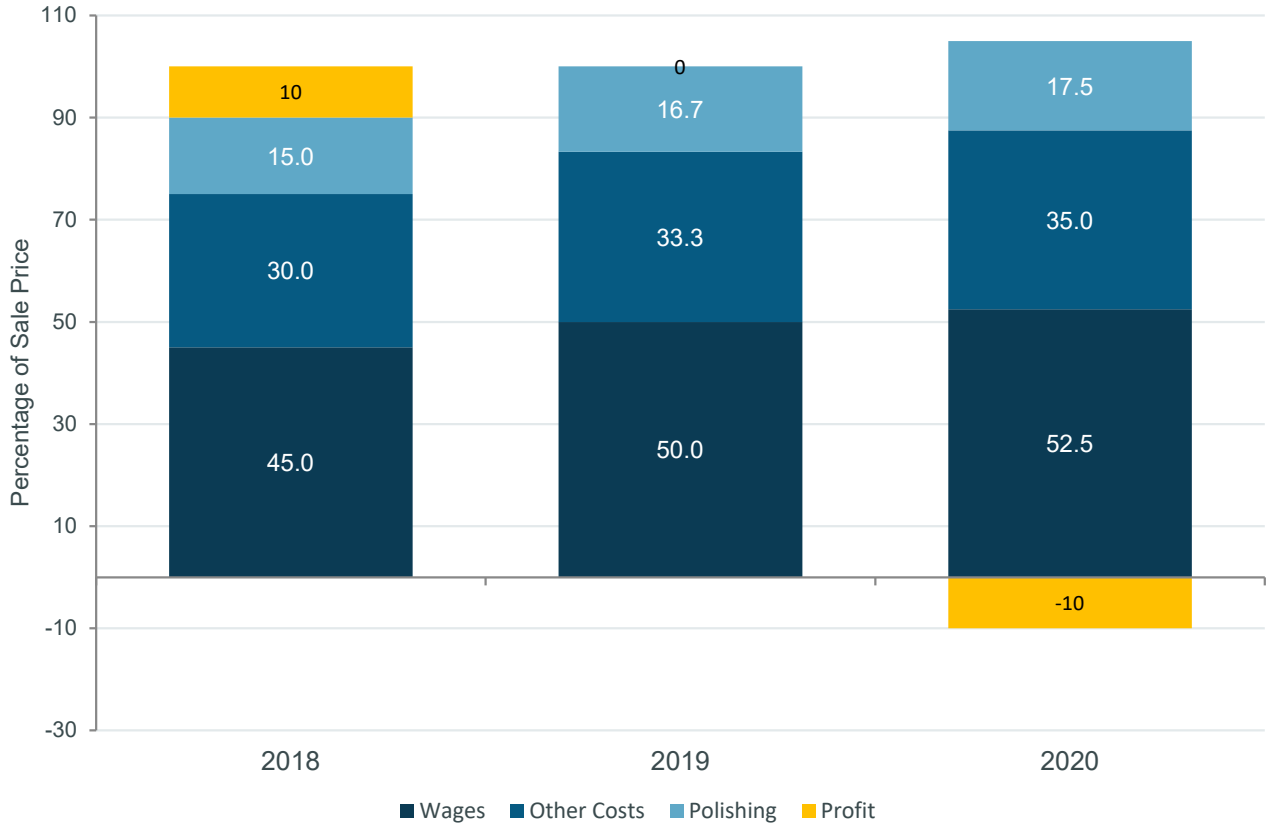
Particulars	**2018**	**2019**	**2020**
Wages	45.0	50.0	52.5
Other costs	30.0	33.3	35.0
Polishing	15.0	16.7	17.5
Total costs	90.0	100.0	105.0
Sale price	100.0	100.0	100.0
Profit or loss	+10.0	—	–5.0

Take the sale price per chair as ₹ 100 and express the other figures in percentages. The percentages so obtained are given below: **Cost per chair/Proceeds per chair**

Illustration 8

Represent the following data by sub-divided bar diagram drawn on the percentage basis. ***Sale price per chair is taken as ₹100.***

Particulars	2018	2019	2020
Wages	45.0	50.0	52.5
Other costs	30.0	33.3	35.0
Polishing	15.0	16.7	17.5
Total costs	90.0	100.0	105.0
Sale price	100.0	100.0	100.0
Profit or loss	+10.0	—	–5.0



One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

What is a Deviation Bar Diagram?

Popularly used for representing net quantities — excess or deficit — such as net profit, net loss, or net exports and imports.

1

Positive & negative values

Unlike most bar diagrams, deviation bars can represent both positive and negative quantities in the same chart.

2

A baseline splits the chart

A horizontal baseline (zero line) runs through the diagram, separating positive from negative values.

3

Above the baseline

Positive values — such as a profit or a surplus — are drawn as bars rising above the baseline.

4

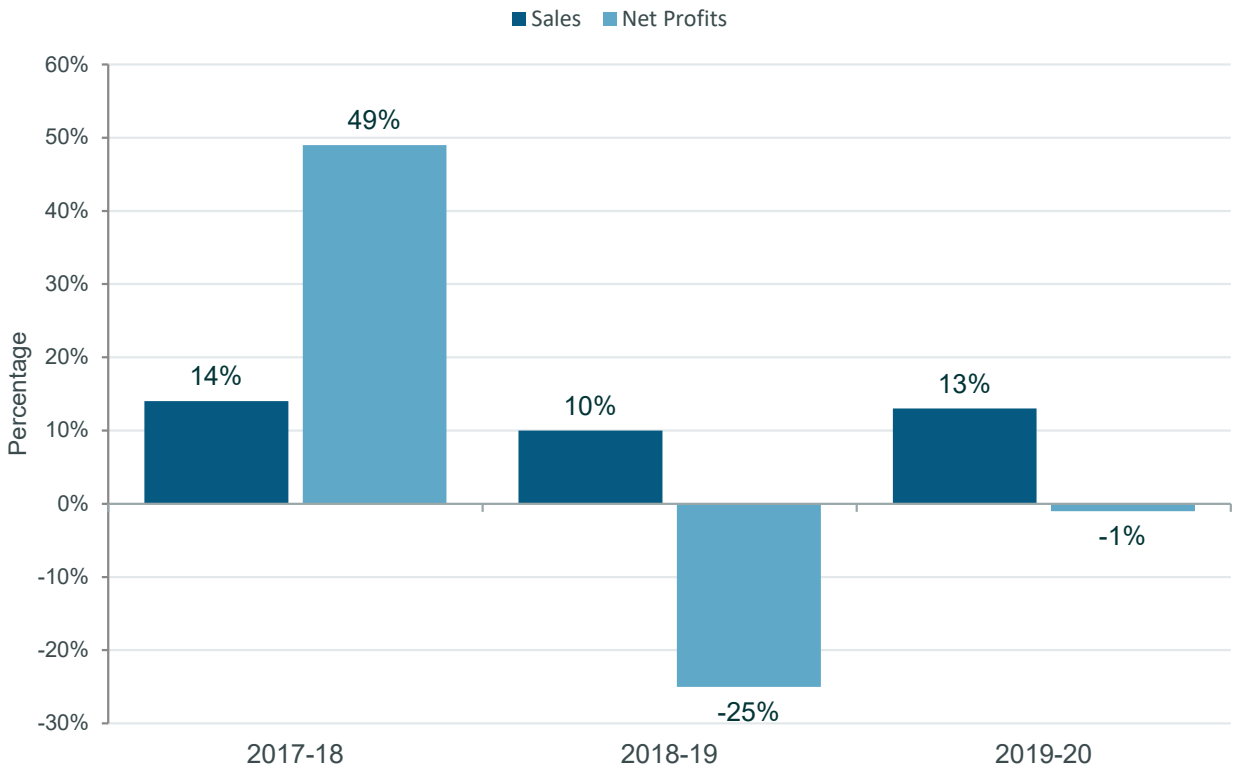
Below the baseline

Negative values — such as a loss or a deficit — are drawn as bars extending below the baseline.

Illustration 9

Present the following data by a suitable diagram showing the Sales and Net Profits of private industrial companies:

Years	Sales	Net Profits
2017-18	14%	49%
2018-19	10%	–25%
2019-20	13%	–1%



One-Dimensional Diagrams

Bar Diagrams — length of the bar represents the value

a

Simple Bar Diagram

One bar per category, showing a single value only.

b

Sub-divided Bar Diagram

Each bar split into segments showing the parts of a total.

c

Multiple Bar Diagram

Two or more bars side by side per category, for comparison.

d

Percentage Bar Diagram

Sub-divided bars converted to %, all bars equal to 100% height.

e

Deviation Bar Diagram

Bars drawn above/below a baseline for net or +/- values.

f

Broken Bar Diagram

A zig-zag break in a bar to save space when one value is very large.

Broken Bar Graph: The Concept

Why a bar sometimes needs to break

1. **Wide variation**
   Some values in a series are very small, others very large — a single scale can't show both clearly.

2. **The problem**
   Fitting the largest bar forces every small bar into just a sliver of the chart.

3. **The fix**
   The tallest bar is deliberately ‘broken’ with a zig-zag, freeing up vertical space for the smaller bars to be seen properly.

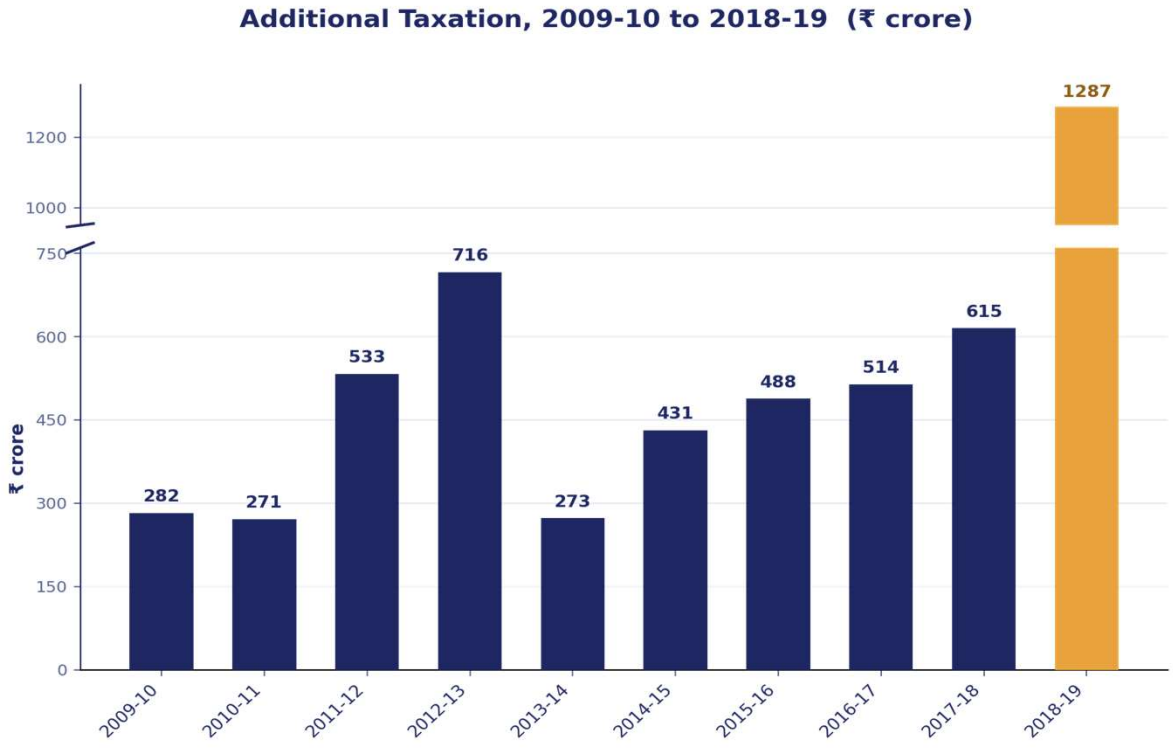
4. **Trade-off**
   The broken bar's true value is still labelled — only its drawn height is compressed, not the data.

Additional Taxation (₹ crore)

Year	₹ crore
2009-10	282
2010-11	271
2011-12	533
2012-13	716
2013-14	273
2014-15	431
2015-16	488
2016-17	514
2017-18	615
2018-19	**1287**

Additional Taxation, 2009-10 to 2018-19

₹ crore · broken-bar diagram



Largest bar (2018-19) is broken to keep the smaller bars readable — actual value 1287 is still labelled above it.

Types of Diagrams

Diagrams are grouped by the number of dimensions used to represent data.

1

One-Dimensional

(Bar Diagrams)

Only the length of the bar matters. 6 sub-types: simple, sub-divided, multiple, percentage, deviation, broken bar.

2

Two-Dimensional

(Area Diagrams)

Length and width both matter — area represents data. 3 sub-types: rectangles, squares, circles/pie.

3

~~Three-Dimensional~~

(Volume Diagrams)

Length, width and height all matter. 3 sub-types: cubes, cylinders, spheres.

4

~~Pictograms & Cartograms~~

(Pictorial / Map)

Data shown via picture symbols (pictograms) or shaded statistical maps (cartograms).

Two-Dimensional Diagrams

Area Diagrams — both length and width represent the value

a

Rectangles

Area is proportional to the value; width and height can both encode data, useful for comparing two variables at once.

b

Squares

Side length is proportional to the square root of the value — keeps figures compact when values vary widely.

c

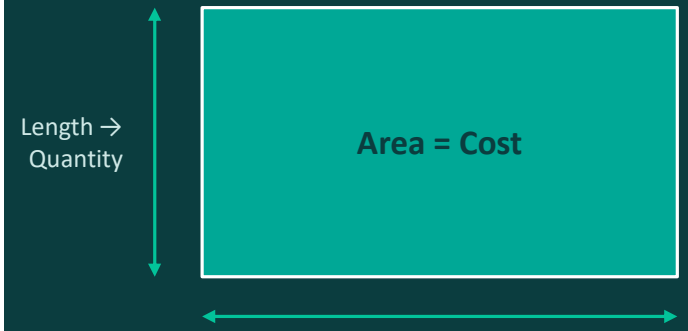
Circles / Pie Diagrams

Circle area (or its radius) represents the total; slices (sectors) show how it splits into components.

What Is a Rectangular Diagram?

Area of a rectangle = Length × Width

A rectangle's area is fixed by both its dimensions, so a single shape can carry two facts at once — e.g. price and quantity, or total cost and its make-up.



Either method below may be used whenever two or more sets of figures are shown as rectangles.

1 · Actual Value Method

Plot the raw figures exactly as given — length and width represent the real totals and components, with no conversion.

2 · Percentage Method (preferred)

Convert every total to 100%, then sub-divide the length into each component's percentage share — this puts groups of different size on the same scale for direct comparison.

Example 11

Represent the following data on details of cost of two commodities by the rectangular diagram:

Given Data

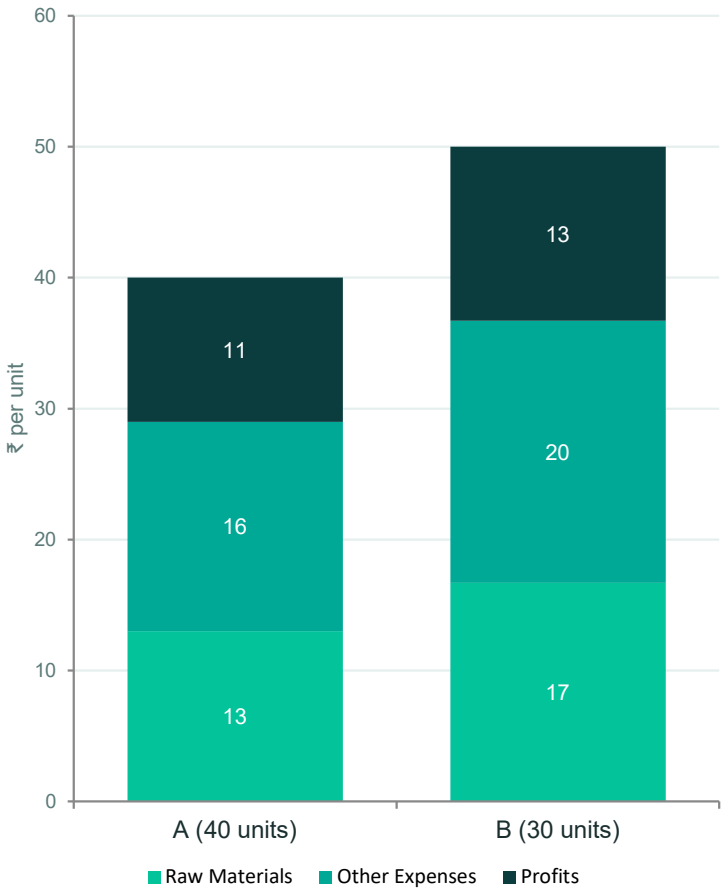
Details	A	B
Price per unit	₹40	₹50
Quantity sold	40 u	30 u
Raw materials	₹520	₹500
Other expenses	₹640	₹600
Profits	₹440	₹400

Per-Unit Solution

Item	A ₹/u	B ₹/u
Raw materials	13.0	16.7
Other expenses	16.0	20.0
Profits	11.0	13.3
Total	**40.0**	**50.0**

Reading it: each rectangle's width (length of the base) equals units sold — 40 for A, 30 for B — and its height stacks raw materials, other expenses, and profit per unit.

Costs & Profits per Unit



Example 12

The following data relate to the monthly expenditure (in rupees) of two families A and B. Represent the above data by a suitable percentage diagram.

Given Data (₹)

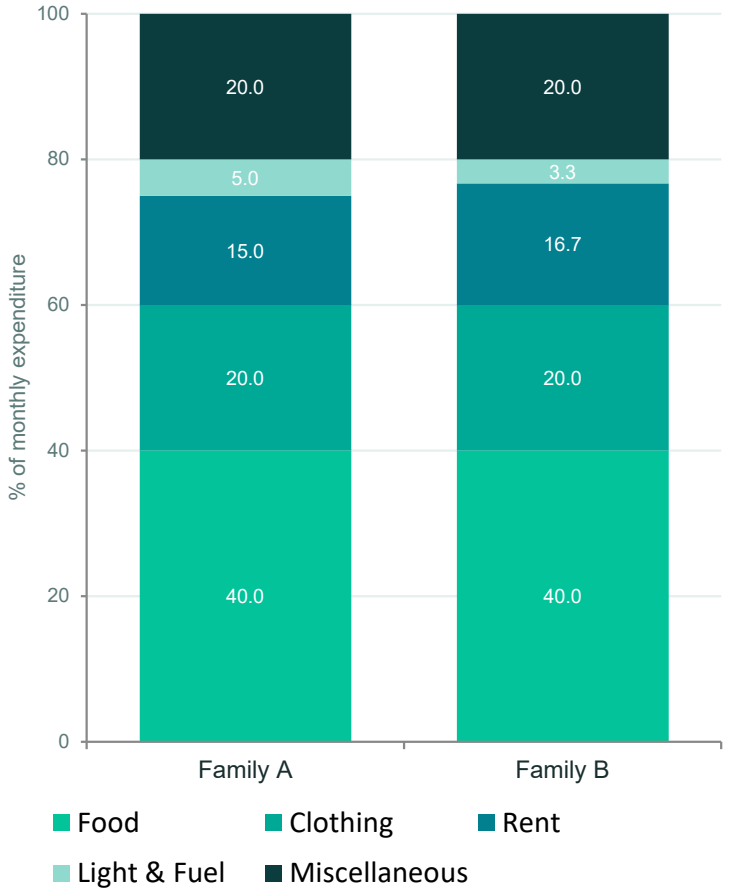
Item	A	B
Food	16,000	12,000
Clothing	8,000	6,000
Rent	6,000	5,000
Light & Fuel	2,000	1,000
Miscellaneous	8,000	6,000
Total	**40,000**	**30,000**

Percentage

Item	A %	B %
Food	40.0	40.0
Clothing	20.0	20.0
Rent	15.0	16.7
Light & Fuel	5.0	3.3
Misc.	20.0	20.0
Total	**100**	**100**

Note: Family A spends 40,000 and Family B spends 30,000 — different totals, so converting each to 100% makes the two directly comparable, stacked as equal-length rectangles.

Percentage Rectangular Diagram



Two-Dimensional Diagrams

Area Diagrams — both length and width represent the value

a

Rectangles

Area is proportional to the value; width and height can both encode data, useful for comparing two variables at once.

b

Squares

Side length is proportional to the square root of the value — keeps figures compact when values vary widely.

c

Circles / Pie Diagrams

Circle area (or its radius) represents the total; slices (sectors) show how it splits into components.

Why Use Squares?

The Problem With Rectangles

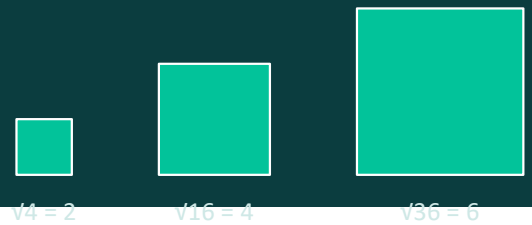
The rectangular method breaks down when values vary widely. If two commodities sold 20 and 240 units, the rectangle widths would sit in a 1 : 12 ratio — producing an unwieldy, hard-to-read diagram.

The Fix

Take the square root of every value first. Because area grows with the square of the side, a modest square can still represent a huge underlying number — compressing wide value ranges into a manageable, comparable set of shapes.

Method

1. Take the square root of each value to be represented.
2. Choose a scale (e.g. divide every square root by a constant) to keep the drawing a convenient size.
3. Draw each square with that side length — larger values naturally produce larger squares.



Example 13

Represent the following data by a suitable diagram: Total Public Sector Outlay (₹ '000 crore)

Five-year Plan	I	II	III	IV	V	VI	VII	VIII
Outlay	1,960	4,672	8,577	16,566	35,595	69,380	1,80,000	3,25,000

Since values swing from 1,960 to 3,25,000, a square diagram — side $\propto \sqrt{\text{outlay}}$ — keeps the picture readable.

Calculations for Drawing the Square Diagram

Plan	Outlay	Square Root	Side of Square (cm)
I	1,960	44.27	0.22
II	4,672	68.35	0.34
III	8,577	92.61	0.64
IV	16,566	128.71	0.64
V	35,595	188.67	0.94
VI	69,380	263.40	1.31
VII	1,80,000	424.26	2.12
VIII	3,25,000	570.08	2.85

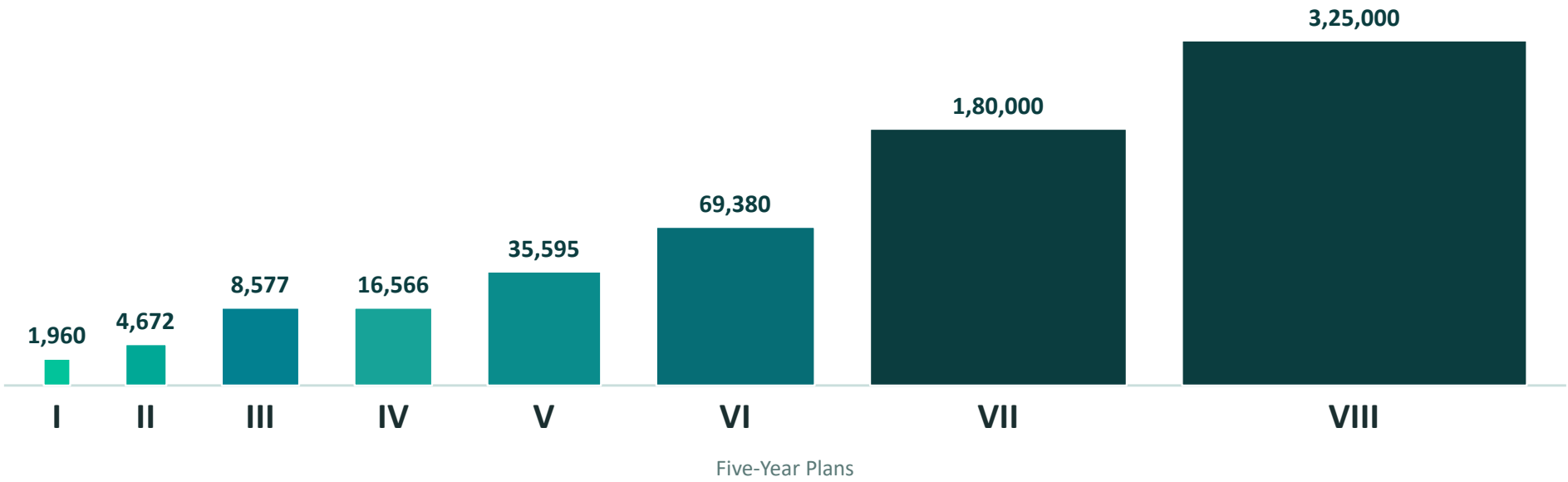
Note

Each square root is divided by a constant scale — here, 200 — to get a workable side length in centimetres.

The side of every square is proportional to the square root of the value it represents, not the value itself.

Illustration 13 — The Square Diagram

Total Public Sector Outlay by Five-Year Plan — side of each square ∝ √outlay



Two-Dimensional Diagrams

Area Diagrams — both length and width represent the value

a

Rectangles

Area is proportional to the value; width and height can both encode data, useful for comparing two variables at once.

b

Squares

Side length is proportional to the square root of the value — keeps figures compact when values vary widely.

c

Circles / Pie Diagrams

Circle area (or its radius) represents the total; slices (sectors) show how it splits into components.

The Idea Behind Circle Diagrams

Reading data through the area of a circle, not its diameter

1. **A two-dimensional diagram**

   Unlike bars, circles show magnitude as area — allowing both a total and its component sectors to be represented in one shape.

2. **Area, not length, carries the meaning**

   The area of a circle is proportional to the square of its radius, so larger values must grow the radius, not the area, linearly.

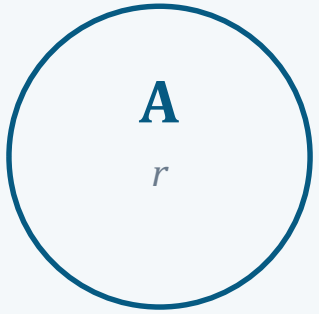
3. **Radius is derived from area**

   Just as squares need a square root construction, circle radii are obtained by dividing the value by π and taking the square root.

4. **A flexible substitute for squares**

   Circles can be used in every situation where squares apply, though both remain harder to compare precisely than bars.

KEY FORMULA



$$\text{Area} = \pi \times r^2$$

$$r^2 = \text{Area} / \pi$$

$$r = \sqrt{(\text{Area} / \pi)}$$

A = Area of the circle · r = Radius

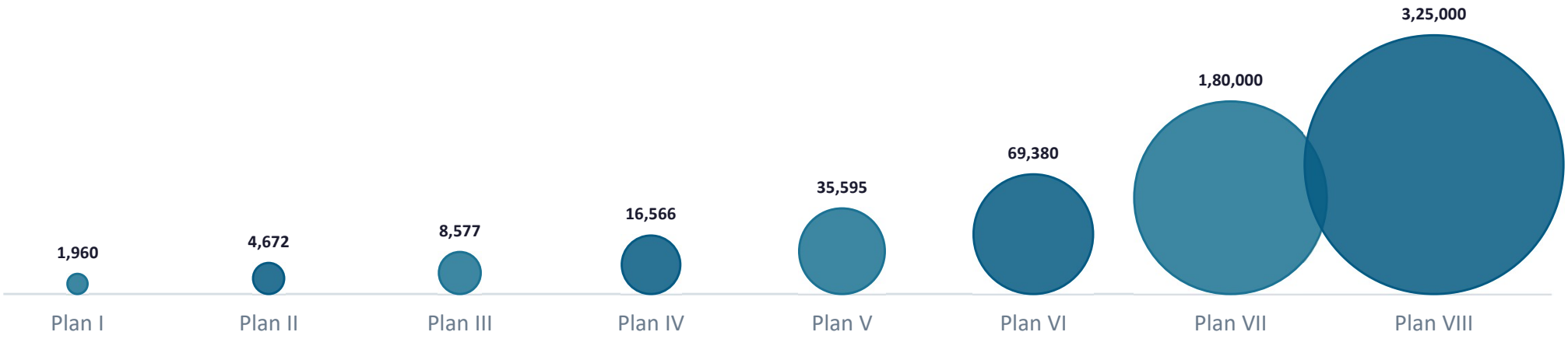
Illustration: Public Sector Outlay

Working out the radius for each Five-Year Plan (₹ crore)

Five-Year Plan	Outlay (n)	n ÷ (22/7)	Radius = √n÷(22/7)
I	1,960.00	624.20	24.98
II	4,672.00	1,487.90	38.57
III	8,577.00	2,731.53	52.26
IV	16,566.00	5,275.80	72.63
V	35,595.00	11,335.99	106.47
VI	69,380.00	22,095.54	148.65
VII	1,80,000.00	57,324.84	239.43
VIII	3,25,000.00	1,03,503.18	321.72

Public Sector Outlay — By Circles

Circle area is proportional to outlay; circles are baseline-aligned



Values in ₹ crore

What Is a Pie Diagram?

1. **Slices of a whole**
   The entire graph looks like a pie, and each component resembles a slice cut from it — ideal for showing how a total (like govt. expenditure) splits across categories.
2. **Compare on a percentage basis**
   Comparing several pies on absolute figures requires larger totals to be drawn as larger circles, which complicates two-dimensional comparison. On a percentage basis, all pies can be equal in size.
3. **One rule, one exception**

   A single pie diagram never runs into this sizing problem — the issue only arises when comparing multiple pies side by side.
4. **Logical sector order**
   Sectors are conventionally arranged by size, largest at 12 o'clock, running clockwise — with catch-all categories like “miscellaneous” or “others” placed last.

THREE STEPS TO BUILD ONE

1. **Convert to degrees**
   Angle = Percentage × 3.6° (since 360° / 100 = 3.6)
2. **Draw the circle**
   Radius sized to fit the available space on the page
3. **Mark the sectors**
   Use a protractor; label inside sectors, or outside with arrows if space is tight

Angle = (Percentage ÷ 100) × 360°

Guideline: keep a pie diagram to five or six categories at most

Illustration: Sixth Plan Public Sector Outlays

Converting percentage outlay into degrees of angle (Percentage × 3.6)

Sector	Percentage	Angle
Energy	27.2	98°
Social Services and Others	16.1	58°
Transport, Communication, etc.	15.9	57°
Industry and Minerals	15.4	56°
Agriculture and Rural Development	12.9	46°
Irrigation, etc.	12.5	45°

Total = 100.0% → 360°

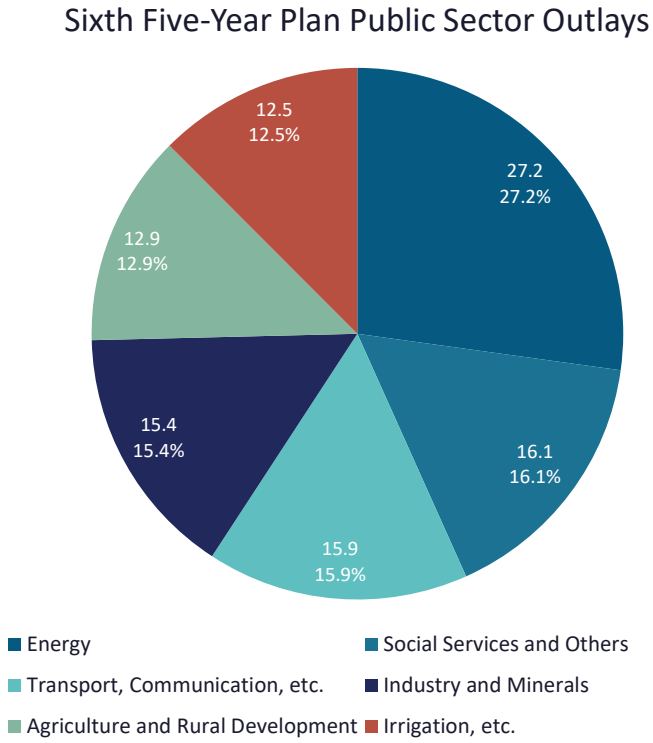
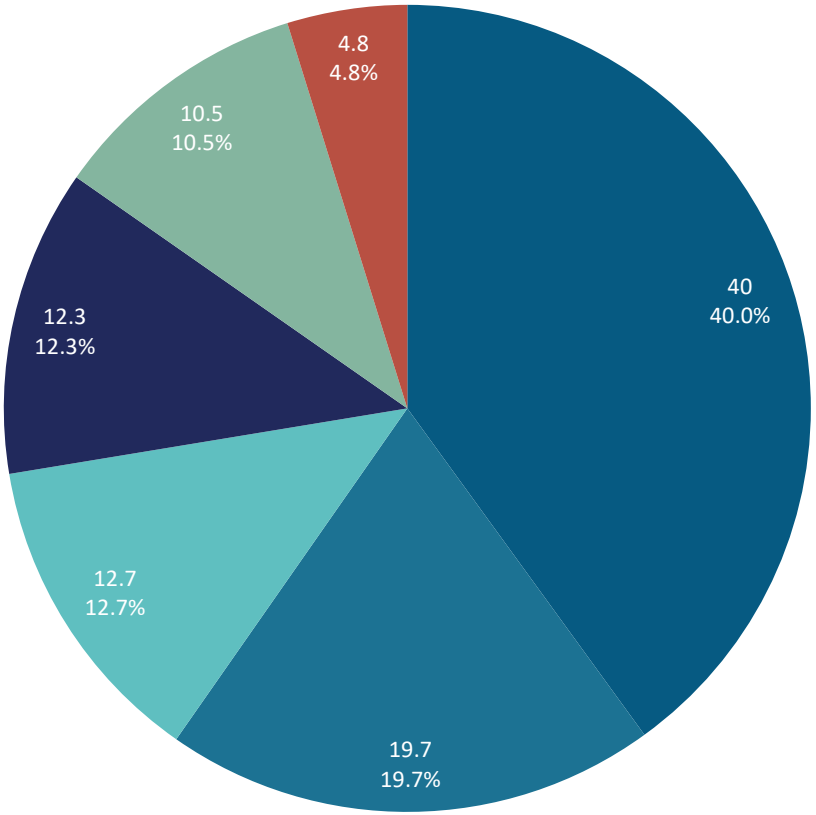


Illustration: Who Bribes Whom

A single-series survey result represented with one pie diagram



SECTORS

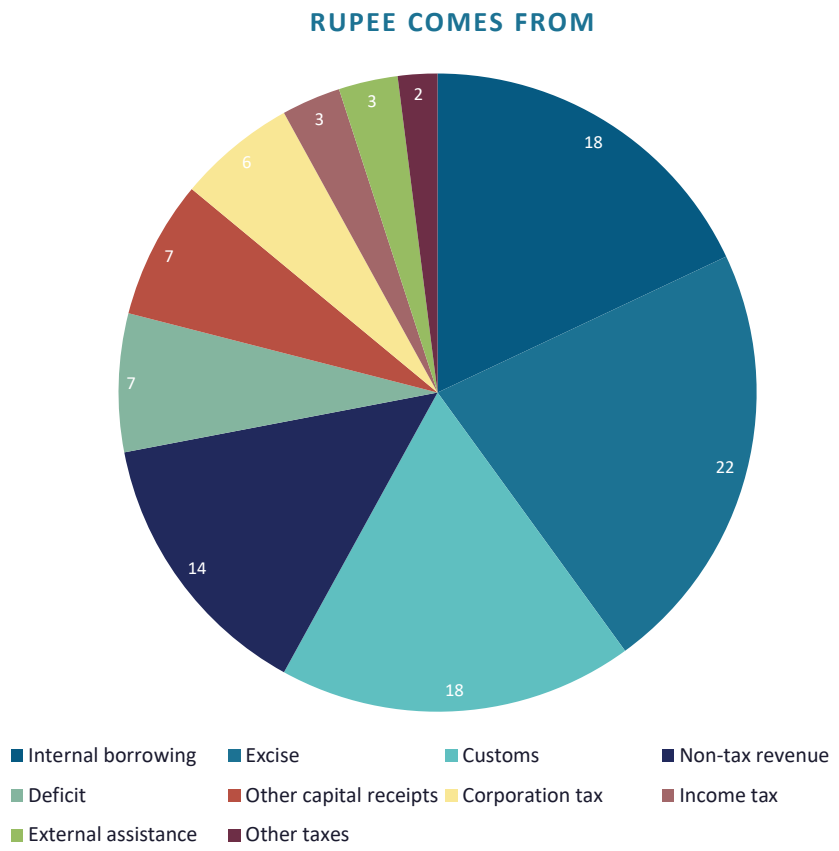
Company to Government	**40.0%**
Company to company	**19.7%**
Company to customers	**12.7%**
Company to employees	**12.3%**
Companies to tax authorities	**10.5%**
Employees to employees	**4.8%**

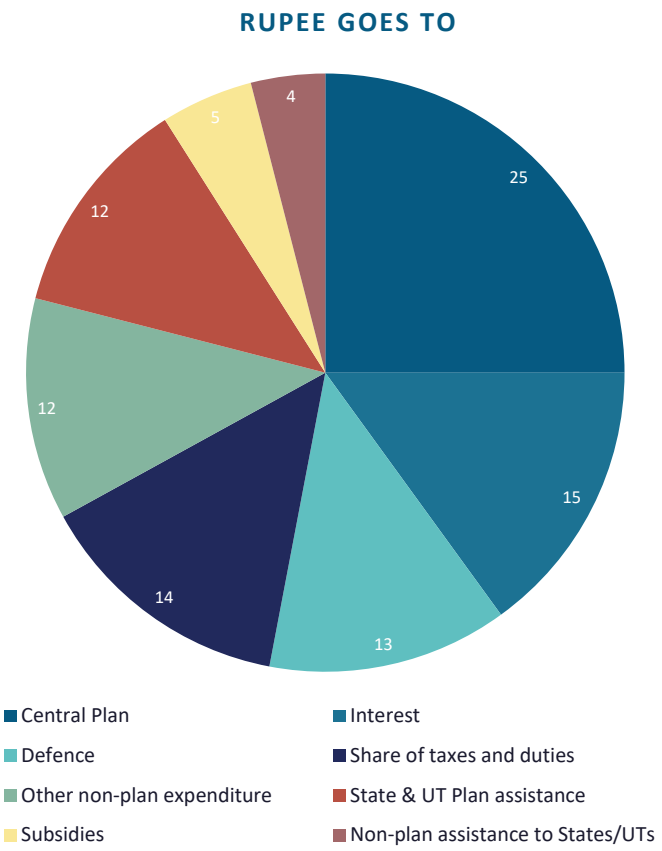
Illustration: How the Rupee Comes & Goes

The following data relate to how the Rupee comes and goes:

Rupee comes	Paise	Rupee goes	Paise
Excise	22	Central Plan	25
Customs	18	Interest	15
Internal borrowing	18	Defence	13
Non-tax revenue	14	Share of taxes and duties	14
Deficit	7	Other non-plan expenditure	12
Other capital receipts	7	State & UT Plan assistance	12
Corporation tax	6	Subsidies	5
Income tax	3	Non-plan assistance to State and UT Govts.	4
External assistance	3	—	—
Other taxes	2	—	—

Illustration: How the Rupee Comes & Goes





Values in paise per rupee

Types of Diagrams

Diagrams are grouped by the number of dimensions used to represent data.

1

One-Dimensional

(Bar Diagrams)

Only the length of the bar matters. 6 sub-types: simple, sub-divided, multiple, percentage, deviation, broken bar.

2

Two-Dimensional

(Area Diagrams)

Length and width both matter — area represents data. 3 sub-types: rectangles, squares, circles/pie.

3

~~Three-Dimensional~~

(Volume Diagrams)

Length, width and height all matter. 3 sub-types: cubes, cylinders, spheres.

4

~~Pictograms & Cartograms~~

(Pictorial / Map)

Data shown via picture symbols (pictograms) or shaded statistical maps (cartograms).

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

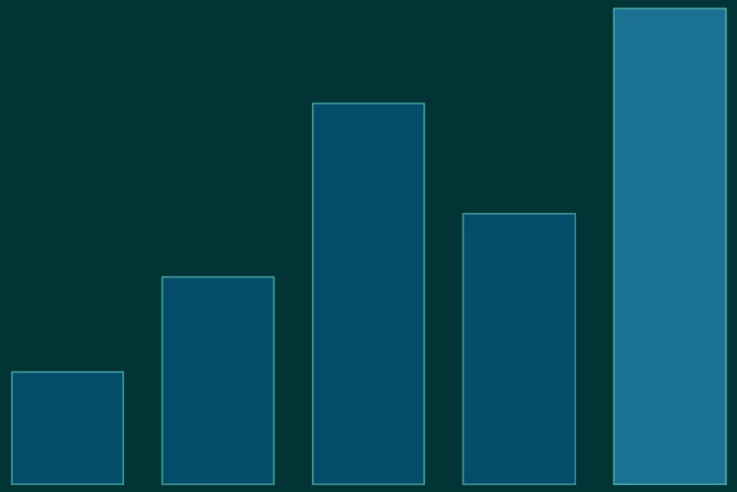
Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

STATISTICAL METHODS · CHAPTER 6.2

Graphs

Presenting Statistical Data Through Visual Communication

A picture is worth 10,000 words.



A Large Variety, Two Broad Heads

A large variety of graphs are used in practice. Broadly, they can be divided under two heads.

T Graphs of Time Series

- Time is plotted on the horizontal (X) axis
- The variable under study is plotted on the vertical (Y) axis
- Also called historigrams — they show how a value changes over successive time periods

F Graphs of Frequency Distributions

- Used to show how observations are spread across classes or values
- Includes histograms, frequency polygons, and ogives
- Reveal the shape, spread, and concentration of a data set at a glance

Constructing charts and graphs is an art, acquired through practice and a few simple rules.

Basic Elements of a Graph

Every graph is built on graph paper using two intersecting reference lines.

1. **Coordinate Axes**

   Two simple lines drawn intersecting each other at right angles.

2. **Point of Origin**

   The point of intersection of the two axes — the ‘zero’ point.

3. **Axis of X (Abscissa)**

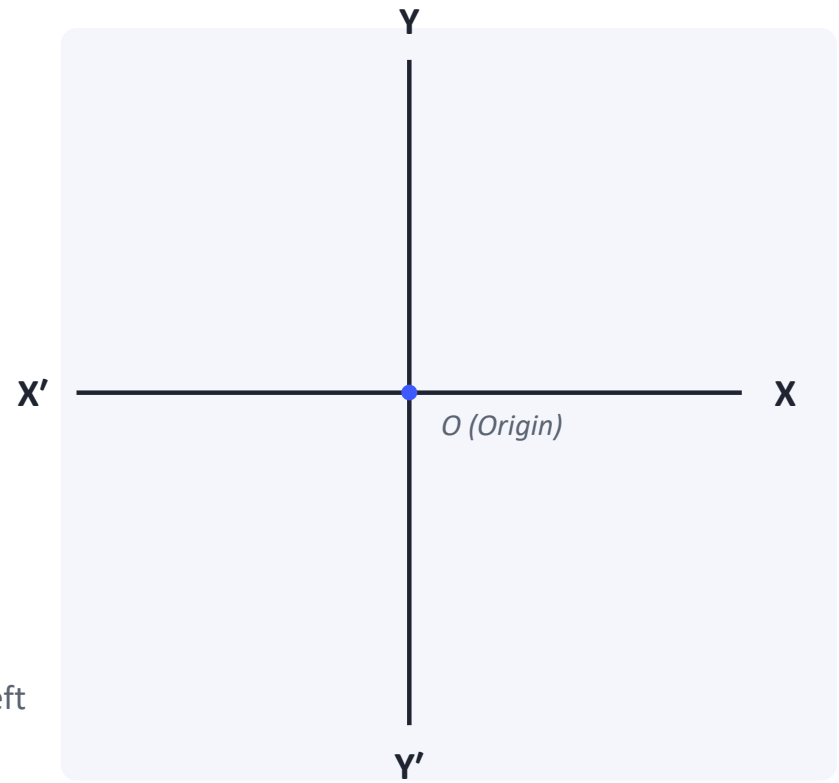
   The horizontal line, also called the X-axis.

4. **Axis of Y (Ordinate)**

   The vertical line, also called the Y-axis.

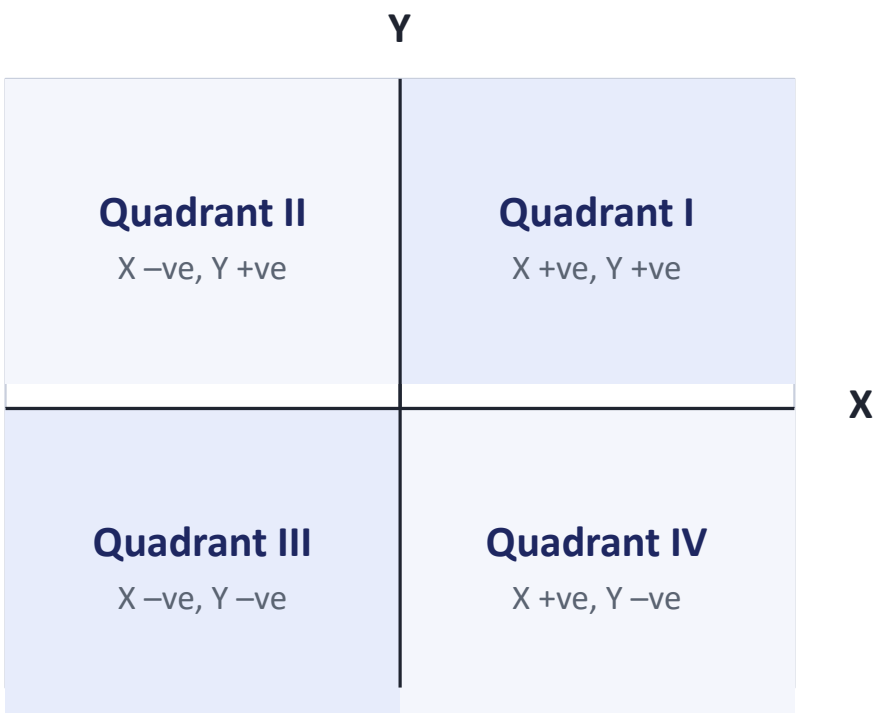
5. **Sign Convention**

   Distances measured right or upward from the origin are positive; left or downward are negative.



The Four Quadrants

The whole plotting area is divided into four quadrants by the coordinate axes.



- **I** Quadrant I: both X and Y are positive.
- **II** Quadrant II: Y is positive, X is negative.
- **III** Quadrant III: both X and Y are negative.
- **IV** Quadrant IV: X is positive, Y is negative.

Since most business data are positive, Quadrant I is the one most frequently used in practice.

Choosing Variables and Scale

Convention guides which variable goes where, and how the scale is chosen.

Independent Variable → X-axis

The independent variable is conventionally placed on the horizontal scale.

Dependent Variable → Y-axis

The dependent variable is placed on the vertical scale.

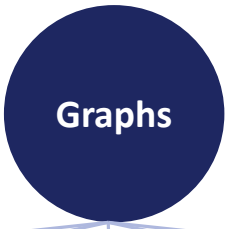
Time Series Convention

For time series, time itself goes on the horizontal axis, and the variable being studied on the vertical axis.

A Convenient, Consistent Scale

Each axis gets a scale representing a convenient unit, chosen so the entire data set fits the available space. The X and Y scales need not be identical.

Various Graphs



1. **Line Graph**
   - Graph of One Variable
   - Graph of Two Variables
   - ~~Graph of Two Scales~~
2. ~~**Range Chart**~~
3. ~~**Band Graph**~~
4. ~~**Ratio Charts**~~
5. **Graphs of Frequency Distribution**
   - Histogram
   - Frequency Polygon
   - ~~Smoothed Frequency Curve~~
   - ~~Cumulative Frequency Curves (Ogives)~~

What Are Line Graphs?

When we observe the values of a variable at different points of time, the series so formed is known as a time series. Plotting time on the X-axis and the variable on the Y-axis, then joining the points with straight lines, gives a line graph.

1 **Simplest to Make**

Requires the least technical skill to construct and understand.

2 **Highly Adaptable**

Suits many uses and presents complex information clearly.

3 **Easy to Compare**

Many variables can be shown on the same graph for comparison.

Two Ways to Construct a Line Graph

Natural (Arithmetic) Scale

Shows absolute changes in the variable from one period to another.

Ratio Scale

Shows the rate of change, or relative changes, from one period to another.

Rules for Constructing Line Graphs (Natural Scale)

1 Time on X, Variable on Y

State the time unit clearly in the title — e.g., calendar or financial year.

2 Y-axis Begins at Zero

Choose a scale so all data fits; equal distances must represent equal magnitude on each axis separately.

3 X-axis Longer than Y-axis

If fluctuations are small or the lowest value is large, use a false base instead.

4 Plot with Dots, Join with Straight Lines

Use dots (not circles or crosses) so they disappear into the line — never curves.

5 Distinguish Multiple Variables

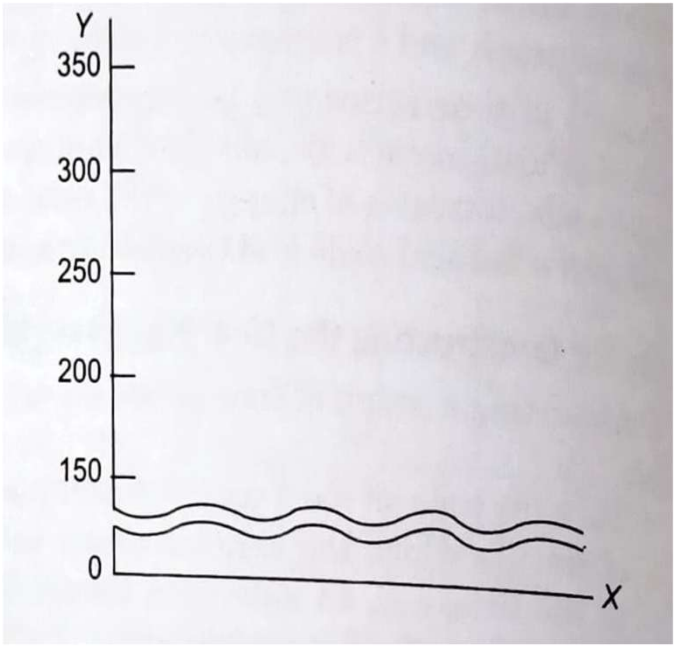
Use thick, thin, dotted lines or colours; always give the graph a clear, suitable title.

6 Letter Horizontally

Years, units, and labels should read without turning the graph sideways.

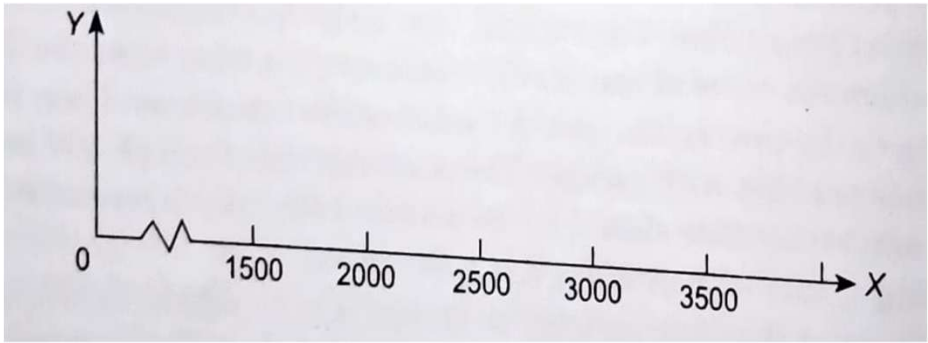
False Baseline

When the lowest value is relatively high and a detailed scale is needed, starting the Y-axis at zero wastes space or hides fluctuations. The solution: break the scale with a wavy (kinked/zigzag) line between zero and the first plotted unit.



Three Objects of a False Baseline

1. Variations in the data are clearly shown.
2. A large part of the graph is not wasted — space is saved.
3. The graph provides better visual communication.



Graph of One Variable

Represent the following data of per capita income graphically.

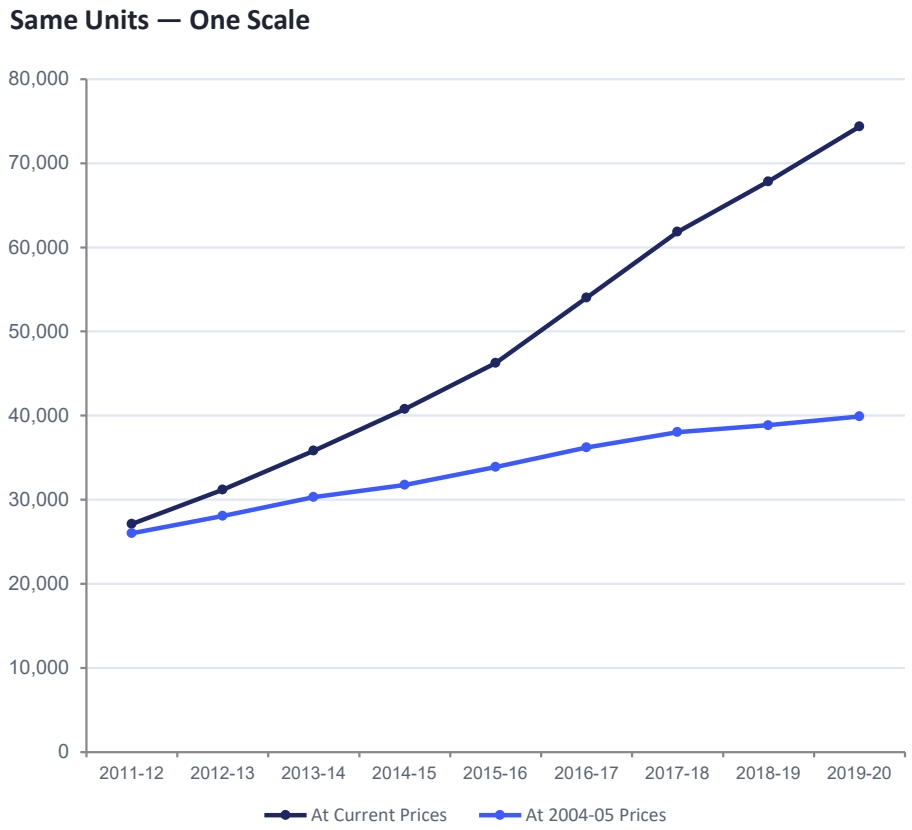
Years	Per Capita Income (At current prices)
2008–09	24,143
2009–10	27,131
2010–11	31,206
2011–12	35,825
2012–13	40,775
2013–14	46,249
2014–15	54,021
2015–16	61,855
2016–17	67,839
2017–18	74,380



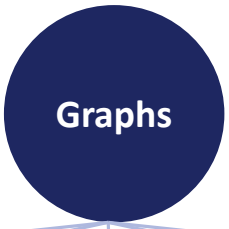
Two or More Variables

Represent the following data by a suitable graph.

Years	Per Capita National Product (₹) At Current Prices	Per Capita National Product (₹) At 2004–05 Prices
2011–2012	27,131	26,015
2012–2013	31,206	28,067
2013–2014	35,825	30,332
2014–2015	40,775	31,754
2015–2016	46,249	33,901
2016–2017	54,021	36,202
2017–2018	61,855	38,048
2018–2019	67,839	38,856
2019–2020	74,380	39,904

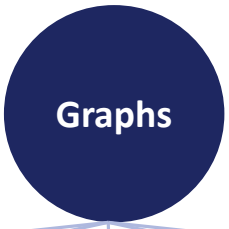


Various Graphs



1. **Line Graph**
   - Graph of One Variable
   - Graph of Two Variables
   - ~~Graph of Two Scales~~
2. ~~**Range Chart**~~
3. ~~**Band Graph**~~
4. ~~**Ratio Charts**~~
5. **Graphs of Frequency Distribution**
   - Histogram
   - Frequency Polygon
   - ~~Smoothed Frequency Curve~~
   - ~~Cumulative Frequency Curves (Ogives)~~

Various Graphs



1. **Line Graph**
   - Graph of One Variable
   - Graph of Two Variables
   - Graph of Two Scales
2. **Range Chart**
3. **Band Graph**
4. **Ratio Charts**
5. **Graphs of Frequency Distribution**
   - Histogram
   - Frequency Polygon
   - Smoothed Frequency Curve
   - Cumulative Frequency Curves (Ogives)

What Is a Histogram?

A set of vertical bars whose areas are proportional to the frequencies represented

Out of several methods of presenting a frequency distribution graphically, the histogram is the **most popular and widely used** technique in practice.

- The variable is always plotted on the X-axis; frequencies depending on it are plotted on the Y-axis.
- Each class is represented by a distance on the scale proportional to its class interval.
- The height of each rectangle (Y-axis) equals the frequency of that class.
- The area of the histogram represents the total frequency distributed across all classes.

Histogram vs. Bar Diagram

Bar Diagram

One-dimensional — only the length of the bar matters, not its width.

Histogram

Two-dimensional — both the length (height) and the width of each rectangle are significant.

Limitation: a histogram cannot be constructed for a distribution with open-end classes.

Constructing a Histogram

The technique differs for equal and unequal class intervals

i Equal Class Intervals

Take frequency on the Y-axis and the variable on the X-axis, then construct adjacent rectangles. The height of each rectangle is directly proportional to its class frequency — no adjustment is required.

Result: a histogram with uniform bar width, comparable directly by height alone.

ii Unequal Class Intervals

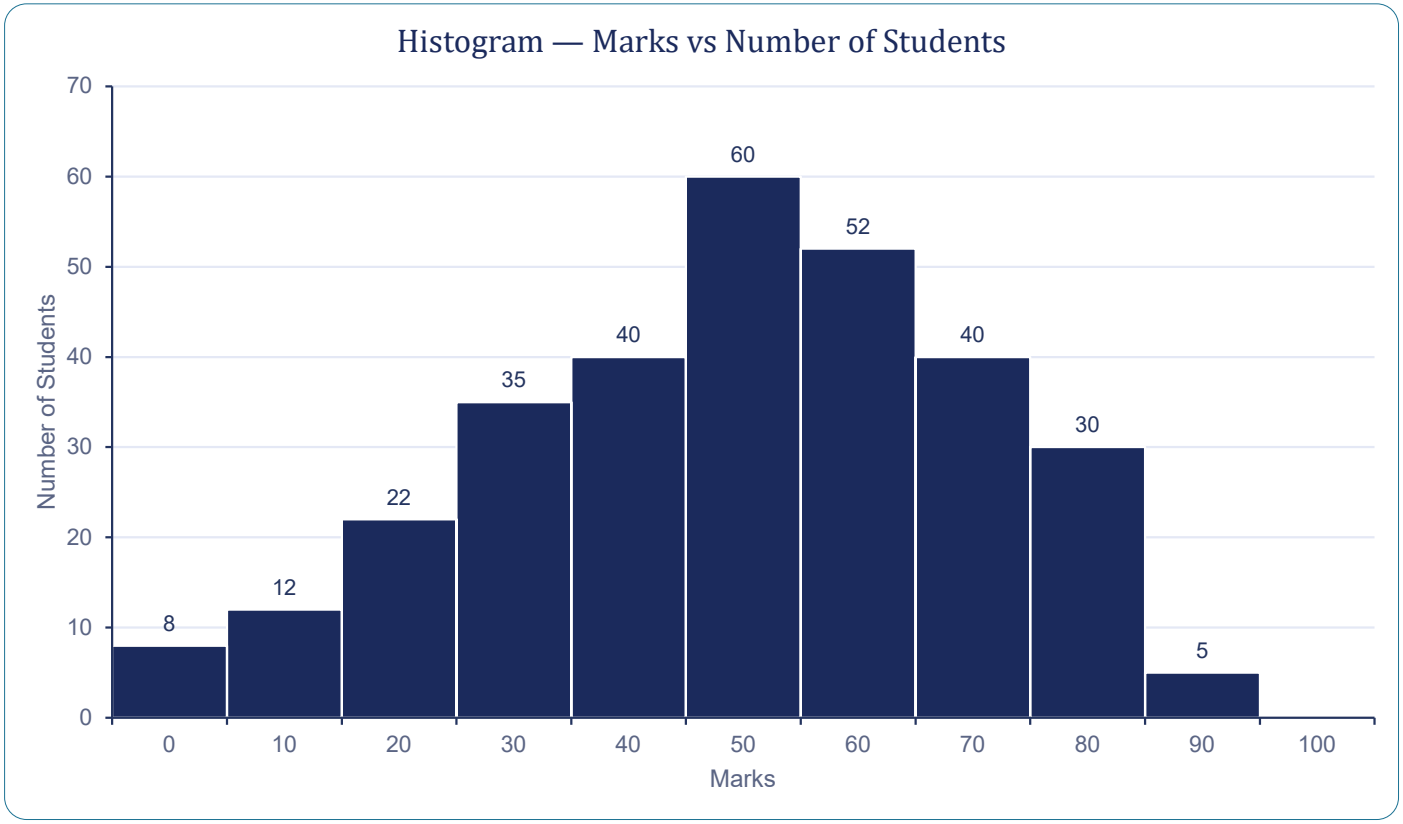
A correction must be made. Find the frequency density of each class — the frequency divided by the width of that class — and plot these density values instead of raw frequencies.

Adjustment rule: take the class with the lowest interval as the base. If another class's interval is twice as wide, divide its height by 2; if three times as wide, divide by 3 — heights stay proportional to frequency ÷ width.

Problem Statement — Illustration 28

Represent the following data by a histogram (equal class intervals)

Marks	Students
0–10	8
10–20	12
20–30	22
30–40	35
40–50	40
50–60	60
60–70	52
70–80	40
80–90	30
90–100	5



Since class intervals are equal throughout, no adjustment in frequencies is required.

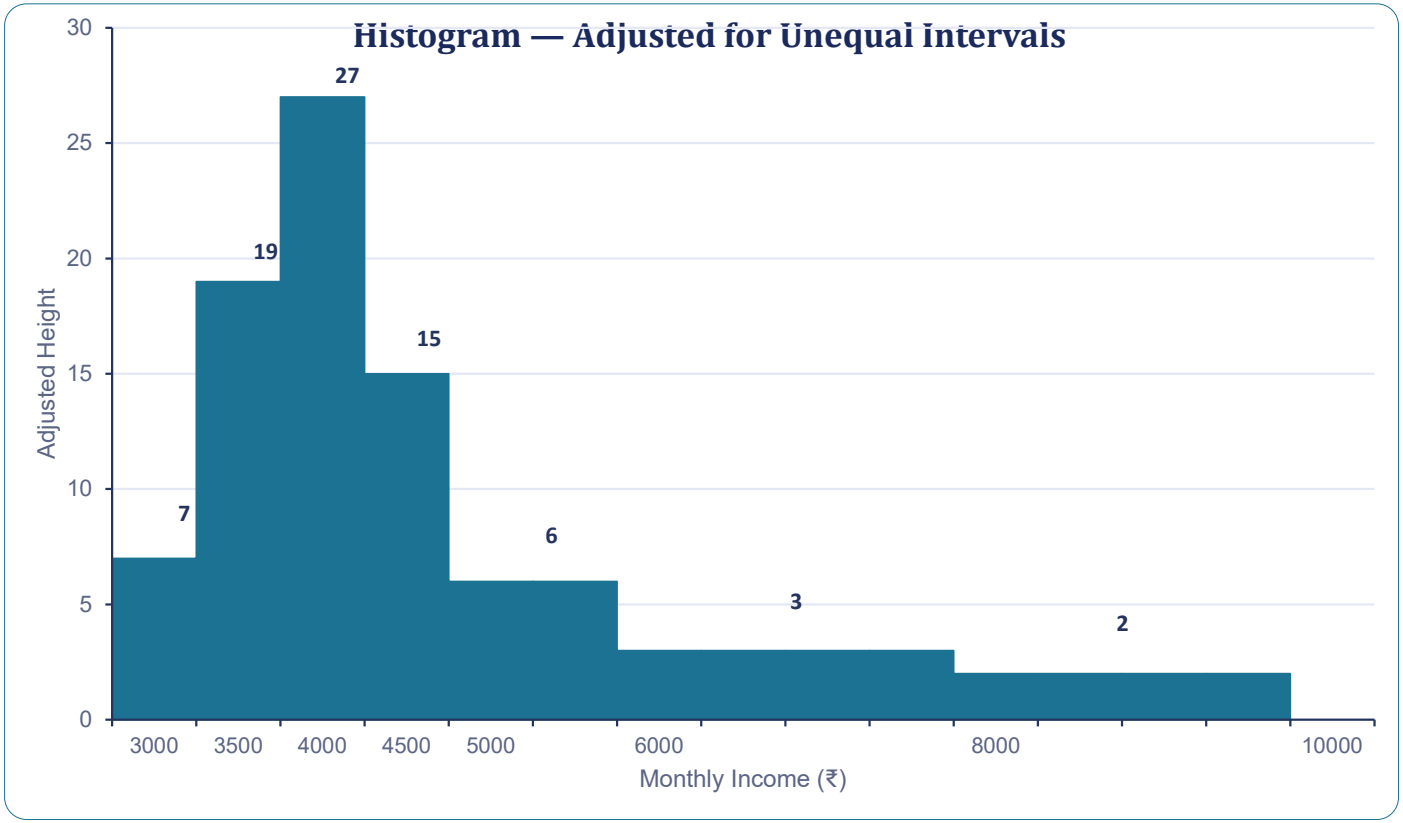
Problem Statement — Illustration 29

Unequal class intervals require a frequency-density adjustment

Income (₹)	Workers
3000–3500	7
3500–4000	19
4000–4500	27
4500–5000	15
5000–6000	6
6000–8000	3
8000–10000	2

Solution note

Lowest interval = 500. Wider classes are built from several 500-wide segments of equal height, so the bar widths stay proportional to each class interval.



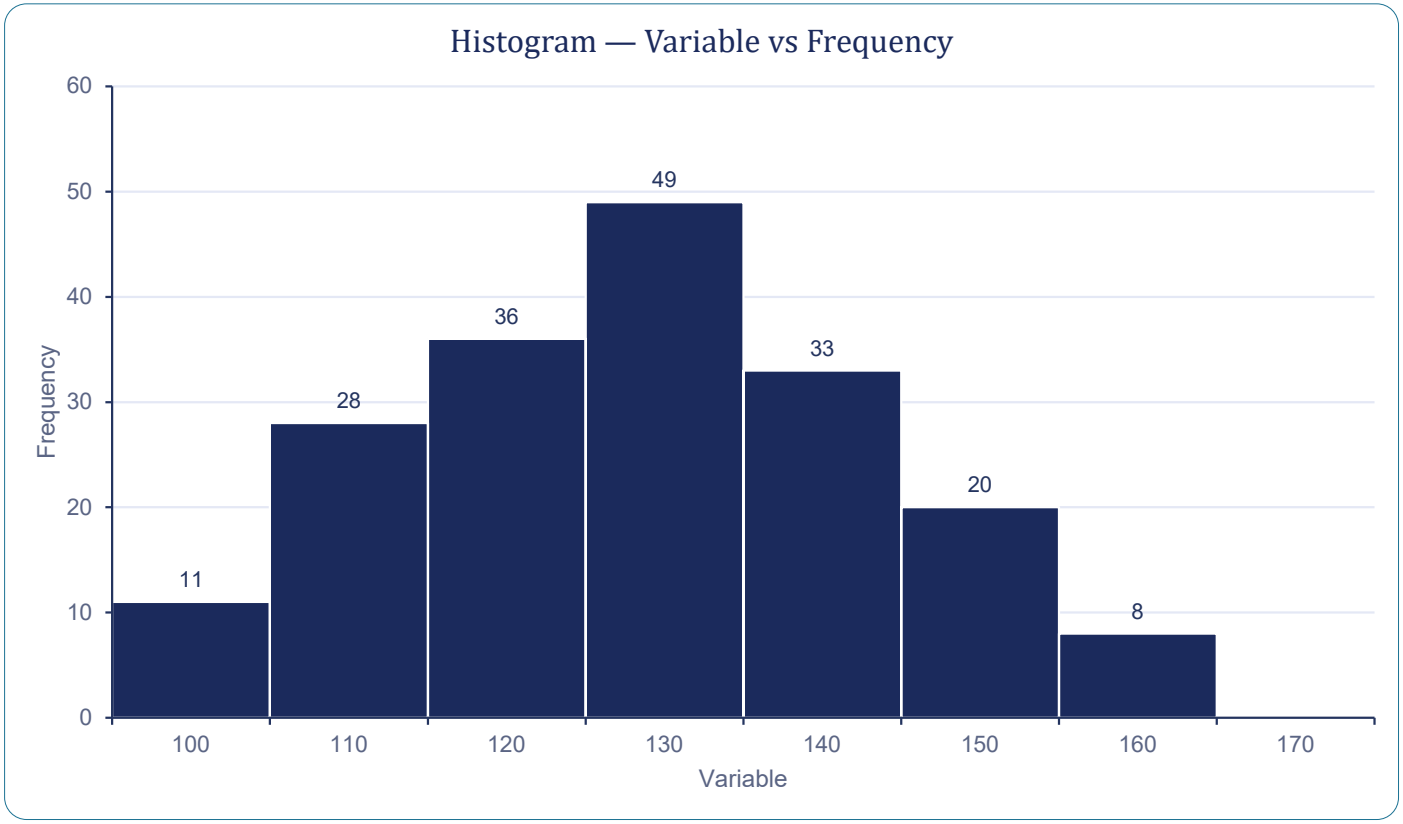
Bar widths are drawn proportional to each class interval — a true histogram, not just equal columns.

Problem Statement — Illustration 30

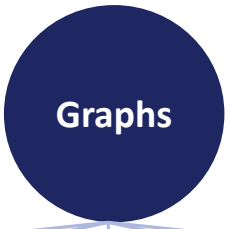
Draw the histogram for the following data

Variable	Frequency
100–110	11
110–120	28
120–130	36
130–140	49
140–150	33
150–160	20
160–170	8

Also applies: when only mid-points are given, first derive the class limits, then build the histogram the same way.



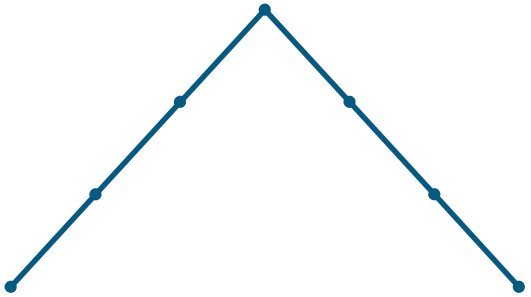
Various Graphs



1. **Line Graph**
   - Graph of One Variable
   - Graph of Two Variables
   - Graph of Two Scales
2. **Range Chart**
3. **Band Graph**
4. **Ratio Charts**
5. **Graphs of Frequency Distribution**
   - Histogram
   - Frequency Polygon
   - Smoothed Frequency Curve
   - Cumulative Frequency Curves (Ogives)

Frequency Polygon

A line graph of a frequency distribution



A frequency polygon is a graph of a frequency distribution. It has more than four sides, and is particularly effective for comparing two or more frequency distributions on the same axes.

Two ways to construct it

1. Draw a histogram, then join the mid-points of each bar's upper edge with straight lines — extending to the baseline at both ends with two zero-frequency hypothetical classes.

2. Take the mid-point of each class interval, plot its frequency, and join the points with straight lines — no histogram needed. The resulting shape is identical to Method 1.

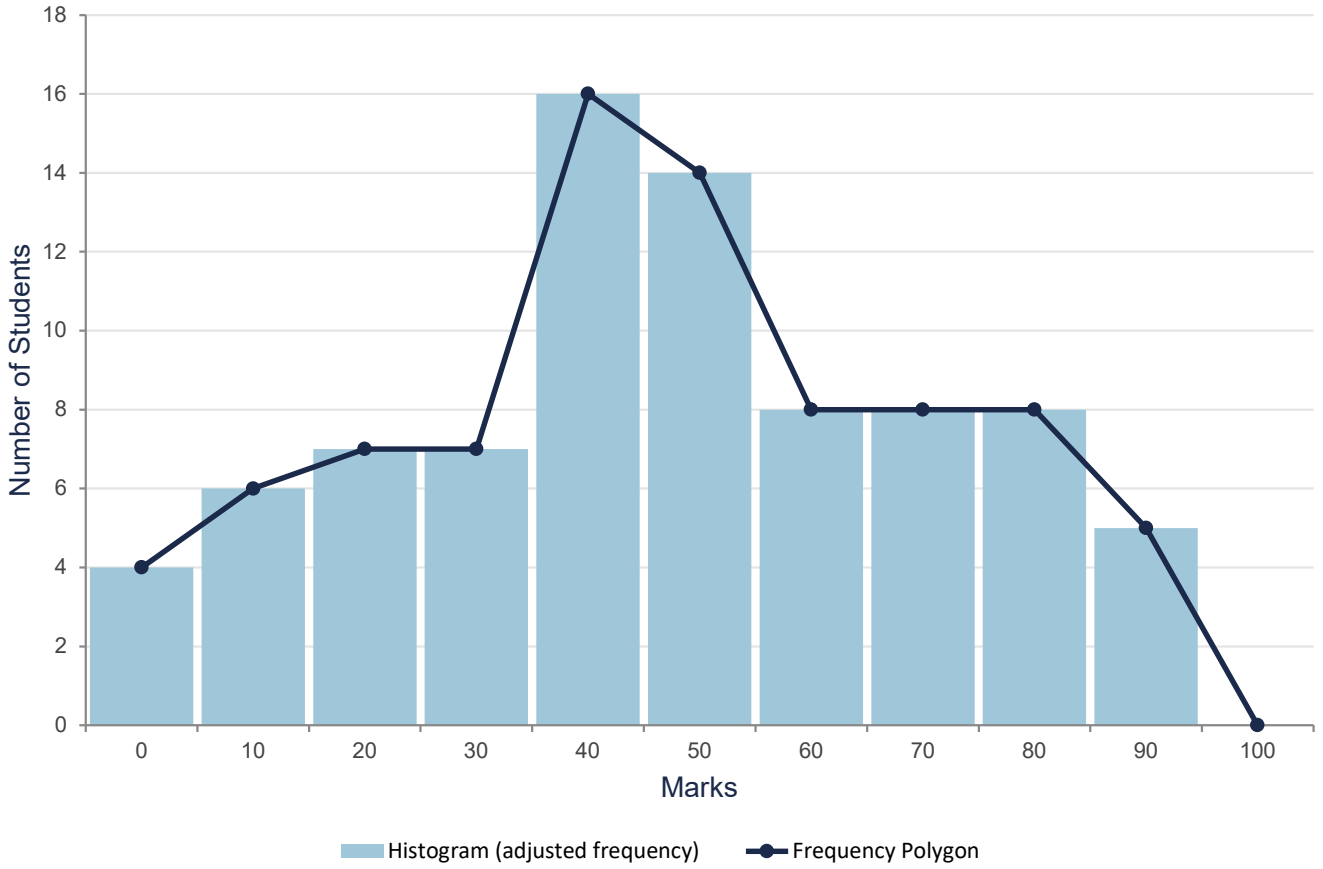
The apex, dropped perpendicular to the X-axis, gives the mode

Smoother and more curve-like as classes and observations increase

Needs adjustment for open-end or unequal class intervals

Histogram & Frequency Polygon

Marks distribution of students (unequal class intervals adjusted)



Marks	Students
0-10	4
10-20	6
20-30	7
30-40	7
40-50	16
50-60	14
60-70	8
70-80	8
80-90	8
90-100	5

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

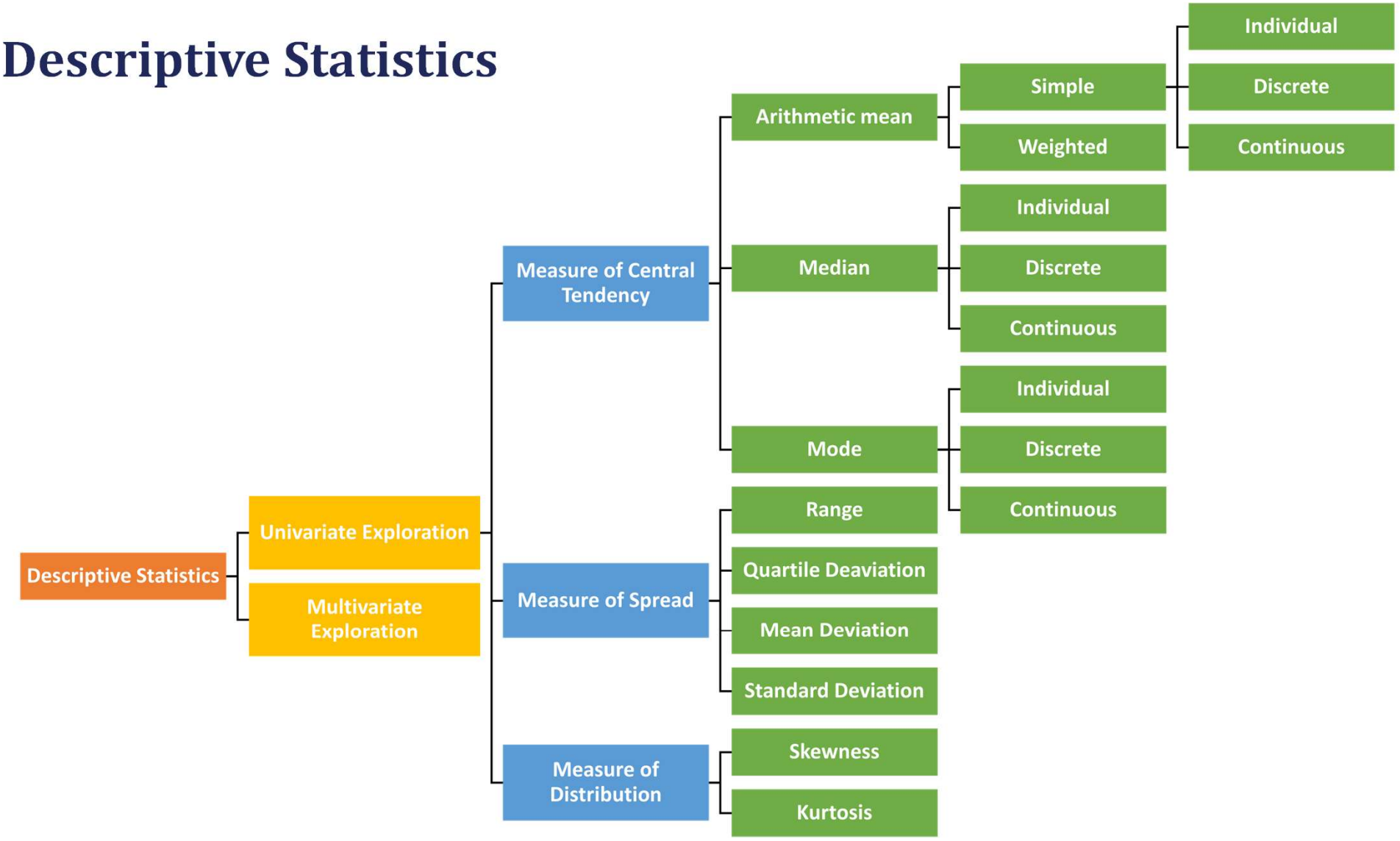
Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

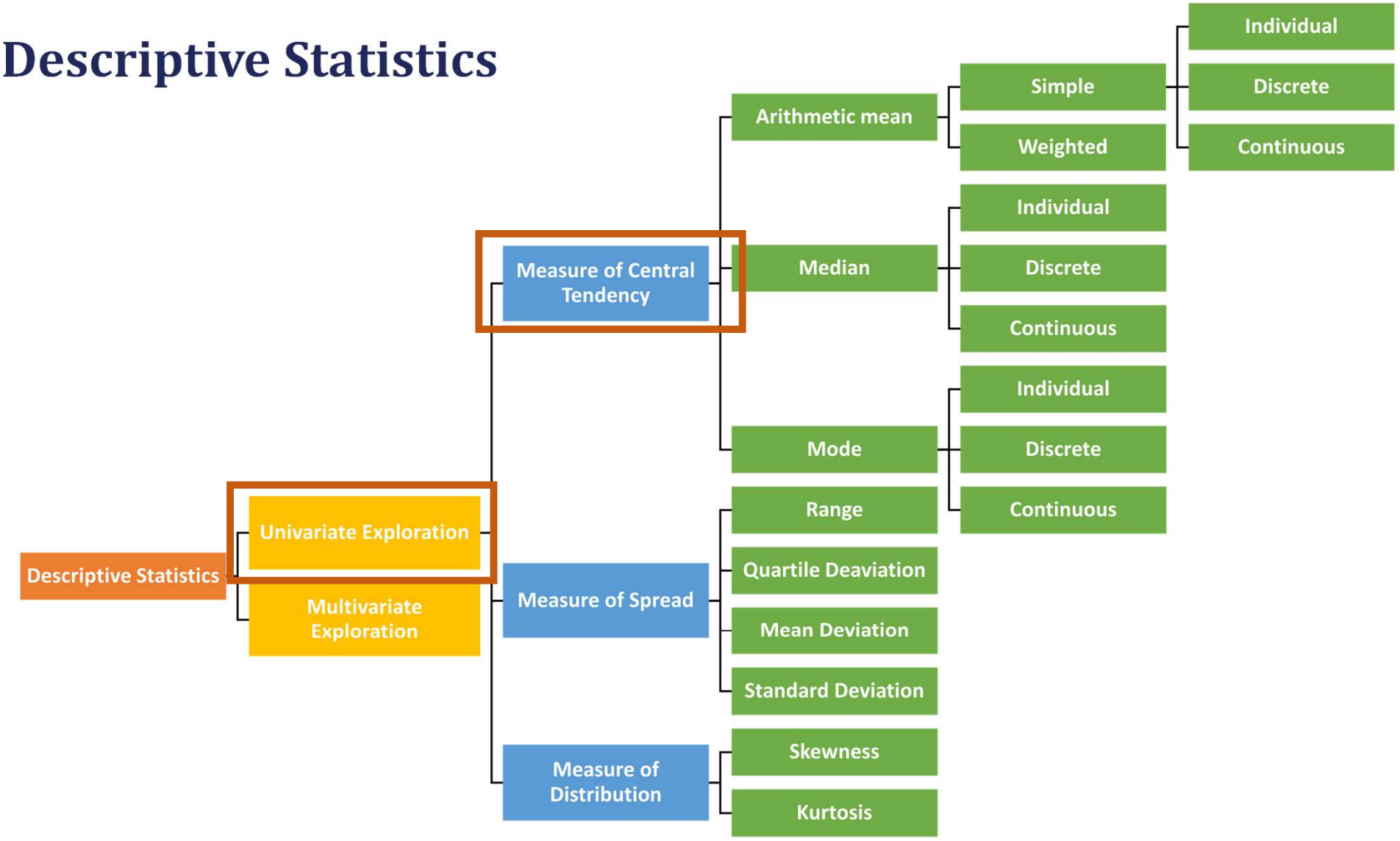
Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

Descriptive Statistics



Descriptive Statistics



Measures of Central Tendency

The most popular and widely used measure of central tendency

Measures of Central Tendency or Averages or Measures of Location.

INTRODUCTION

Statistical analysis often condenses an unwieldy mass of data into one single, representative value — its central or expected value. This is what statisticians call an 'average,' though its precise meaning differs from everyday use (average height, income, etc.).

Key Idea

An average lies between the largest and smallest values — a measure of central tendency.

OBJECTIVES OF AVERAGING

I **Get a Single Representative Value**

Condenses a whole mass of data into one figure, giving a bird's-eye view of the entire group so millions of values can be represented meaningfully.

II **Facilitate Comparison**

Reduces data to a single measure so groups, places, or time periods can be compared — e.g. pass percentages across colleges or years.

Types of Measures of Central Tendency/Averages

The following are the important types of averages

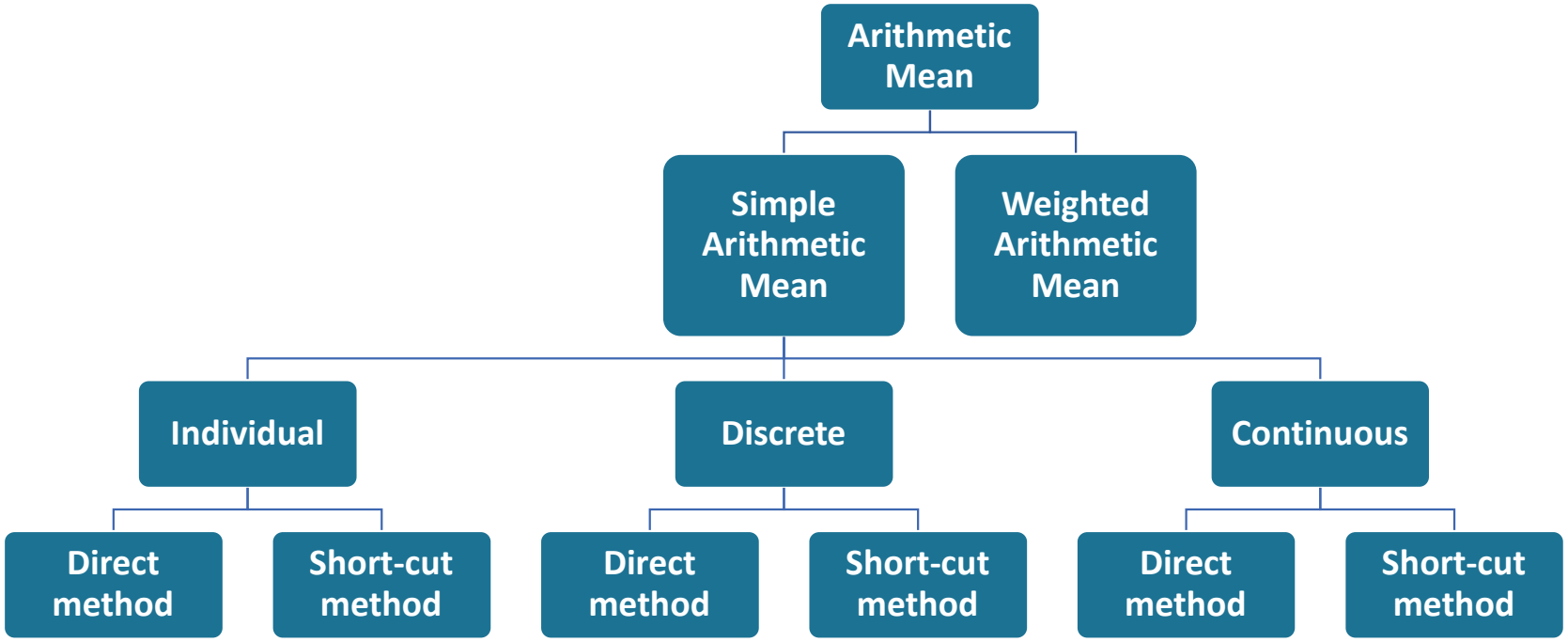
1. **Arithmetic Mean**
2. **Median**
3. **Mode**
4. ~~**Geometric Mean**~~
5. ~~**Harmonic Mean**~~

Beyond These

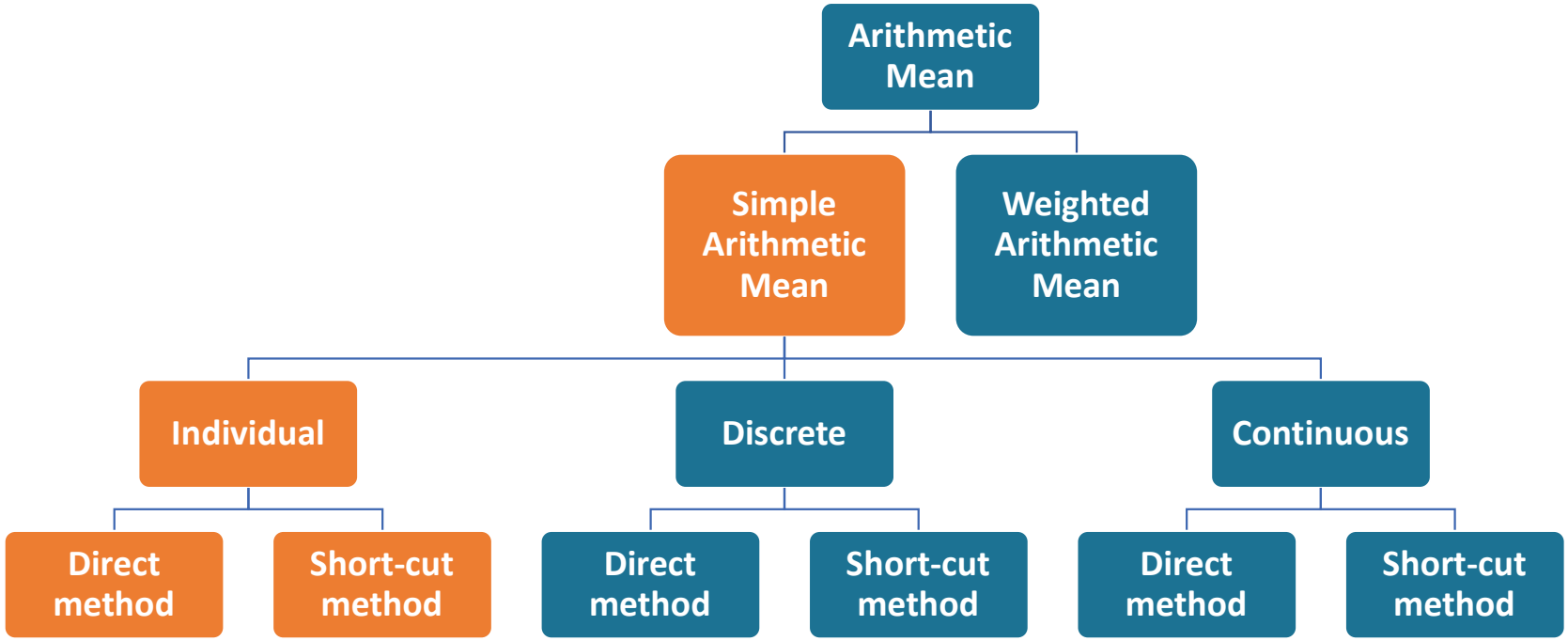
There are less important averages, such as moving average and progressive average. These have a very limited field of application and are, therefore, not so popular.

Measures of Central Tendency : Arithmetic Mean

The arithmetic mean — what most laymen call an ‘average’ — is obtained by adding together all the items and dividing this total by the number of items. It is the most popular and widely used way of representing an entire data set by a single value.



Measures of Central Tendency : Arithmetic Mean



Simple Arithmetic Mean - Individual Observations — Direct Method

Used when frequencies are not given

$$\overline{X} = (X_1 + X_2 + X_3 + \ldots + X_n) / N = \Sigma X / N$$

$\overline{X}$ = Arithmetic Mean **ΣX** = sum of all values of the variable X **N** = number of observations

Two steps to calculate the mean

1. Add together all the values of the variable X and obtain the total, i.e., ΣX.
2. Divide this total by the number of observations, i.e., N.

Simple Arithmetic Mean - Individual Observations — Direct Method

Example1- Monthly income (₹) of 10 employees in an office

Employee	Monthly Income (X)
1	14,780
2	15,760
3	26,690
4	27,750
5	24,840
6	24,920
7	16,100
8	17,810
9	27,050
10	26,950
ΣX	**2,22,650**

$$\bar{X} = \Sigma X / N = 2{,}22{,}650 / 10$$

Arithmetic Mean = ₹ 22,265

Simple Arithmetic Mean - Individual Observations — Shortcut method

Deviations are taken from an arbitrary (assumed) origin

$$\bar{X} = A + \Sigma d / N \qquad \text{where } d = (X - A)$$

A = the assumed mean; d = deviation of each item from the assumed mean

Steps

1. Take an assumed mean, A.
2. Take the deviations of items from the assumed mean and denote these deviations by d.
3. Obtain the sum of these deviations, i.e., Σd.
4. Apply the formula: $\bar{X} = A + \Sigma d / N$.

Simple Arithmetic Mean - Individual Observations — Shortcut method

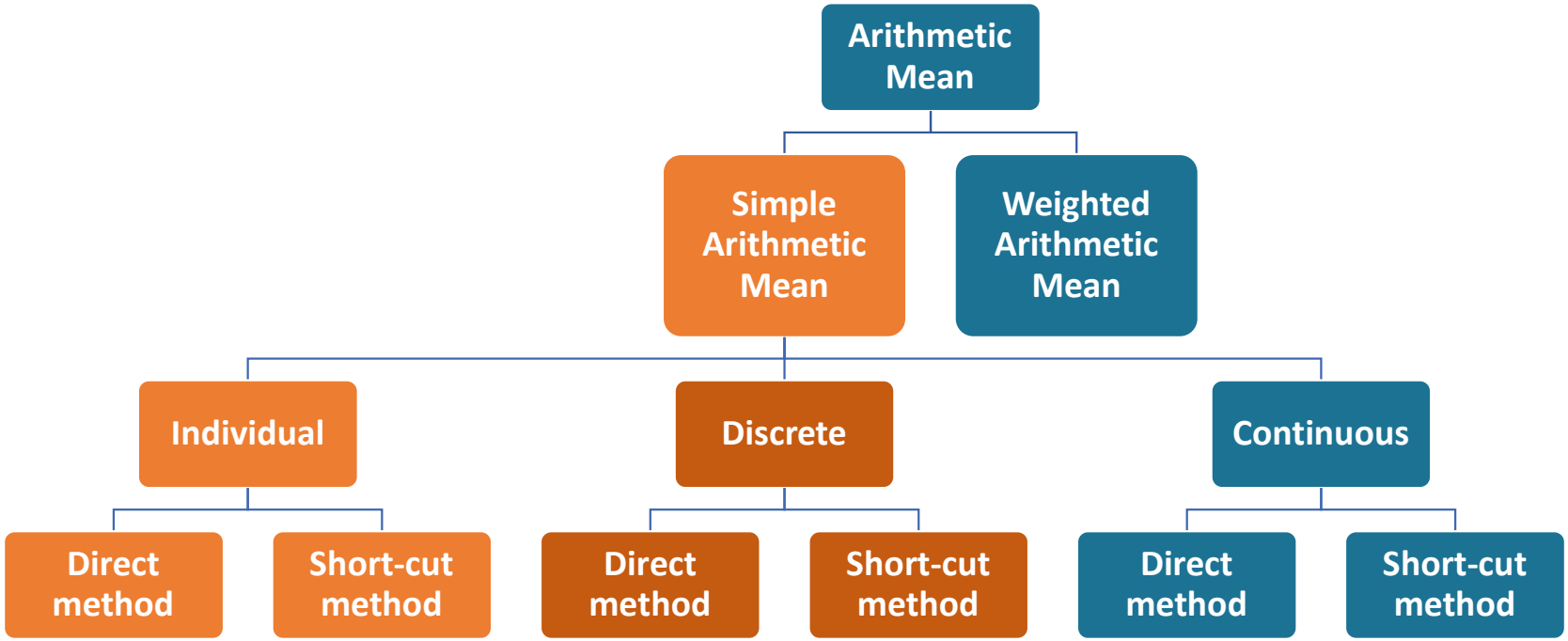
Taking ₹ 22,000 as the assumed mean

Employee	Income (X)	d = (X − 22000)
1	14,780	−7,220
2	15,760	−6,240
3	26,690	+4,690
4	27,750	+5,750
5	24,840	+2,840
6	24,920	+2,920
7	16,100	−5,900
8	17,810	−4,190
9	27,050	+5,050
10	26,950	+4,950
Σd		2650

$$\bar{X} = A + \Sigma d/N = 22000 + 2650/10 = 22000 + 265$$

Arithmetic Mean = ₹ 22,265

Measures of Central Tendency : Arithmetic Mean



Simple Arithmetic Mean - Discrete Series – Direct method

Marks obtained by 60 students of a class

$$\bar{X} = \Sigma fX / N$$

Marks (X)	Students (f)	fX
20	8	160
30	12	360
40	20	800
50	10	500
60	6	360
70	4	280
	N = 60	ΣfX = 2,460

$\bar{X}$ = 2,460 / 60 = 41

Simple Arithmetic Mean - Discrete Series – Short-cut Method

Marks obtained by 60 students of a class

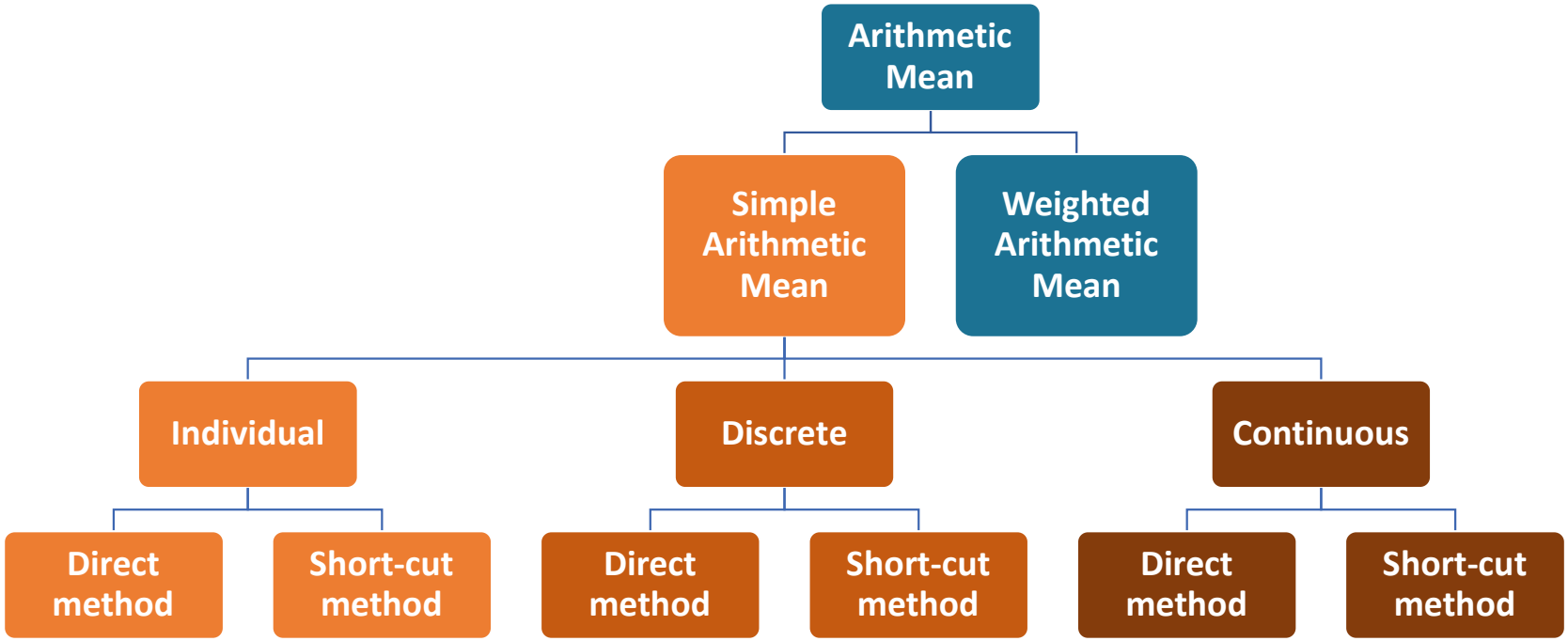
$$\bar{X} = A + \Sigma fd / N$$

Same data, assumed mean A = 40

Marks (X)	Students (f)	d = (X−40)	fd
20	8	−20	−160
30	12	−10	−120
40	20	0	0
50	10	+10	+100
60	6	+20	+120
70	4	+30	+120
	N = 60		Σfd = 60

$\bar{X}$ = 40 + 60/60 = 41

Measures of Central Tendency : Arithmetic Mean



Simple Arithmetic Mean - Continuous Series — Direct Method

Marks of 100 students, grouped into class intervals

$$\bar{X} = \Sigma fm / N \qquad (m = \text{mid-point of each class})$$

Marks	Mid-point (m)	Students (f)	fm
0–10	5	5	25
10–20	15	10	150
20–30	25	25	625
30–40	35	30	1,050
40–50	45	20	900
50–60	55	10	550
	N = 100		Σfm = 3,300

$\bar{X}$ = 3,300 / 100 = 33

Simple Arithmetic Mean - Continuous Series — Short-cut & Step-Deviation

Same data, assumed mean A = 35, class width(interval) i = 10

Short-cut: $\bar{X} = A + \Sigma fd / N$

Step-deviation: $\bar{X} = A + (\Sigma fd'/N) \times i$

Marks	m	f	d=(m−35)	fd	d′=(m−35)/10	fd′
0–10	5	5	−30	−150	−3	−15
10–20	15	10	−20	−200	−2	−20
20–30	25	25	−10	−250	−1	−25
30–40	35	30	0	0	0	0
40–50	45	20	+10	+200	+1	+20
50–60	55	10	+20	+200	+2	+20
		N=100		Σfd=−200		Σfd′=−20

Short-cut: 35 − 200/100 = 33

Step-deviation: 35 − (20/100)×10 = 33

Correcting Incorrect Values

When a wrong item was used while calculating the mean

Illustration 6

Mean marks of 100 students were found to be 40. Later, a score of 53 was discovered to have been misread as 83. Find the correct mean.

$N = 100, \bar{X} = 40 \quad \Rightarrow \quad \Sigma X = N\bar{X} = 4{,}000$

Correct $\Sigma X = 4{,}000 - 83 + 53 = 3{,}970$

Correct $\bar{X} = 3{,}970 / 100$

Correct Mean = 39.7

Illustration 7

Mean of 100 observations was found to be 40. Two items, 30 and 27, were wrongly taken instead of the correct 3 and 72. Find the correct mean.

$\Sigma X = 100 \times 40 = 4{,}000$

Less incorrect items (30+27=57): $4{,}000 - 57 = 3{,}943$

Add correct items (3+72=75): $3{,}943 + 75 = 4{,}018$

Correct $\bar{X} = 4{,}018 / 100$

Correct Mean = 40.18

Arithmetic Mean in Open-end Classes

When the lower limit of the first class or the upper limit of the last class is unknown

The mean cannot be found unless an assumption is made about the unknown limit. The assumption depends on the class interval following the first class and preceding the last class.

Marks	Students	Marks	Students
Below 10	4	30–40	15
10–20	6	40–50	8
20–30	10	Above 50	7

Uniform interval of 10 ⇒ assume first class 0–10, last class 50–60

Marks	Students	Marks	Students
Below 10	4	60–100	7
10–30	6	Above 100	3
30–60	10		

Interval widens 10→20→30 ⇒ assume first class 0–10, last class 100–150

If class widths vary irregularly, avoid trying to determine the unknown limits — use the median or mode instead. Because of the difficulty of ascertaining the limits of open-end distributions, the arithmetic mean should not be used in such cases.

Combined Mean of Two or More Groups

$$\bar{X}_{12} = (N_1 \bar{X}_1 + N_2 \bar{X}_2) / (N_1 + N_2)$$

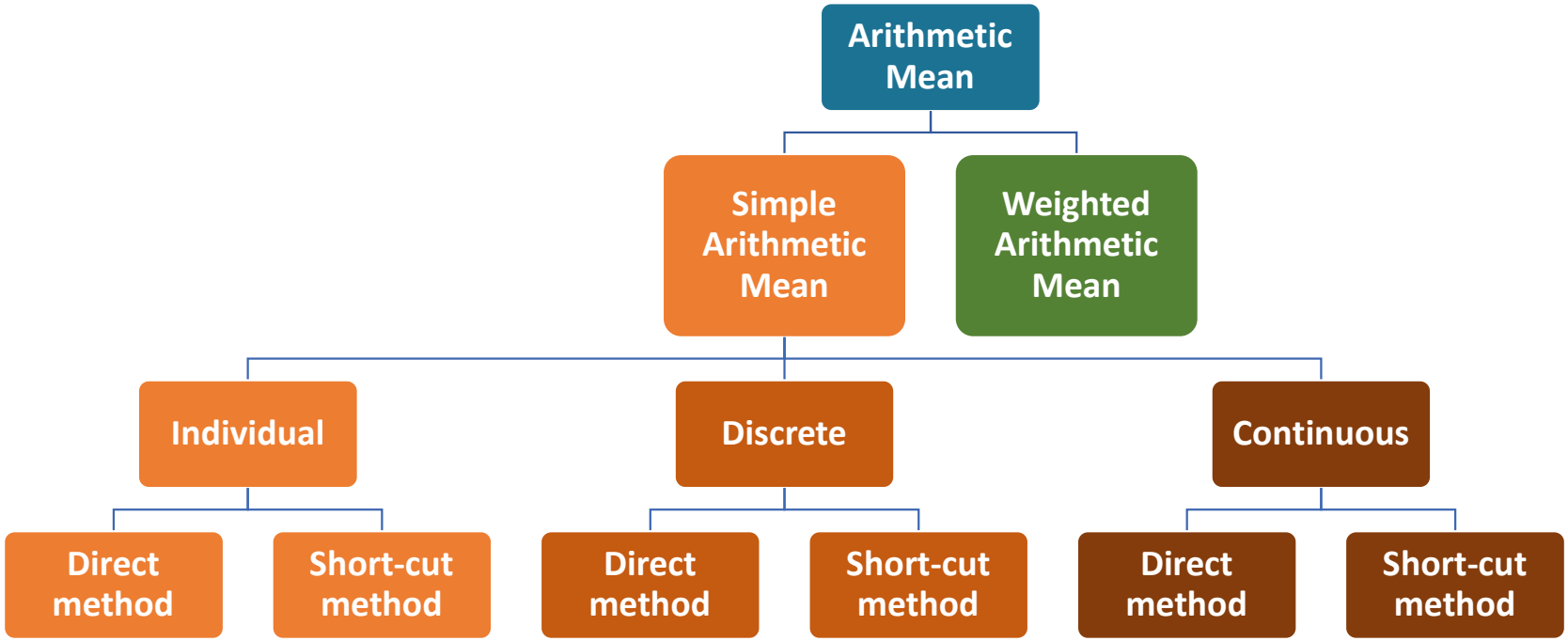
The mean height of 25 male workers in a factory is 61 inches and the mean height of 35 female workers in the same factory is 58 inches. Find the combined mean height of 60 workers in the factory.

N_1 (male)	$\bar{X}_1$	N_2 (female)	$\bar{X}_2$
25	61 in	35	58 in

$$\bar{X}_{12} = (25 \times 61 + 35 \times 58) / 60 = (1{,}525 + 2{,}030) / 60 = 3{,}555/60$$

Combined Mean Height = 59.25 in

Measures of Central Tendency : Arithmetic Mean



Weighted means

The simple arithmetic mean gives equal importance to every item. When items carry different relative importance, we assign each a weight (W) and compute the Weighted Arithmetic Mean instead.

$$\bar{X}_w = \frac{\Sigma WX}{\Sigma W}$$

X = variable values (X_1, X_2, ..., X_n)
W = weights attached to each value

Two steps:

- Multiply each weight by its X, obtain the total ΣWX
- Divide that total by the sum of weights, ΣW

For a frequency distribution

$$\bar{X}_w = \Sigma W(fX) \;/\; \Sigma W$$

$$\bar{X}_w = W_1(f_1X_1)+W_2(f_2X_2)+...+W_n(f_nX_n) \;/\; (W_1+W_2+...+W_n)$$

Simple and weighted means comparison

01

Equal when weights are equal

$$\bar{X} = \bar{X}_w \quad if \quad W_1 = W_2$$

Simple arithmetic mean equals the weighted mean whenever every weight is the same.

02

Simple mean is smaller

$$\bar{X} < \bar{X}_w \quad if \quad (W_2-W_1)(X_1-X_2) < 0$$

True only when greater weights are attached to greater values, and smaller weights to smaller values.

03

Simple mean is greater

$$\bar{X} > \bar{X}_w \quad if \quad (W_2-W_1)(X_1-X_2) < 0$$

True only when a smaller weight is attached to higher values, and a greater weight to smaller values.

Weighted means: Example

A contractor employs three types of workers—male, female and children. To a male, he pays ₹ 300 per day, to a female worker ₹ 250 per day and to a child worker ₹ 140 per day. What is the average wage per day paid by the contractor?.

Wages / day (X)	No. of workers (W)	WX
₹300	20	6,000
₹250	15	3,750
₹140	5	700
Total	**ΣW = 40**	**ΣWX = 10,450**

WEIGHTED MEAN WAGE

$\bar{X}_w = \Sigma WX / \Sigma W = 10{,}450 / 40$

₹ 261.25

per day

Simple mean of (300+250+140)/3 = ₹230 would be correct only if equal numbers of each worker type were employed — which they are not, so weighting is essential.

Weighted means: Example

A train runs 240 miles at a speed of 60 m.p.h., another 480 miles at a speed of 80 m.p.h., then due to repairs of the track travels for 15 minutes at a speed of 10 m.p.h., and finally covers the remaining distance of 275 miles at a speed of 55 miles p/h. What is the average speed in miles per hour?.

*Time taken in covering 240 miles at a speed of 60 m.p.h., = 240 minutes. [(240/60)*60 minutes]*
*Time taken in covering 480 miles at a speed of 80 m.p.h., = 360 minutes. [(480/80)*60 minutes)*
Distance covered in 15 minutes at a speed of 10 m.p.h., = 2.5 miles.
*Time taken in covering 275 miles at a speed of 55 miles p/h = 300 minutes. [(275/55)*60 minutes]*

Speed, mph (X)	Time, min (W)	WX
60	240	14,400
80	360	28,800
10	15	150
55	300	16,500
Total	**ΣW = 915**	**ΣWX = 59,850**

AVERAGE SPEED

$X_w = \Sigma WX / \Sigma W = 59{,}850 / 915$

65.41 mph

weighted by time taken, not distance

Weighted means: Example

Comment on the performance of the students of the three colleges given below using simple and weighted averages.

College Course of study	College A		College B		College C	
	Pass %	**Number of students**	**B'** Pass %	**Number of students**	**C'** Pass %	**Number of students**
M.A.	71	3	82	2	81	2
M.Com.	83	4	76	3	76	3.5
B.A.	73	5	73	6	74	4.5
B.Com.	74	2	76	7	58	2
B.Sc.	65	3	65	3	70	7
M.Sc.	66	3	60	7	73	2

Comparing three colleges: simple vs. weighted mean

Calculation of simple arithmetic mean.

College Course of study	**A** X Pass %	**No of students**	**B** X Pass %	**No of students**	**C** X Pass %	**No of students**
M.A.	71	3	82	2	81	2.0
M.Com.	83	4	76	3	76	3.5
B.A.	73	5	73	6	74	4.5
B.Com.	74	2	76	7	58	2.0
B.Sc.	65	3	65	3	70	7.0
M.Sc.	66	3	60	7	73	2.0
	ΣX = 432		**ΣX = 432**		**ΣX = 432**	

Comparing three colleges: simple vs. weighted mean

Calculation of simple arithmetic mean.

College A	College B	College C
Simple mean $\Sigma X / N = 432/6 = 72$	Simple mean $\Sigma X / N = 432/6 = 72$	Simple mean $\Sigma X / N = 432/6 = 72$

Comparing three colleges: simple vs. weighted mean

Calculation of weighted arithmetic mean.

College Course of study	College A			College B			College C		
	X Pass %	W No of students	WX	X Pass %	W No of students	WX	X Pass %	W No of students	WX
M.A.	71	3	213	82	2	164	81	2.0	162
M.Com.	83	4	332	76	3	228	76	3.5	266
B.A.	73	5	365	73	6	438	74	4.5	333
B.Com.	74	2	148	76	7	532	58	2.0	116
B.Sc.	65	3	195	65	3	195	70	7.0	490
M.Sc.	66	3	198	60	7	420	73	2.0	146
	ΣX = 432	**ΣW = 20**	**ΣWX = 1,451**	**ΣX = 432**	**ΣW = 28**	**ΣWX = 1,977**	**ΣX = 432**	**ΣW = 21**	**ΣWX = 1,513**

Comparing three colleges: simple vs. weighted mean

Calculation of weighted arithmetic mean.

College A	College B	College C
Weighted mean	Weighted mean	Weighted mean
$X_w = \Sigma WX / \Sigma W$ $X_w = 1{,}451 / 20$ $X_w = 72.55$	$X_w = \Sigma WX / \Sigma W$ $X_w = 1{,}977 / 28$ $X_w = 70.61$	$X_w = \Sigma WX / \Sigma W$ $X_w = 1{,}513 / 21$ $X_w = 72.05$

Comparing three colleges: simple vs. weighted mean

College A

Simple mean

$\Sigma X / N = 432/6 = 72$

Weighted mean

$X_w = \Sigma WX / \Sigma W$
$X_w = 1{,}451 / 20$
$X_w = 72.55$

College B

Simple mean

$\Sigma X / N = 432/6 = 72$

Weighted mean

$X_w = \Sigma WX / \Sigma W$
$X_w = 1{,}977 / 28$
$X_w = 70.61$

College C

Simple mean

$\Sigma X / N = 432/6 = 72$

Weighted mean

$X_w = \Sigma WX / \Sigma W$
$X_w = 1{,}513 / 21$
$X_w = 72.05$

Conclusion

The simple mean (72) is identical for all three colleges, wrongly suggesting equal performance. The weighted mean — which accounts for how many students actually sat each course — is highest for College A (72.55), showing its students perform best overall.

Types of Measures of Central Tendency/Averages

The following are the important types of averages

1. **Arithmetic Mean**
2. **Median**
3. **Mode**
4. ~~**Geometric Mean**~~
5. ~~**Harmonic Mean**~~

Beyond These

There are less important averages, such as moving average and progressive average. These have a very limited field of application and are, therefore, not so popular.

Measures of Central Tendency : Median

What Is the Median?

The middle value of a distribution — it splits ordered data into two equal halves.



MEDIAN

50% of values below	50% of values above

Unlike the mean (built from every value), the median is a positional average — its value depends only on where each item sits, not its size.

N is ODD

Median = an actual value in the series



Equal items sit on either side of the median

N is EVEN

Median = average of the two middle values



Median = a derived (calculated) figure

How to Find the Median

3 simple steps

1. **Arrange**

   Sort data in ascending (or descending) order.

2. **Locate**

   Find the (N+1)/2th item — use cumulative frequency for discrete series.

3. **Read off**

   That value (or the average of two middle values) is the median.

Measures of Central Tendency : Median: Three Types of observations

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Median : Individual Observations

Steps

i Arrange data in ascending order.

ii Odd N — median is an actual item.

Median = Size of **(N+1)/2** th item

ODD N = 7

Example1 — Median Wage of 7 Workers

Wages (in ₹): 14,100 14,150 16,080 17,120 15,200 16,160 17,400

Sl. No.	Wages Arranged in Ascending Order
1	14,100
2	14,150
3	15,200
5	16,160
6	17,120
7	17,400

Median = Size of **(N+1)/2** th item = (7+1)/2 = 4th item = **₹ 16,080**

MEDIAN WAGE **₹ 16,080**

Measures of Central Tendency : Median : Individual Observations

Steps

i Arrange data in ascending order.

ii Even N — average the two middle items.

Median = Size of **(N+1)/2** th item

EVEN N = 10

Example — Median Monthly Income of 10 Employees

Monthly Income (in ₹): 14,391 15,384 25,591 15,407 16,672 26,522 16,777 26,753 27,850 37,490

Sl. No.	Income Arranged in Ascending Order
1	14,391
2	15,384
3	15,407
4	16,672
5	**16,777**
6	**25,591**
7	26,522
8	26,753
9	27,850
10	37,490

Median = Size of **(N+1)/2** th item = 11/2 = 5.5th item

Size of 5.5th item = (16,777 + 25,591) / 2 = 42,368 / 2

MEDIAN INCOME **₹ 21,184**

Measures of Central Tendency : Median: Three Types of observations

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Median : Discrete Observations

Steps

i. Arrange the data in ascending or descending order of magnitude.

ii. Find out the cumulative frequencies (cf). Add previous frequency

iii. Apply the formula: Median = Size of (N+1)/2 th item. Where N is cf of last record

iv. Look at the cf column and find the total that is equal to, or next higher than, (N+1)/2. The value of the variable corresponding to it is the median.

Median = Size of **(N+1)/2** th item

Example— Median Income

Income (₹): 15,000 15,500 16,800 18,000 18,500 17,800
Persons: 24 26 20 16 6 30

Income	*f*	*Cf*
15,000	*24*	*24*
15,500	*26*	*50*
16,800	*20*	*70*
17,800	*30*	*100*
18,000	*16*	*116*
18,500	*6*	*122*

Median = Size of **(N+1)/2** th item
= (122+1)/2 = 61.5th item

The cf just ≥ 61.5 is 70, corresponding to income ₹ 16,800 — the median.

MEDIAN INCOME **₹ 16,800**

Measures of Central Tendency : Median: Three Types of observations

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Median : Continuous Observations

$$\text{Median} = L + (N/2 - cf) \times i$$

$N/2$	Lower limit of the median class
L	Lower limit of the median class
cf	Cumulative frequency of the class preceding the median class
f	Simple frequency of the median class
i	Class interval of the median class

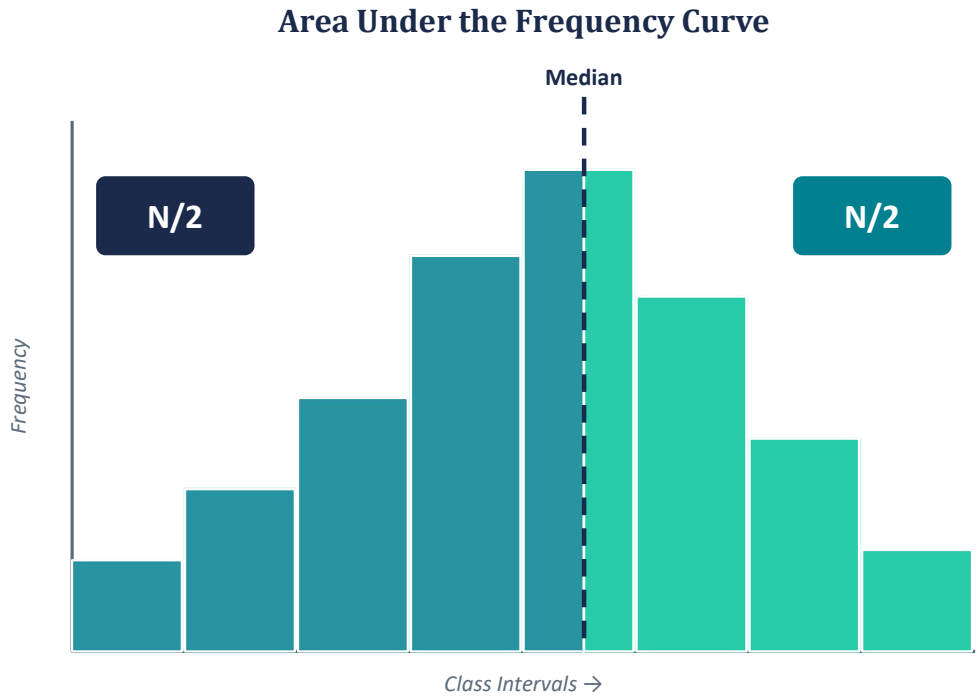
Key rule: use N/2 as the rank (not (N+1)/2). Median divides the area of the curve into two equal halves, so no +1 adjustment is made in continuous series.

Median Rank in a Continuous Series

Use N/2 as the rank — not (N+1)/2

Key Rule

Median class rank = N / 2



Area (left of median) = **Area** (right of median) = N/2

Measures of Central Tendency : Median : Continuous Observations

Example: Calculate the median for the following frequency distribution:

Marks	Number of Students
45–50	10
40–45	15
35–40	26
30–35	30
25–30	42
20–25	31
15–20	24
10–15	15
5–10	7

Arrange in ascending order

Marks	f	cf
5–10	7	7
10–15	15	22
15–20	24	46
20–25	31	77
25–30	42	119
30–35	30	149
35–40	26	175
40–45	15	190
45–50	10	200

Measures of Central Tendency : Median : Continuous Observations

Median Marks of Students: Exclusive classes

Marks	f	cf
5–10	7	7
10–15	15	22
15–20	24	46
20–25	31	77
25–30	**42**	**119**
30–35	30	149
35–40	26	175
40–45	15	190
45–50	10	200

Solution

N = 200 → Median = size of N/2 th item = 100th item

Median lies in the class 25–30 (cf just ≥ 100 is 119)

L = 25, N/2 = 100, cf = 77, f = 42, i = 5

$$\text{Median} = L + (N/2 - cf)/f \times i$$

$$= 25 + (100 - 77)/42 \times 5$$

$$= 25 + 2.74 = 27.74$$

MEDIAN MARKS **27.74**

Measures of Central Tendency : Median : Continuous Observations

Median Weight of Apples (Inclusive Classes): Inclusive classes are converted to exclusive by − 0.5 from lower limits and + 0.5 to upper limits.

Weight (gm)	f
410–419	14
420–429	20
430–439	42
440–449	54
450–459	45
460–469	18
470–479	7

Weight (gm)	f	cf
409.5–419.5	14	14
419.5–429.5	20	34
429.5–439.5	42	76
439.5–449.5	54	130
449.5–459.5	45	175
459.5–469.5	18	193
469.5–479.5	7	200

Measures of Central Tendency : Median : Continuous Observations

Median Weight of Apples (Inclusive Classes): Inclusive classes are converted to exclusive by − 0.5 from lower limits and + 0.5 to upper limits.

Weight (gm)	f	cf
409.5–419.5	14	14
419.5–429.5	20	34
429.5–439.5	42	76
439.5–449.5	**54**	**130**
449.5–459.5	45	175
459.5–469.5	18	193
469.5–479.5	7	200

Solution

N = 200 → Median = size of N/2 th item = 100th item

Median lies in class 439.5–449.5 (cf just ≥ 100 is 130)

L = 439.5, N/2 = 100, cf = 76, f = 54, i = 10

$$\text{Median} = 439.5 + (100 - 76)/54 \times 10$$

$$= 439.5 + 4.44 = 443.94$$

MEDIAN WEIGHT **443.94 gm**

Measures of Central Tendency : Median : Continuous Observations

Median from Cumulative ('Less Than') Data

Marks	f
Less than 5	29
Less than 10	195
Less than 15	241
Less than 20	117
Less than 25	52
Less than 30	10
Less than 35	6
Less than 40	3
Less than 45	2

Marks	f	cf
0–5	29	29
5–10	195	224
10–15	**241**	**465**
15–20	117	582
20–25	52	634
25–30	10	644
30–35	6	650
35–40	3	653
40–45	2	655

Solution

Cumulative ('less than') frequencies are converted to simple frequencies first.

N = 655 → Median = size of N/2 th item = 327.5th item

Median lies in the class 10–15

L = 10, N/2 = 327.5, cf = 224, f = 241, i = 10

$$\text{Median} = 10 + (327.5 - 224)/241 \times 10$$

$$= 10 + 4.29 = 14.29$$

MEDIAN **14.29**

Measures of Central Tendency : Median : Continuous Observations

Median from Mid-values

Class group	f
115	6
125	25
135	48
145	72
155	116
165	60
175	38
185	22
195	3

1 **Find the class width (i)**

Mid-values increase in equal steps.

e.g. 125 − 115 = 10 → i = 10

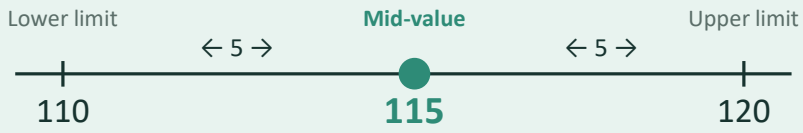
2 **Find the half-width**

The mid-value splits the class into two equal halves: i / 2 = 10 / 2 = 5

3 **Subtract & add the half-width**

Lower limit = Mid-value − i/2

Upper limit = Mid-value + i/2



Lower = Mid − i/2

Upper = Mid + i/2

(i = class width)

Mid-value	− 5	+ 5	Class group
115	110	120	110–120
125	120	130	120–130
135	130	140	130–140
145	140	150	140–150
155	150	160	150–160

Measures of Central Tendency : Median : Continuous Observations

Median from Mid-values

Class group	f
115	6
125	25
135	48
145	72
155	116
165	60
175	38
185	22
195	3

Class group	f	cf
110–120	6	6
120–130	25	31
130–140	48	79
140–150	72	151
150–160	**116**	**267**
160–170	60	327
170–180	38	365
180–190	22	387
190–200	3	390

Solution

Mid-values are converted into exact class groups (width 10) before finding cf.

N = 390 → Median = size of N/2 th item = 195th item

Median lies in the class 150–160 (cf just ≥ 195 is 267)

L = 150, N/2 = 195, cf = 151, f = 116, i = 10

$$\text{Median} = 150 + (195 - 151)/116 \times 10$$

$$= 150 + 3.79 = 153.79$$

MEDIAN **153.79**

Measures of Central Tendency : Median: Three Types of observations

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Median: Three Types of Series (Summary)

Individual Observations

Raw, ungrouped data

1. Arrange data in ascending or descending order
2. Count total items, N
3. If N is odd, median = value of the (N+1)/2 th item
4. If N is even, median = average of the two middle items: N/2 th and (N/2 + 1) th

Median = Size of (N+1)/2 th item

e.g. 4, 6, 8, 10, 12 → N = 5 (odd)
Median = 3rd item = 8

Discrete Series

Individual values with frequencies

1. Arrange values in ascending order
2. Find cumulative frequencies (cf) — add previous frequency
3. Apply: Median = size of (N+1)/2 th item
4. Find the cf just ≥ (N+1)/2; its value is the median

Median = Size of (N+1)/2 th item

e.g. N = 122 → 61.5th item
cf just ≥ 61.5 → Median = ₹16,800

Continuous Series

Grouped, class-interval data

1. Find cf for each class
2. Use N/2 as the rank (not (N+1)/2, since frequencies lose individuality)
3. Locate the median class: cf just ≥ N/2
4. Interpolate within that class using the formula

Median = L + (N/2 − cf)/f × i

e.g. L=25, N/2=100, cf=77, f=42, i=5
Median = 25 + 2.74 = 27.74

Types of Measures of Central Tendency/Averages

The following are the important types of averages

1. **Arithmetic Mean**
2. **Median**
3. **Mode**
4. ~~**Geometric Mean**~~
5. ~~**Harmonic Mean**~~

Beyond These

There are less important averages, such as moving average and progressive average. These have a very limited field of application and are, therefore, not so popular.

Mode: Introduction

The mode is the value in a series of observations that occurs most frequently — the value around which the other items cluster most densely.

Also known as the most typical, or "fashionable," value of a distribution.

Why "density," not just frequency?

A single exact value that repeats most often can fluctuate wildly between samples. The mode is better understood as the value with the greatest frequency density in its immediate neighbourhood.

QUICK EXAMPLE

Find the mode of:

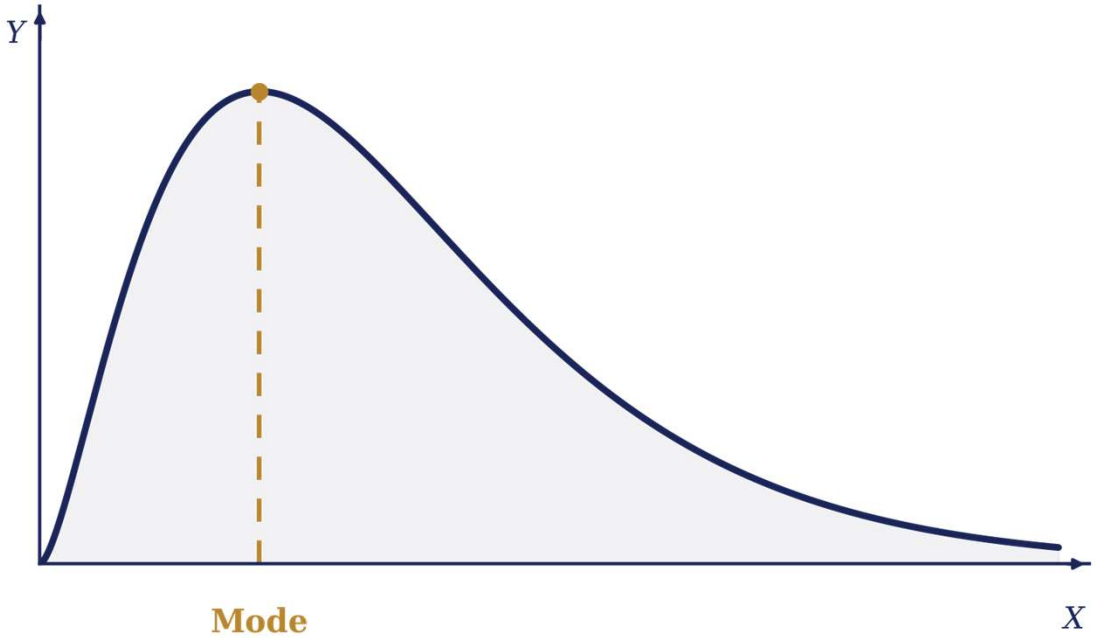
3	5	8	5	4	5	9	3

5 occurs 3 times — more than any other value.

MODE = 5

The value the data cluster around most densely.

The Mode on a Frequency Curve



- The value of the variable where the curve reaches its maximum height is the mode.

Mode: A Caution: Highest Frequency ≠ Majority

College president election — 816 votes polled among three candidates



Mr. Y wins

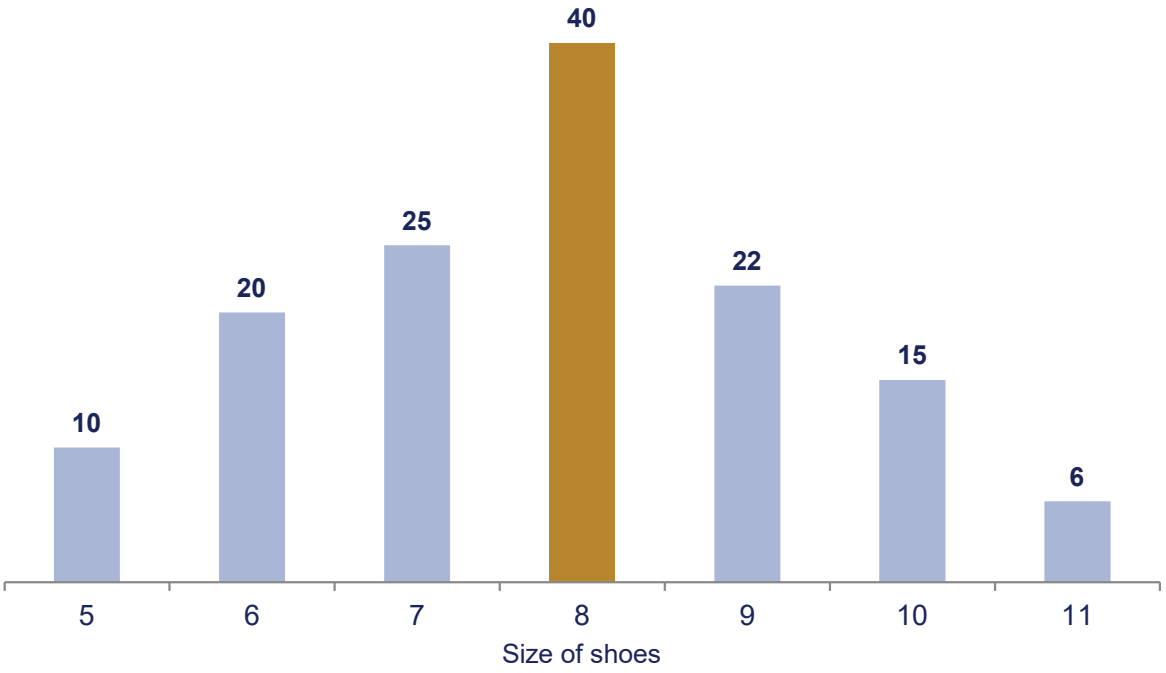
with the highest vote count — the modal choice among the three.

But not a majority: 538 votes (268 + 270) were cast against him — more than the 278 he received.

The mode identifies the most frequent value — it does not guarantee that value dominates the whole set.

Mode : Worked Example — Shoe Sizes

The mode is the easiest average to compute: it is simply the value with the highest frequency.



MODAL SIZE 8

40 people wear this size — the maximum number of persons among all sizes.

A retailer stocking the most-demanded size cares about the mode — not the mean or median size.

Why Mode Matters

Mean and median can both misrepresent data that mode reveals clearly

1 Extreme values

The mean is distorted by outliers — a few very high or low figures pull the average away from what's typical.

2 Uneven spread

The median can be unrepresentative when items cluster unevenly, such as most values sitting near one end of a range.

3 Common experience

For most common wage, height, or shoe size, mode is the natural average — the value people actually experience most.

Mode is the easiest average to compute, and the only one built entirely on frequency — the most typical value in the data.

Calculating Measures of Central Tendency : Mode

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Calculating Measures of Central Tendency : Mode

TYPE 1 · INDIVIDUAL OBSERVATIONS

Calculate the mode from following data of the marks obtained by 10 students.

Example — Marks obtained by 10 students

10	27	24	12	27	27	20	18	15	30

Grouping the values by frequency:

Marks	Total Students
10	1
12	1
15	1
18	1
20	1
24	1
27	3
30	1

RESULT

27

27 occurs 3 times — more than any other mark.

Modal marks = 27

Note: determining mode for individual observations essentially involves grouping the raw data first.

Calculating Measures of Central Tendency : Mode

TYPE 1 · INDIVIDUAL OBSERVATIONS

When two or more values share the same maximum frequency, **no single value can be called the modal value.** Such a series is bimodal (or multimodal), and the mode is said to be ill-defined.

Illustration — Income data (₹)

15110	16120	17130	16120	15110
18140	17130	16120	17130	18140

Frequency of each value:

15110	2
16120	**3**
17130	**3**
18140	2

RESULT

₹16,120 & ₹17,130

both occur 3 times — the same, maximum frequency.

Mode is ill-defined (bimodal)

A single, clear-cut mode cannot be identified from this data — a limitation worth flagging whenever it occurs.

Measures of Central Tendency : Mode

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Mode: Finding Mode by Inspection

TYPE 2 · DISCRETE OBSERVATIONS

In a discrete series, the mode can often be spotted simply by looking at which value the items cluster around most heavily.

Size of garment	28	29	30	31	32	33
No. of persons	10	20	40	65	50	15

RESULT

Modal size = 31

Size 31 is worn by the maximum number of persons — 65 — so it is clearly the value around which the data is most concentrated.

Measures of Central Tendency : Mode

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Measures of Central Tendency : Mode: Two Ways to Locate the Mode

TYPE 3 · CONTINUOUS SERIES

Form (i) — using differences (Δ)

$$M_0 = L + \Delta_1 / (\Delta_1 + \Delta_2) \times i$$

$\Delta_1 = | f_1 - f_0 |$ (modal vs. preceding class)

$\Delta_2 = | f_1 - f_2 |$ (modal vs. succeeding class)

L = lower limit of modal class · i = class interval

Form (ii) — using frequencies directly

$$M_0 = L + (f_1 - f_0) / (2f_1 - f_0 - f_2) \times i$$

f_1 = frequency of the modal class

f_0 = frequency of the preceding class

f_2 = frequency of the succeeding class

Two steps, always in this order:

1. Prepare a grouping table and analysis table (or use inspection) to ascertain the modal class.
2. Substitute L, Δ_1/Δ_2 (or f_0, f_1, f_2) and i into the formula.

Caution: Class intervals must be uniform. If unequal, make them equal first, assuming frequencies are spread evenly — otherwise the result is misleading.

Measures of Central Tendency : Mode: Example

TYPE 3 · CONTINUOUS SERIES

Calculate mode for following data

Marks	Students
Above 0	3
Above 10	5
Above 20	7
Above 30	10
Above 40	12
Above 50	15
Above 60	12
Above 70	6
Above 80	2
Above 90	8

Next step: Before the mode formula can be used, this cumulative data must be converted into a simple (class-wise) frequency distribution.

Marks	Students
0–10	3
10–20	5
20–30	7
30–40	10
40–50	12
50–60	15
60–70	12
70–80	6
80–90	2
90–100	8

Measures of Central Tendency : Mode: Example

TYPE 3 · CONTINUOUS SERIES

Substituting Into the Mode Formula (50–60)

$$M_0 = L + \Delta_1 / (\Delta_1 + \Delta_2) \times i$$

Marks	Students	
40–50	12	f_0
50–60	15	f_1
60–70	12	f_2

L

50

Lower limit of the modal class 50–60

$\Delta_1 = | f_1 - f_0 |$

$|15 - 12| = 3$

Modal class f (15) vs. preceding class f (12)

$\Delta_2 = | f_1 - f_2 |$

$|15 - 12| = 3$

Modal class f (15) vs. succeeding class f (12)

i

10

Width of the modal class

$$M_0 = 50 + (3/3+3) \times 10 = 50 + 5$$

MODAL MARKS $M_0 = 55$

Measures of Central Tendency : Mode: Two Ways to Locate the Mode

TYPE 3 · CONTINUOUS SERIES

Find the value of mode from the given data

Weight (kg)	Students
93–97	2
98–102	5
103–107	12
108–112	17
113–117	14
118–122	6
123–127	3
128–132	1

Next step: its inclusive limits must first be converted to exclusive class limits.

Weight (kg)	Students
92.5–97.5	2
97.5–102.5	5
102.5–107.5	12
107.5–112.5	17
112.5–117.5	14
117.5–122.5	6
122.5–127.5	3
127.5–132.5	1

Measures of Central Tendency : Mode: Two Ways to Locate the Mode

TYPE 3 · CONTINUOUS SERIES

Weight (kg)	Students	
92.5–97.5	2	
97.5–102.5	5	
102.5–107.5	12	f_0
107.5–112.5	17	f_1
112.5–117.5	14	f_2
117.5–122.5	6	
122.5–127.5	3	
127.5–132.5	1	

Substituting Into the Mode Formula

$$\text{Mode} = L + \Delta_1 / (\Delta_1 + \Delta_2) \times i$$

L

107.5

Lower real limit of the modal class

$\Delta_1 = | f_1 - f_0 |$

$|17 - 12| = 5$

Modal class f (17) vs. preceding class f (12)

$\Delta_2 = | f_1 - f_2 |$

$|17 - 14| = 3$

Modal class f (17) vs. succeeding class f (14)

i

5

Width of the modal class

Mode = 107.5 + (5/5+3) × 5 = 107.5 + 3.125

MODAL WEIGHT **110.625 kg**

Measures of Central Tendency : Mode

Individual Observations

Raw, ungrouped data

Discrete Series

Individual values with frequencies

Continuous Series

Grouped, class-interval data

Relationship among Mean, Median & Mode

In a moderately skewed distribution, the median lies about one-third of the way from mean to mode.



Typical order in a positively skewed distribution

Karl Pearson's Empirical Relationship

Mode = Mean − 3 (Mean − Median)

Mode = 3 Median − 2 Mean

Median = Mode + ⅔ (Mean − Mode)

Illustration 29

a Mean = 35.4, Mode = 32.1 — find median.

32.1 = 3(median) − 2(35.4)

3(median) = 32.1 + 70.8 = 102.9

Median = 102.9 / 3 = 34.3

b Median = 20.6, Mode = 26 — find mean.

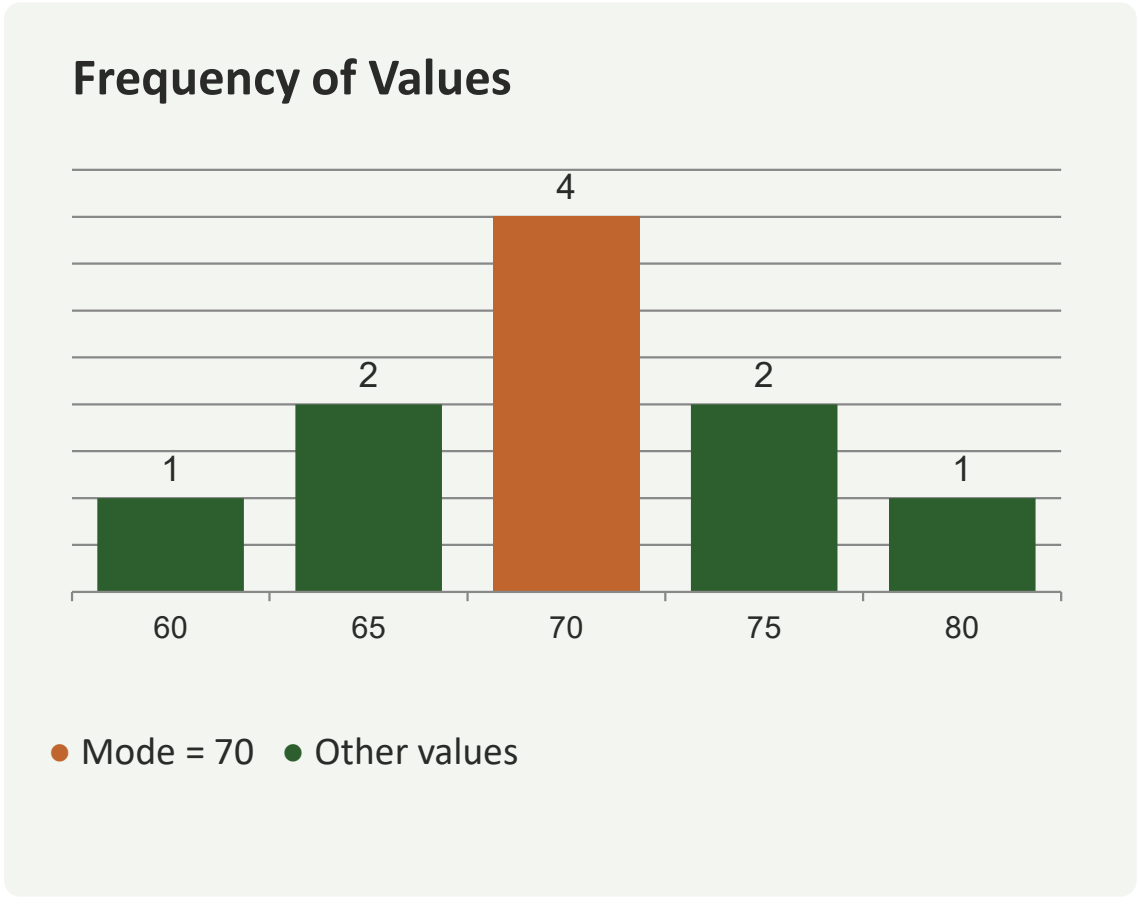
26 = 3(20.6) − 2(mean) = 61.8 − 2(mean)

2(mean) = 61.8 − 26 = 35.8

Mean = 17.9

Symmetric Distribution: Mean, Median, Mode

Dataset — Test Scores, n = 10: 60, 65, 65, 70, 70, 70, 70, 75, 75, 80



MEAN
70

MEDIAN
70

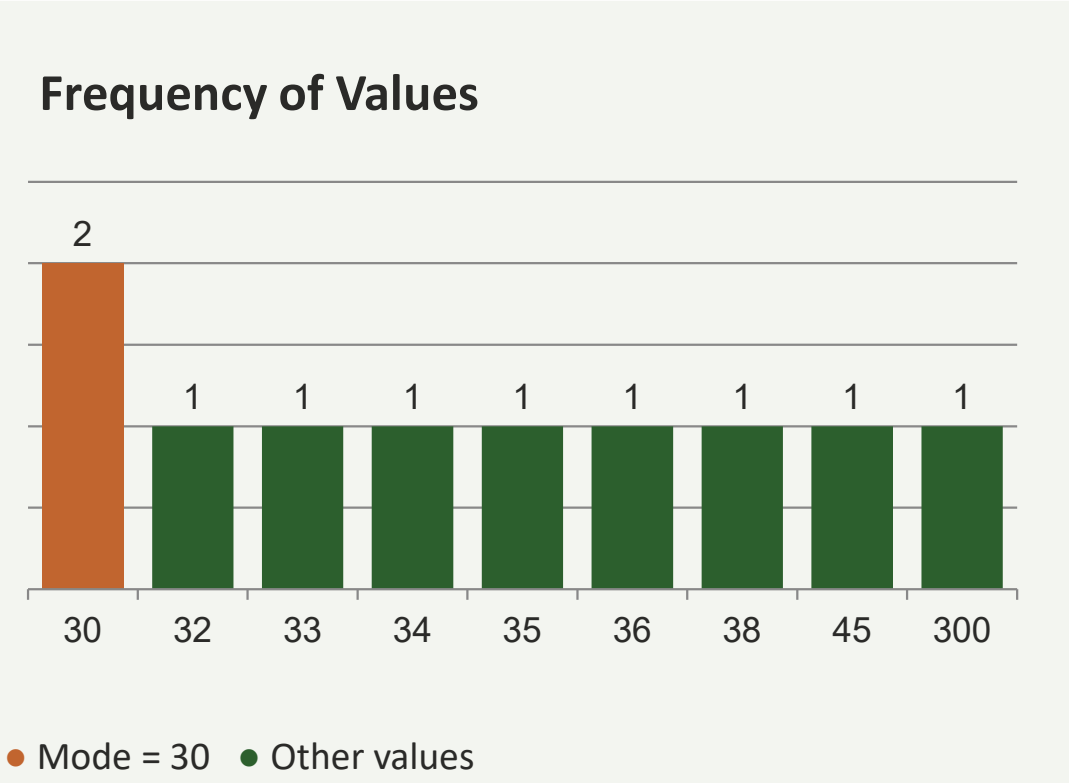
MODE
70

Mean ≈ Median ≈ Mode

All three measures land on the same value (70). The data is balanced around its center — safe to use the mean, and safe to assume something close to a normal distribution.

Right-Skewed Distribution (Positively Skewed): Mean, Median, Mode

Dataset , n = 10: 30, 30, 32, 33, 34, 35, 36, 38, 45, 300



MEAN
61.3

MEDIAN
34.5

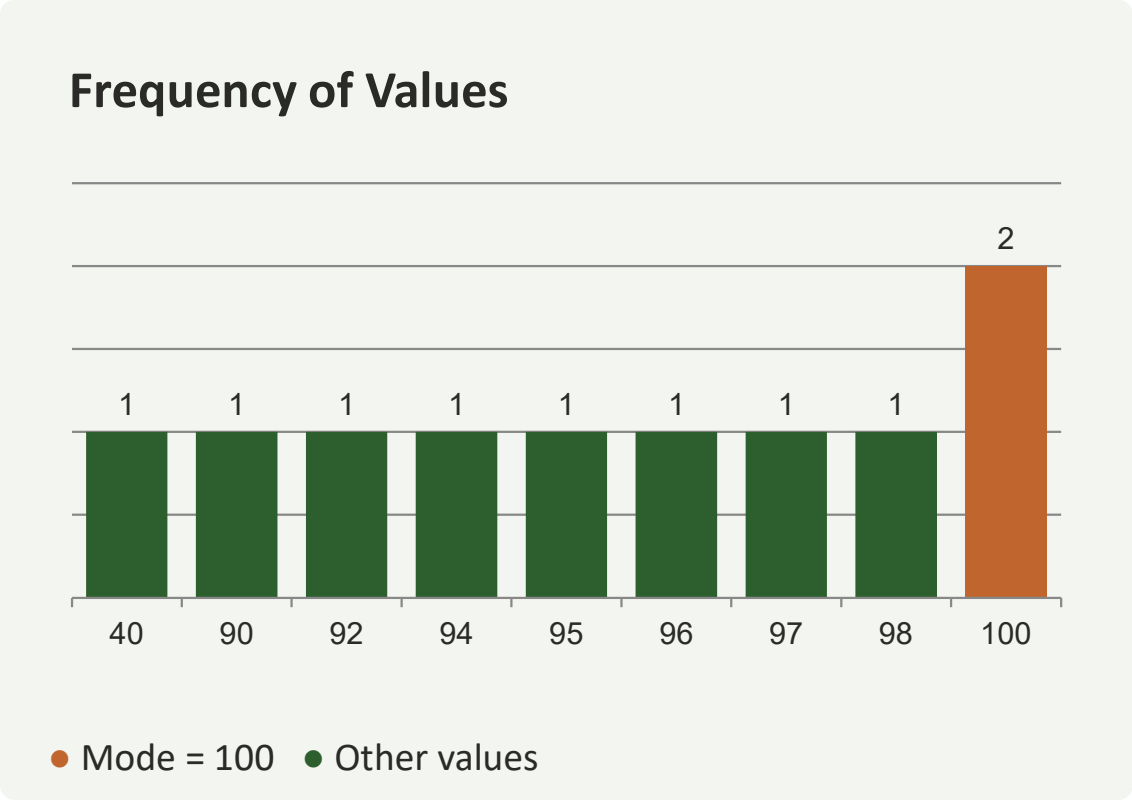
MODE
30

Mode < Median < Mean

A few very high incomes pull the mean up (61.3) far above where most people actually earn. The median (34.5) reflects the typical earner much better — report the median; consider a log-transform.

Left-Skewed Distribution (Negatively Skewed): Mean, Median, Mode

Dataset —n = 10: 40, 90, 92, 94, 95, 96, 97, 98, 100, 100



MEAN
90.2

MEDIAN
95.5

MODE
100

Mean < Median < Mode

One very low score (40) drags the mean down (90.2) below where most students actually scored. The mode (100) marks where most of the class really landed.

Types of Measures of Central Tendency/Averages

The following are the important types of averages

1. **Arithmetic Mean**
2. **Median**
3. **Mode**
4. ~~**Geometric Mean**~~
5. ~~**Harmonic Mean**~~

Beyond These

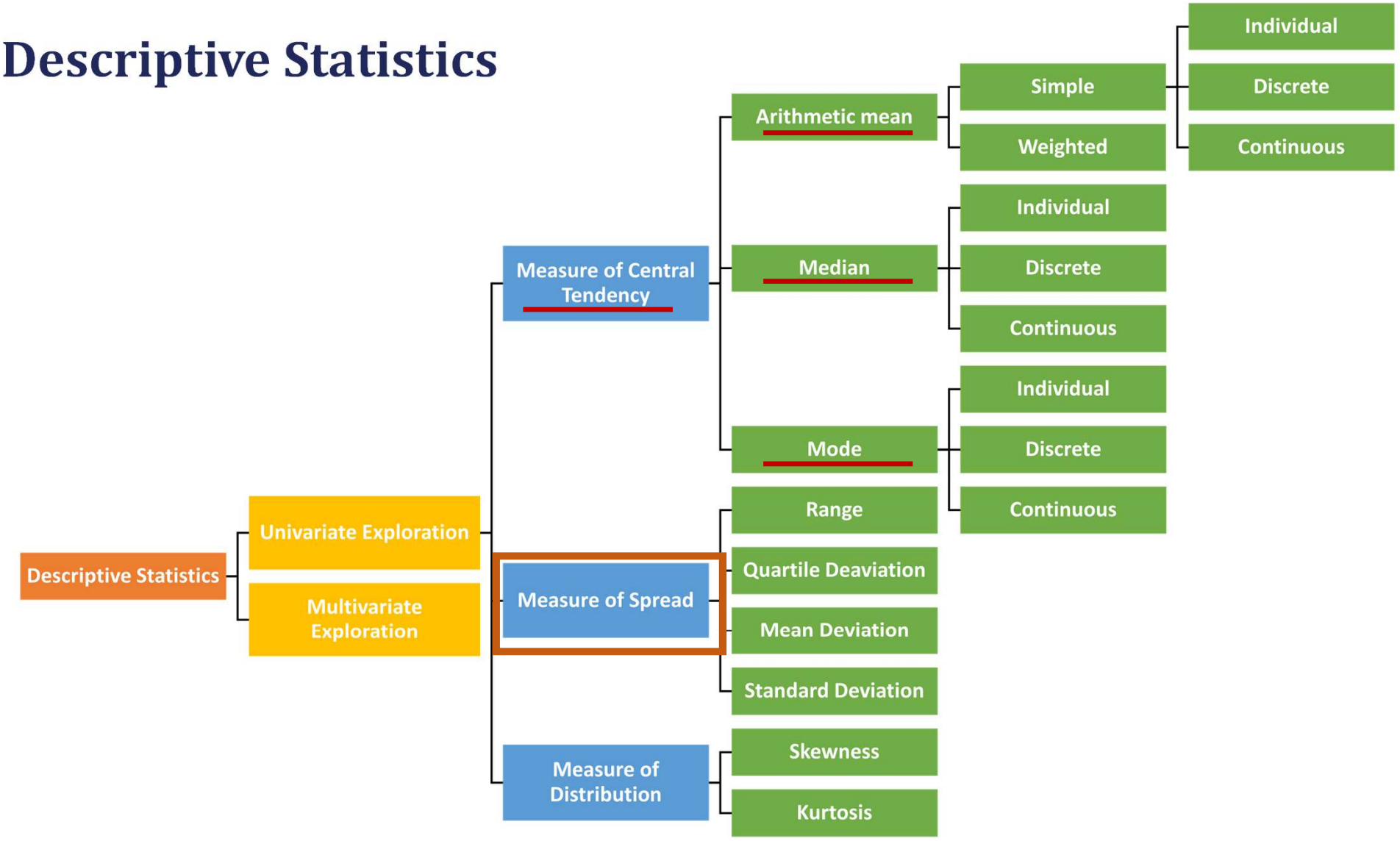
There are less important averages, such as moving average and progressive average. These have a very limited field of application and are, therefore, not so popular.

Measures of Central Tendency

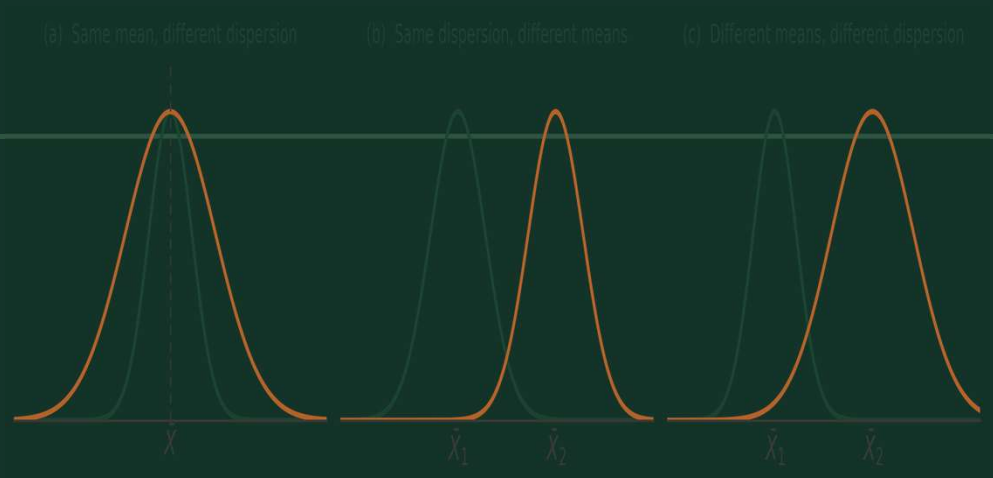
Measure	Individual Series	Discrete Series	Continuous Series
Arithmetic Mean	Direct: $\bar{X} = \Sigma X / N$ Short-cut: $\bar{X} = A + \Sigma d/N$	Direct: $\bar{X} = \Sigma fX / N$ Short-cut: $\bar{X} = A + \Sigma fd/N$	Direct: $\bar{X} = \Sigma fm / N$ Short-cut: $\bar{X} = A + \Sigma fd/N$ Step-deviation: $\bar{X} = A + (\Sigma fd'/N) \times i$
Combined Mean	$\bar{X}_{12} = (N_1\bar{X}_1 + N_2\bar{X}_2) / (N_1+N_2)$		
Weighted Mean	$\bar{X}_w = \Sigma WX / \Sigma W$ (Arithmetic)		
Median	Size of (N+1)/2 th item	Size of (N+1)/2 th item	Size of N/2 th item Median = $L + [(N/2 - c.f.)/f] \times i$
Mode	By inspection, or value occurring most frequently	Grouping method — value around which frequencies concentrate	$M_0 = L + [\Delta_1 / (\Delta_1+\Delta_2)] \times i$ $\Delta_1 = \lvert f_1 - f_0 \rvert$ $\Delta_2 = \lvert f_1 - f_2 \rvert$

N = *number of observations* · ***A*** = *assumed mean* · ***d*** = $X-A$ · ***i*** = *class interval* · ***f*** = *frequency* . ***L***= *lower limit of the class* . ***c.f.*** *cumulative freq. before median class* . ***m*** *mid-point of class* . ***i*** *class interval* . f_0, f_1, f_2 freq. of preceding, modal, and succeeding class

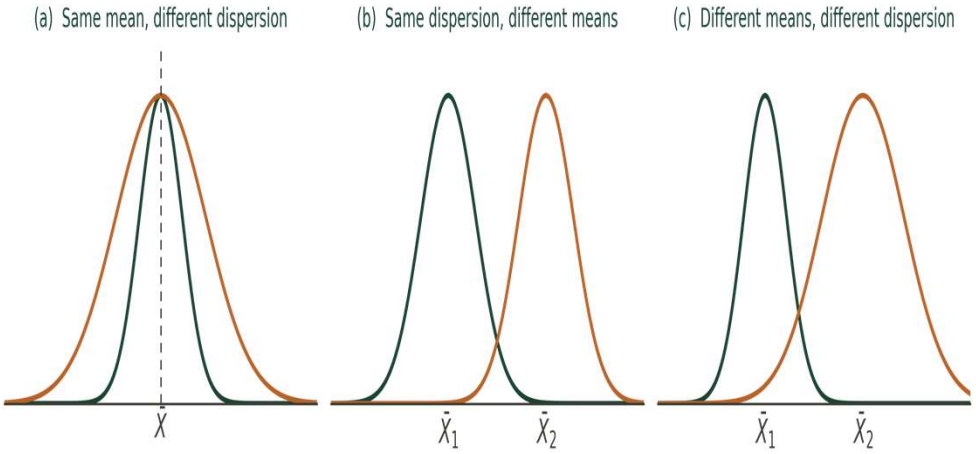
Descriptive Statistics



Measures of Spread/Dispersion



Averages Alone Can Mislead



	Series A	Series B	Series C
	100	100	1
	100	105	489
	100	102	2
	100	103	3
	100	90	5
Total	**500**	**500**	**500**
$\bar{x}$	**100**	**100**	**100**

All three series share $\bar{x}$ = 100 — yet A has zero spread, B a little, and C an enormous one.

The mean alone cannot distinguish these distributions. Two groups can have identical average wages, salaries, or incomes and still differ completely in how that value is spread across individuals — which is exactly what dispersion measures.

Measure of Spread or Dispersion

Dispersion, defined

"The degree to which numerical data tend to spread about an average value."

— Spiegel

Why it matters

- Two series can share the same mean yet differ hugely in spread
- A small dispersion means the average is typical & reliable
- A large dispersion makes the average unreliable, even misleading

3 purposes of measuring variation

1. Test the reliability of an average
2. Control the variability itself
3. Compare two or more series

A good measure of variation should be…

- Simple to understand
- Rigidly defined
- Amenable to algebra
- Not swayed by extremes
- Easy to compute
- Based on every item
- Sampling-stable

Measure of Spread or Dispersion

Absolute measures

Expressed in the original unit (₹, kg, tonnes). Comparable only when two data sets share the same unit and similar average size.

Relative measures

A ratio of an absolute measure to its average — a "coefficient of variation." Used when units or average sizes differ (e.g. Quintals of sugar vs tones of sugarcane, managers vs. worker's salary).

5 methods of studying variation

1. Range
2. Interquartile Range & Quartile Deviation
3. Mean (Average) Deviation
4. Standard Deviation
5. Lorenz Curve

Range & Quartile Deviation depend only on a few positional values — quick, but ignore most of the data.

Mean & Standard Deviation use every observation — more reliable, and rest on the same base throughout.

Rule: use Arithmetic Mean as base with Standard Deviation; use A.M. or Median as base with Mean Deviation.

Measures of Spread / Dispersion

1. **Range**
   Difference between the highest and lowest values in the data set.
   Positional

2. **Interquartile Range & Quartile Deviation**
   Spread of the middle 50% of values, between Q1 and Q3.
   Positional

3. ~~**Mean (Average) Deviation**~~
   ~~Average of the absolute differences from the mean.~~
   Calculated

4. **Standard Deviation**
   Square root of the average squared deviation from the mean.
   Calculated

5. ~~**Lorenz Curve**~~
   ~~Cumulative graph comparing distribution against perfect equality.~~
   Graphic

Range

The simplest method of studying dispersion

Definition

Range is defined as the difference between the value of the largest item and the value of the smallest item included in the distribution.

Key Insight

If two distributions have about the same average, the one with the smaller range has less dispersion — its average is more typical of the group.

Formulas

$$\text{Range} = L - S$$

Coefficient of Range

$$\frac{(L - S)}{(L + S)}$$

L = value of the largest item

S = value of the smallest item

Range: Individual & Discrete Series

The following are the prices of shares of XYZ Co. Ltd., Monday to Saturday. Calculate range and its coefficient

Daily Closing Price (₹)

Day	Price (₹)
Monday	200
Tuesday	210
Wednesday	208
Thursday	**160 (S)**
Friday	220
Saturday	**250 (L)**

Solution

$L = 250$ and $S = 160$

$\text{Range} = L - S = 250 - 160 = ₹\ 90$

$$\text{Coefficient of Range} = (L-S)/(L+S)$$
$$= (250-160)/(250+160) = 90/410$$

Coefficient of Range = 0.22

Range: Continuous Series & Merits

Calculate coefficient of range from the following data

Marks obtained by Students

Marks	No. of Students
10–20	8
20–30	10
30–40	12
40–50	8
50–60	4

L = 60, S = 10

Range = 60 − 10 = 50

Coeff. of Range
= 50 / 70 = 0.714

Two ways to find L and S in grouped data: (1) upper limit of highest class − lower limit of lowest class, or (2) mid-point of highest class − mid-point of lowest class.

Merits

- Amongst all methods of studying dispersion, Range is the simplest to understand and easiest to compute.
- It takes minimum time to calculate. If a quick (rather than very accurate) picture of variability is needed, Range is a good choice.

Measures of Spread / Dispersion

5 methods of studying variation

1. **Range**
   Difference between the highest and lowest values in the data set.
   Positional

2. **Interquartile Range & Quartile Deviation**
   Spread of the middle 50% of values, between Q1 and Q3.
   Positional

3. ~~**Mean (Average) Deviation**~~
   ~~Average of the absolute differences from the mean.~~
   Calculated

4. **Standard Deviation**
   Square root of the average squared deviation from the mean.
   Calculated

5. ~~**Lorenz Curve**~~
   ~~Cumulative graph comparing distribution against perfect equality.~~
   Graphic

Measures of Spread / Dispersion

5 methods of studying variation

1. **Range**
   Difference between the highest and lowest values in the data set.
   Positional

2. **Interquartile Range & Quartile Deviation**
   Spread of the middle 50% of values, between Q1 and Q3.
   Positional

3. ~~**Mean (Average) Deviation**~~
   ~~Average of the absolute differences from the mean.~~
   Calculated

4. **Standard Deviation**
   Square root of the average squared deviation from the mean.
   Calculated

5. ~~**Lorenz Curve**~~
   ~~Cumulative graph comparing distribution against perfect equality.~~
   Graphic

Why Not Just Use the Range?

The range has real limitations as a measure of dispersion

01

Based on two extreme items

The range depends only on the smallest and largest values, ignoring everything in between.

02

Ignores the scatter within the range

It says nothing about how observations are actually distributed between the extremes.

03

Sensitive to outliers

A single unusually high or low value can distort the range and mislead interpretation.

The idea: discard the extremes and study the range of the middle 50% of the data — this is the ***interquartile range.***

Defining the Interquartile Range

The range which includes the middle 50 per cent of the distribution

One quarter of the observations at the lower end, and another quarter at the upper end, are excluded.

In other words, the interquartile range represents the difference between the third quartile and the first quartile.

FORMULA

$$\text{Interquartile Range} = Q_3 - Q_1$$

Q_3 = third quartile (75th percentile) *Q_1 = first quartile (25th percentile)*

Q1

First quartile — 25% of observations lie below this point.

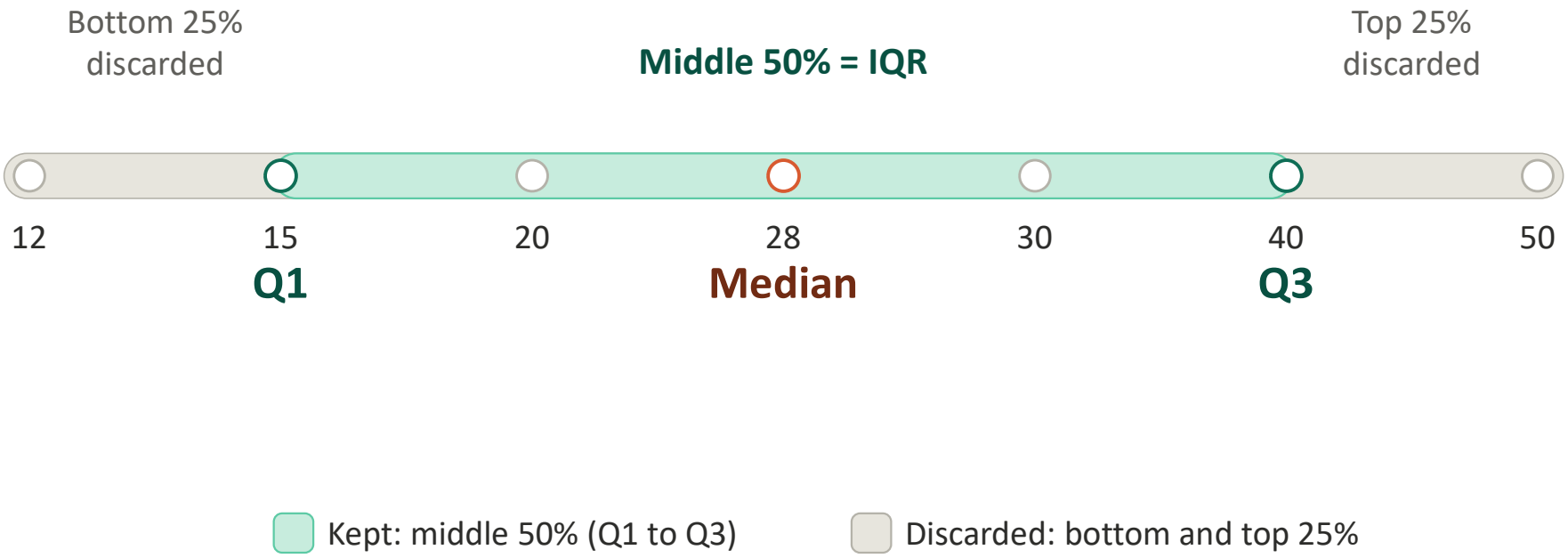
Q3

Third quartile — 75% of observations lie below this point.

Together, Q_1 and Q_3 bracket the central 50% of the data — the part least affected by extreme values.

Interquartile range (IQR)

The middle 50% of a dataset, bounded by the first quartile (Q1) and third quartile (Q3)



Quartile Deviation (Semi-Interquartile Range)

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2}$$

Half of the interquartile range — the average distance of each quartile from the median.

What Q.D. tells you

Symmetrical distribution: Q1 and Q3 are equidistant from the median, so Median ± Q.D. covers exactly the middle 50% of observations.

Asymmetrical distribution: Q1 and Q3 are not equidistant from the median — so Median ± Q.D. covers only approximately 50%.

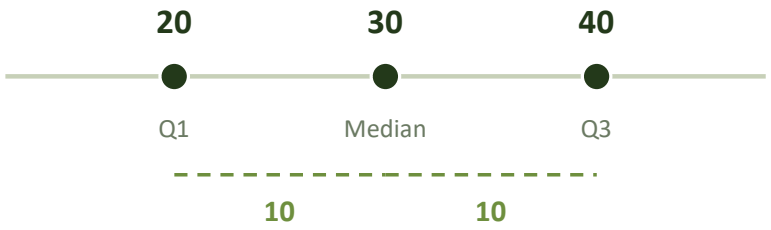
Symmetric vs Asymmetric: An Example

Whether Q1 and Q3 sit equidistant from the median changes what Median ± Q.D. actually covers

SYMMETRIC

Symmetric Example

Q1 = 20 Median = 30 Q3 = 40



Q.D. = (40 – 20) / 2 = 10

Distances match (10 = 10) — Median ± Q.D. = 20 to 40, exactly Q1 to Q3.

ASYMMETRIC

Asymmetric Example

Q1 = 20 Median = 25 Q3 = 45



Q.D. = (45 – 20) / 2 = 12.5

Distances differ (5 ≠ 20) — Median ± Q.D. = 12.5 to 37.5, which only approximates Q1 to Q3.

Quartile Deviation (Semi-Interquartile Range)

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2}$$

Half of the interquartile range — the average distance of each quartile from the median.

What Q.D. tells you

Symmetrical distribution: Q1 and Q3 are equidistant from the median, so Median ± Q.D. covers exactly the middle 50% of observations.

Asymmetrical distribution: Q1 and Q3 are not equidistant from the median — so Median ± Q.D. covers only approximately 50%.

READING THE MAGNITUDE

Small Q.D.
High uniformity — low variation among the central 50% of items.

Large Q.D.
Wide variation among the central items — the middle 50% is spread out.

Coefficient of Quartile Deviation

Quartile deviation is an **absolute** measure of dispersion — it is expressed in the same units as the data, which makes it hard to compare across different distributions.

The **coefficient of quartile deviation** is the corresponding relative measure, so it can compare the degree of variation across distributions with different units or scales.

FORMULA

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

This equals $(Q_3 - Q_1)/2$ divided by $(Q_3 + Q_1)/2$ — the quartile deviation expressed relative to the average of the two quartiles.

Interquartile Range & Quartile Deviation

1

Interquartile Range

$$IQR = Q_3 - Q_1$$

2

Quartile Deviation (Q.D.)

$$Q.D. = (Q_3 - Q_1) / 2$$

3

Coefficient of Q.D.

$$(Q_3 - Q_1) / (Q_3 + Q_1)$$

Interquartile Range & Quartile Deviation: Individual Observations

Find out the value of quartile deviation and its coefficient from the following data:
Roll Number : 1 2 3 4 5 6 7
Marks : 20 28 40 12 30 15 50

Ascending marks (N = 7):

12	**15**	20	28	30	**40**	50

1	**Locate Q_1**	Q_1 = size of (N+1)/4 th item = (8)/4 = 2nd item → Q_1 = 15
2	**Locate Q_3**	Q_3 = size of 3(N+1)/4 th item = 3(8)/4 = 6th item → Q_3 = 40
3	**range**	Range = (Q_3 − Q_1) = (40 − 15) = 25
4	**Quartile Deviation**	Q.D. = (Q_3 − Q_1)/2 = (40 − 15)/2 = 12.5
5	**Coefficient of Q.D.**	(Q_3−Q_1)/(Q_3+Q_1) = 25/55 = 0.455

Interquartile Range & Quartile Deviation: Discrete Observations

Illustration 4: Compute coefficient of quartile deviation from the following data:

Marks	10	20	30	40	50	60
Number of students	4	7	15	8	7	2

Marks	10	20	30	40	50	60
Frequency	4	7	15	8	7	2
c.f.	4	11	26	34	41	43

Q_1 position

(N+1)/4 = 44/4 = 11th item → falls in c.f. 11 → $Q_1 = 20$

Q_3 position

3(N+1)/4 = 132/4 = 33rd item → falls in c.f. 34 → $Q_3 = 40$

Quartile Deviation

Q.D. = (40 − 20)/2 = 10

Coefficient of Q.D.

(40−20)/(40+20) = 20/60 = 0.333

Interquartile Range & Quartile Deviation: Continuous Observations

Find Q.D. and its coefficient from grouped wage data (200 wage earners)

Wages (₹/hour)	f	cf
Less than 35	14	14
35 – 37	62	76
38 – 40	99	175
41 – 43	18	193
Over 43	7	200

INTERPOLATION FORMULA

$$Q_k = L + \frac{kN/4 - cf}{f} \times i$$

L = lower boundary of the class · cf = cumulative frequency before it · f = class frequency · i = class width. Use k = 1 for Q_1, k = 3 for Q_3.

1. **Locate Q_1 class** — N/4 = 200/4 = 50 → lies in class 35–37 (L=35, cf = 14, f = 62) → $Q_1 = 35 + (50-14)/62 \times 2 = 36.16$
2. **Locate Q_3 class** — 3N/4 = 150 → lies in class 38–40 (L=38, cf = 76, f = 99) → $Q_3 = 38 + (150-76)/99 \times 2 = 39.49$
3. **Quartile Deviation** — Q.D. = $(Q_3 - Q_1)/2 = (39.49 - 36.16)/2 = 1.67$
4. **Coefficient of Q.D.** — $(Q_3 - Q_1)/(Q_3 + Q_1) = (39.49-36.16)/(39.49+36.16) = 3.33/75.65 = 0.044$

Merits & Limitations of Quartile Deviation

Merit

1. **Handles open-end distributions**
   Works well even when the first or last class is open-ended, since extreme values aren't needed.

2. **Resistant to extreme values**
   Useful for erratic or badly skewed data, where other measures get warped by outliers.

Limitation

3. **Ignores 50% of the data**
   The lowest and highest quarters are discarded entirely, so it isn't based on every item.

4. **Not capable of algebraic treatment**
   It cannot be manipulated mathematically and is heavily affected by sampling fluctuations.

5. **A positional average, not true dispersion**
   It measures a distance on a scale rather than scatter around an average — more a partition than a dispersion measure.

Measures of Spread / Dispersion

5 methods of studying variation

1. **Range**
   Difference between the highest and lowest values in the data set.
   Positional

2. **Interquartile Range & Quartile Deviation**
   Spread of the middle 50% of values, between Q1 and Q3.
   Positional

3. ~~**Mean (Average) Deviation**~~
   ~~Average of the absolute differences from the mean.~~
   Calculated

4. **Standard Deviation**
   Square root of the average squared deviation from the mean.
   Calculated

5. ~~**Lorenz Curve**~~
   ~~Cumulative graph comparing distribution against perfect equality.~~
   Graphic

What Is Standard Deviation?

Its significance lies in being free from the defects of earlier methods and satisfying nearly every property of a good measure of dispersion.

1. **By Far the Most Important Measure**
   Standard deviation is the most widely used and reliable measure for studying dispersion in a distribution.

2. **Root Mean Square Deviation**
   It is the square root of the mean of the squared deviations of items from the arithmetic mean.

3. **Reflects the Degree of Variability**
   A small σ means high uniformity and homogeneity of the series; a large σ means the opposite.

4. **The Best Judge of a Representative Mean**
   Among comparable series with similar means, the one with the smallest σ has the most representative mean.

Calculating σ for Individual Observations

Two approaches, depending on whether the actual mean is a convenient whole number.

Actual Mean Method

$$\sigma = \sqrt{(\Sigma x^2 / N)}$$

where $x = (X - \bar{X})$

1. Calculate the actual mean, $\bar{X}$.
2. Find deviations $x = (X - \bar{X})$ for every item.
3. Square the deviations and total them: Σx^2.
4. Divide by N and take the square root.

Assumed Mean Method

$$\sigma = \sqrt{[\Sigma d^2/N - (\Sigma d/N)^2]}$$

where $d = (X - A)$

1. Choose a convenient assumed mean, A.
2. Find deviations $d = (X - A)$ and total them: Σd.
3. Square the deviations and total them: Σd^2.
4. Substitute Σd^2, Σd and N into the formula.

Calculating σ for Individual Observations: Actual Mean Method

Blood serum cholesterol levels of 10 persons: 240, 260, 290, 245, 255, 288, 272, 263, 277, 251

X	$x = (X - \bar{X})$	x^2
240	-24.1	580.81
260	-4.1	16.81
290	25.9	670.81
245	-19.1	364.81
255	-9.1	82.81
288	23.9	571.21
272	7.9	62.41
263	-1.1	1.21
277	12.9	166.41
251	-13.1	171.61
$\Sigma X = 2641$		**$\Sigma x^2 = 2689$**

SOLUTION

$\bar{X} = (2641 / 10)$, $\bar{X} = 264.1$

$$\sigma = \sqrt{(\Sigma x^2 / N)}$$

$$= \sqrt{(2689 / 10)}$$

$$\sigma = 16.398$$

mg / 100 ml (approx.)

Calculating σ for Individual Observations: Assumed Mean Method

Blood serum cholesterol levels of 10 persons: 240, 260, 290, 245, 255, 288, 272, 263, 277, 251 (assumed mean A = 264)

X	d = (X – 264)	d^2
240	-24	576
260	-4	16
290	26	676
245	-19	361
255	-9	81
288	24	576
272	8	64
263	-1	1
277	13	169
251	-13	169
ΣX = 2641	**Σd = +1**	**Σd^2 = 2689**

SOLUTION

N = 10, $\Sigma d = +1$, $\Sigma d^2 = 2689$

$$\sigma = \sqrt{[\Sigma d^2/N - (\Sigma d/N)^2]}$$

$$\sigma = \sqrt{(2689/10 - (1/10)^2)}$$

$$= \sqrt{(268.9 - 0.01)}$$

$\sigma = 16.398$ *mg / 100 ml (approx.)*

Calculating σ for Discrete Series

For calculating standard deviation in discrete series, any of the following methods may be applied.

01 Actual Mean Method

Deviations are taken from the actual mean $\bar{X}$, squared, and multiplied by frequencies.

$$\sigma = \sqrt{\left(\Sigma f x^2 / N \right)}, \quad x = (X - \bar{X})$$

Rarely used in practice — fractional means make calculations tedious.

02 Assumed Mean Method

Deviations are taken from a conveniently chosen assumed mean A instead of the actual mean.

$$\sigma = \sqrt{\left(\Sigma f d^2/N - (\Sigma f d/N)^2 \right)}, \quad d = (X - A)$$

The most commonly used method in practice.

03 Step Deviation Method

Deviations of mid-points from the assumed mean are further divided by class width (interval) i.

$$\sigma = \sqrt{\left(\Sigma f d^2/N - (\Sigma f d/N)^2 \right)} \times i, \quad d = (X-A)/i$$

Simplifies arithmetic considerably; most efficient for grouped data.

Calculating σ by the Actual Mean Method

Size of item vs. frequency — deviations x = (X − X̄) taken directly from the actual mean

X	f	fX	x *(X − X̄)*	x^2	fx^2
3.5	3	10.50	-3.59	12.89	38.66
4.5	7	31.50	-2.59	6.71	46.95
5.5	22	121.00	-1.59	2.53	55.61
6.5	60	390.00	-0.59	0.35	20.88
7.5	85	637.50	+0.41	0.17	14.30
8.5	32	272.00	+1.41	1.99	63.63
9.5	8	76.00	+2.41	5.81	46.47
N = 217		**ΣfX = 1538.50**			**Σfx² = 286.50**

STEP 1 — FIND THE MEAN

X̄ = ΣfX / N = 1538.50 / 217 = 7.09

STEP 2 — FIND σ

σ = √(Σfx² / N)

Σfx² = 286.50, N = 217

= √(286.50 / 217) = √1.320

σ = 1.149

Calculating σ by Assumed Mean Method

Size of item vs. frequency — assumed mean A = 6.5

X	f	d = (X−6.5)	fd	fd^2
3.5	3	-3	-9	27
4.5	7	-2	-14	28
5.5	22	-1	-22	22
6.5	60	0	0	0
7.5	85	+1	+85	85
8.5	32	+2	+64	128
9.5	8	+3	+24	72
N = 217			**Σfd = +128**	**Σfd^2 = 362**

SOLUTION

$$\sigma = \sqrt{\left(\Sigma fd^2/N - (\Sigma fd/N)^2\right)}$$

$\Sigma fd^2 = 362$, $\Sigma fd = 128$, $N = 217$

$= \sqrt{(1.668 - 0.348)}$

$\sigma = 1.149$

Calculating σ by Step Deviation Method

Illustration 12: The annual salaries of a group of employees are given in the following table:

Salaries (in ₹ '000)	45	50	55	60	65	70	75	80
Number of persons	3	5	8	7	9	7	4	7

Annual salaries (₹ '000) of a group of employees — assumed mean A = 60, i = 5

X	f	d=(X−60)/5	fd	fd²
45	3	-3	-9	27
50	5	-2	-10	20
55	8	-1	-8	8
60	7	0	0	0
65	9	+1	+9	9
70	7	+2	+14	28
75	4	+3	+12	36
80	7	+4	+28	112
N = 50			**Σfd = 36**	**Σfd² = 240**

SOLUTION

$\sigma = \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i$

$\Sigma fd^2=240$, $N=50$, $\Sigma fd=36$, $i=5$

$= \sqrt{(4.8 - 0.5184)} \times 5$

$\sigma = 10.35$

Calculating σ for a Continuous Series

In a continuous series, discrete-series methods still apply — but the step deviation method is used most often in practice.

FORMULA

$$\sigma = \sqrt{\left(\Sigma fd^2/N - (\Sigma fd/N)^2 \right)} \times i$$

$d = (m - A) / i$
i = class interval
A = assumed mean
m = mid-point

STEPS

1. Find the mid-points (m) of the various classes.
2. Take deviations of the mid-points from the assumed mean A; denote as d.
3. Wherever possible, take out a common factor (class interval i) from this column.
4. Multiply frequency (f) by deviation (d) for each class and obtain Σfd.
5. Square the deviations, multiply by frequency, and obtain Σfd^2.

Only difference from discrete series: mid-points of classes must first be found.

Calculating σ for a Continuous Series

Calculate mean, median and standard deviation from the following data. Profits of Companies (₹ lakh)

Profits (₹ lakh)	0–10	10–20	20–30	30–40	40–50	50–60
No. of companies	12	17	23	39	16	3

Σ companies (N) **110** | Class interval (i) **10** | Assumed mean (A) **35**

Profits	f	m.p. (m)	(m−35)/10 = d’	fd	(fd)²	cf
0–10	12	5	-3	-36	108	12
10–20	17	15	-2	-34	68	29
20–30	23	25	-1	-23	23	52
30–40	39	35	0	0	0	91
40–50	16	45	1	16	16	107
50–60	3	55	2	6	12	110
Total	**N = 110**			**Σfd’= −71**	**Σfd’² = 227**	

RESULTS

$\bar{X} = A + (\Sigma fd/N) \times i$

Mean $35 + (-71/110) \times 10$

₹ 28.55 lakh

Calculating σ for a Continuous Series

Calculate mean, median and standard deviation from the following data. Profits of Companies (₹ lakh)

Profits (₹ lakh)	0–10	10–20	20–30	30–40	40–50	50–60
No. of companies	12	17	23	39	16	3

Σ companies (N) **110** | Class interval (i) **10** | Assumed mean (A) **35**

Profits	f	m.p. (m)	(m−35)/10 = d’	Fd’	(fd’)²	cf
0–10	12	5	-3	-36	108	12
10–20	17	15	-2	-34	68	29
20–30	23	25	-1	-23	23	52
30–40	39	35	0	0	0	91
40–50	16	45	1	16	16	107
50–60	3	55	2	6	12	110
Total	**N = 110**			**Σfd’ = −71**	**Σfd’² = 227**	

RESULTS

Mean **₹ 28.55 lakh**

Median

$$L + (N/2 - cf)/f \times i$$

N/2=55 , L=30, cf=52, f=39

₹ 30.77 lakh

Calculating σ for a Continuous Series

Calculate mean, median and standard deviation from the following data. Profits of Companies (₹ lakh)

Profits (₹ lakh)	0–10	10–20	20–30	30–40	40–50	50–60
No. of companies	12	17	23	39	16	3

Σ companies (N) **110** | Class interval (i) **10** | Assumed mean (A) **35**

Profits	f	m.p. (m)	(m−35)/10 = d	Fd’	(fd’)²	cf
0–10	12	5	-3	-36	108	12
10–20	17	15	-2	-34	68	29
20–30	23	25	-1	-23	23	52
30–40	39	35	0	0	0	91
40–50	16	45	1	16	16	107
50–60	3	55	2	6	12	110
Total	**N = 110**			**Σfd’ = −71**	**Σfd’² = 227**	

RESULTS

Mean **₹ 28.55 lakh**

Median **₹ 30.77 lakh**

$$\sigma = \sqrt{\left(\frac{\Sigma fd^2}{N} - \left(\frac{\Sigma fd}{N}\right)^2\right)} \times i$$

$$\sqrt{(227/110 - (-71/110)^2)} \times 10$$
$$= \sqrt{(2.064 - 0.417)} \times 10$$
$$= 1.283 \times 10$$

σ = ₹ 12.83 lakh

Calculating σ for a Continuous Series

Find the standard deviation from the following data.

Age under	10	20	30	40	50	60	70	80
No. dying	15	30	53	75	100	110	115	125

This is a cumulative ("less-than") series — first convert to ordinary class intervals (0–10, 10–20, …) with class frequencies before applying step deviation.

N (total) **125**

Class interval (i) **10**

Age	f	m.p. (m)	(m−35)/10 = d	fd	fd²
0–10	15	5	-3	-45	135
10–20	15	15	-2	-30	60
20–30	23	25	-1	-23	23
30–40	22	35	0	0	0
40–50	25	45	1	25	25
50–60	10	55	2	20	40
60–70	5	65	3	15	45
70–80	10	75	4	40	160
Total	**N = 125**			**Σfd = 2**	**Σfd² = 488**

SOLUTION

$$\sigma = \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i$$

$$= \sqrt{(488/125 - (2/125)^2)} \times 10$$

$$= \sqrt{(3.904 - 0.0003)} \times 10$$

$$= 1.978 \times 10$$

$$\sigma = \mathbf{19.76}$$

Merits & Limitations of Standard Deviation

Merit

1. **Rigidly Defined & Based on Every Item**
   Its value is fixed and mathematically determined, taking every observation in the series into account.

2. **Capable of Algebraic Treatment**
   Amenable to further mathematical manipulation — for instance, the combined σ of two or more series can be computed directly.

Limitation

3. **Harder to Calculate & Understand**
   Involves squaring, summing and extracting a square root — less intuitive than simpler measures for a non-technical audience.

4. **Overweights Extreme Values**
   Because deviations are squared, values far from the mean influence σ disproportionately more than values close to it.

5. **Unsuitable for Open-End Distributions**
   Cannot be computed without first assuming values for the open classes at either end of the series.

Measures of Spread / Dispersion

5 methods of studying variation

1. **Range**
   Difference between the highest and lowest values in the data set.
   Positional

2. **Interquartile Range & Quartile Deviation**
   Spread of the middle 50% of values, between Q1 and Q3.
   Positional

3. ~~**Mean (Average) Deviation**~~
   ~~Average of the absolute differences from the mean.~~
   Calculated

4. **Standard Deviation**
   Square root of the average squared deviation from the mean.
   Calculated

5. ~~**Lorenz Curve**~~
   ~~Cumulative graph comparing distribution against perfect equality.~~
   Graphic

Measures of Spread

Same number "10" — three different meanings

R Range

Max – Min

"Highest value is 10 units more than the lowest."



Range = 55–45 = 10

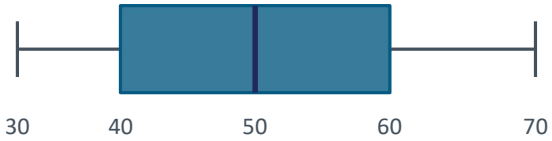
Uses only 2 points — very sensitive to outliers.

Example: This week's temps ranged 45–55°C.

Q Quartile Deviation

(Q3 – Q1) / 2

"Middle 50% of data lies within ±10 of the median."



QD = (60–40)/2 = 10

Ignores extremes — robust to outliers.

Example: House prices — QD reflects the typical range

S Standard Deviation

√(Σ(x–mean)² / n)

"Each value differs from the mean by ~10, on average."



Uses every point — most common in ML & stats.

Example: Salary avg = 50k, SD = 10k → most earn 40k–60k.

Quartiles, Deciles & Percentiles

Beyond the median — measures that divide a series into equal parts

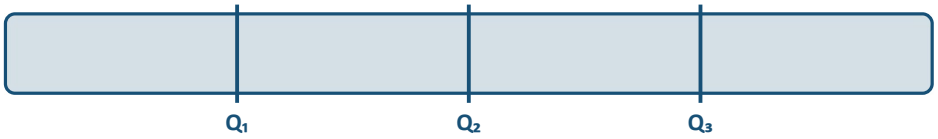
The median splits a series into 2 equal halves. Quartiles, deciles, and percentiles generalize this idea, splitting a series into 4, 10, and 100 equal parts.

Q **Quartiles** — divides the series into 4 equal parts *3 quartiles: Q_1, Q_2, Q_3*

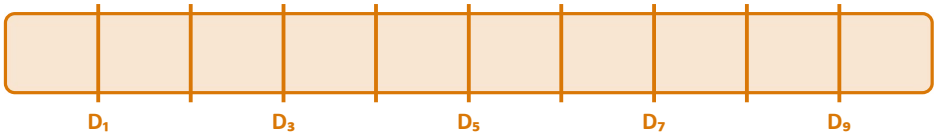
D **Deciles** — divides the series into 10 equal parts. *9 deciles: D_1, D_2, …, D_9*

P **Percentiles** — divides the series into 100 equal parts. *99 percentiles: P_1, P_2, …, P_{99}*

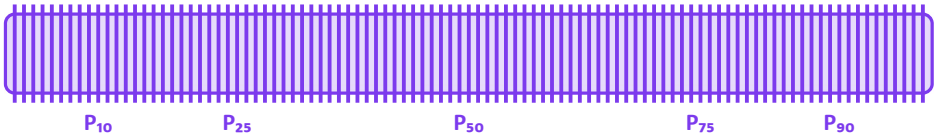
Quartiles — 4 equal parts



Deciles — 10 equal parts



Percentiles — 100 equal parts



$Q_2 = D_5 = P_{50}$ = Median

Quartiles, Deciles & Percentiles Individual Series

Position formulas: $Q_1 = (N+1)/4$, $Q_2 = (N+1)/2$, $Q_3 = 3(N+1)/4$ · $D_k = k(N+1)/10$ · $P_k = k(N+1)/100$

Marks of 9 students (ascending)
12, 18, 25, 30, 35, 40, 45, 50, 60 **N = 9**

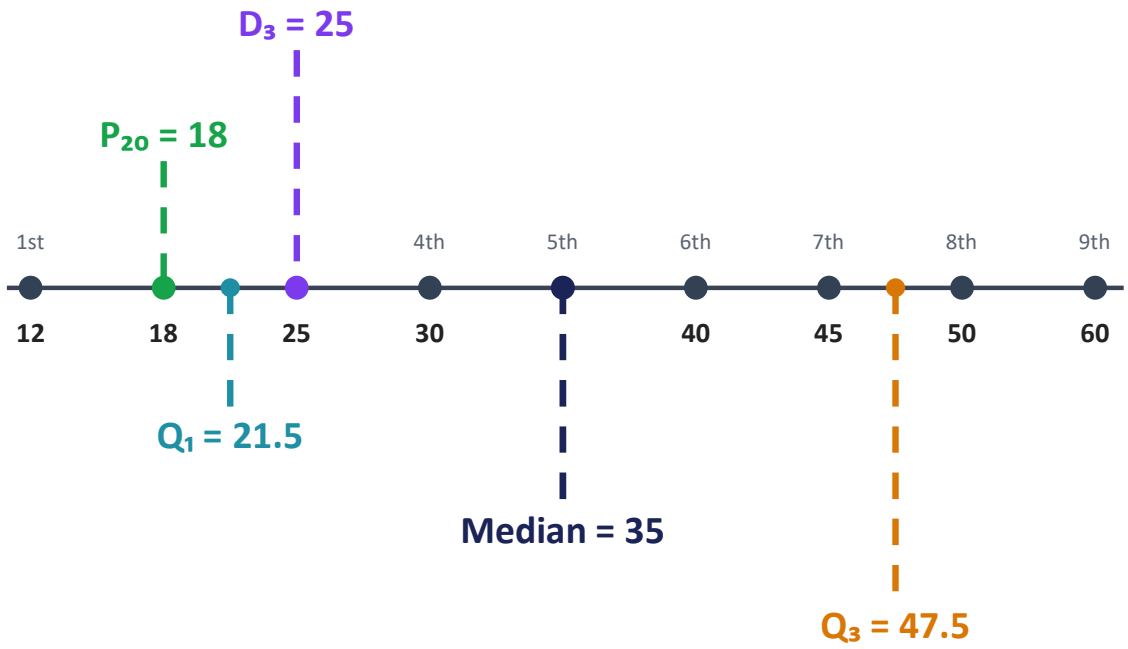
P_{20} position = 20(9+1)/100 = 2nd item
→ exactly the 2nd value = 18

Q_1 position = (9+1)/4 = 2.5th item
→ between 2nd (18) & 3rd (25)
Q_1 = (18 + 25)/2 = 21.5

D_3 position = 3(9+1)/10 = 3rd item
→ exactly the 3rd value = 25

Median position = (9+1)/2 = 5th item
→ exactly the 5th value = 35

Q_3 position = 3×10/4 = 7.5th item
→ between 7th (45) & 8th (50)
Q_3 = 45 + 50) = 47.5



Quartiles, Deciles & Percentiles: Discrete Series

Position formulas: $Q_1 = (N+1)/4$, $Q_2 = (N+1)/2$, $Q_3 = 3(N+1)/4$ · $D_k = k(N+1)/10$ · $P_k = k(N+1)/100$

X	f	cf
10	4	4
20	6	10
30	10	20
40	8	28
50	2	30
N = 30		

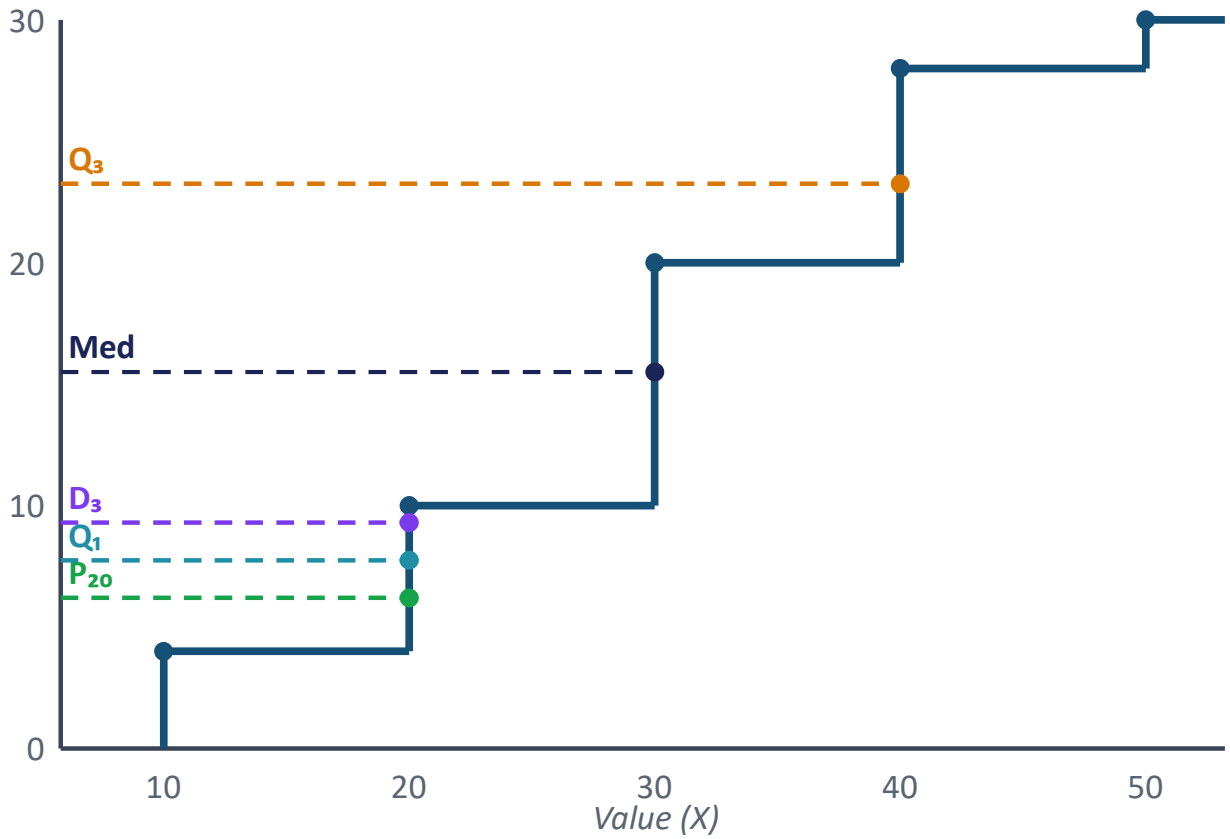
P_{20} pos = 20(30+1)/100 = 6.2th → P_{20} = 20

Q_1 pos = (30+1)/4 = 7.75th → Q_1 = 20

D_3 pos = 3(30+1)/10 = 9.3th → D_3 = 20

Median pos = 31/2 = 15.5th → Median = 30

Q_3 pos = 3×31/4 = 23.25th → Q_3 = 40



P_{20}, Q_1, and D_3 all fall on the same jump (10→10 at X=20), since their positions 6.2, 7.75, and 9.3 all lie between cf = 4 and cf = 10.

Quartiles, Decile & Percentile: Continuous Series

Locating Q_1, Q_3, D_3, P_{20} on the cumulative-frequency curve (ogive)

Class	f	cf
0–5	7	7
5–10	18	25
10–15	25	50
15–20	30	80
20–25	20	100
N = 100		

Quartiles, Decile & Percentile: Continuous Series

Formula: L + ((target − cf) / f) × I where target is $\mathbf{Q_1}=(N)/4$ · $\mathbf{Q_3}=3(N)/4$ · $D_k = k(N)/10$ · $P_k = k(N)/100$

Class	f	cf
0–5	7	7
5–10	18	25
10–15	25	50
15–20	30	80
20–25	20	100

Q_1 — Lower Quartile

Class: 5–10

Position: N/4 = 100/4 = 25th item

L=5, cf=7, f=18, i=5

$Q_1 = 5 + ((25-7)/18) \times 5$ **$Q_1 = 5 + 5.00 = 10$**

Q_3 — Upper Quartile

Class: 15–20

Position: 3N/4 = 300/4 = 75th item

L=15, cf=50, f=30, i=5

$Q_3 = 15 + ((75-50)/30) \times 5$ **$Q_3 = 15 + 4.17 = 19.17$**

D_3 — Third Decile

Class: 10–15

Position: 3N/10 = 300/10 = 30th item

L=10, cf=25, f=25, i=5

$D_3 = 10 + ((30-25)/25) \times 5$ **$D_3 = 10 + 1.00 = 11$**

P_{20} — 20th Percentile

Class: 5–10

Position: 20N/100 = 2000/100 = 20th item

L=5, cf=7, f=18, i=5

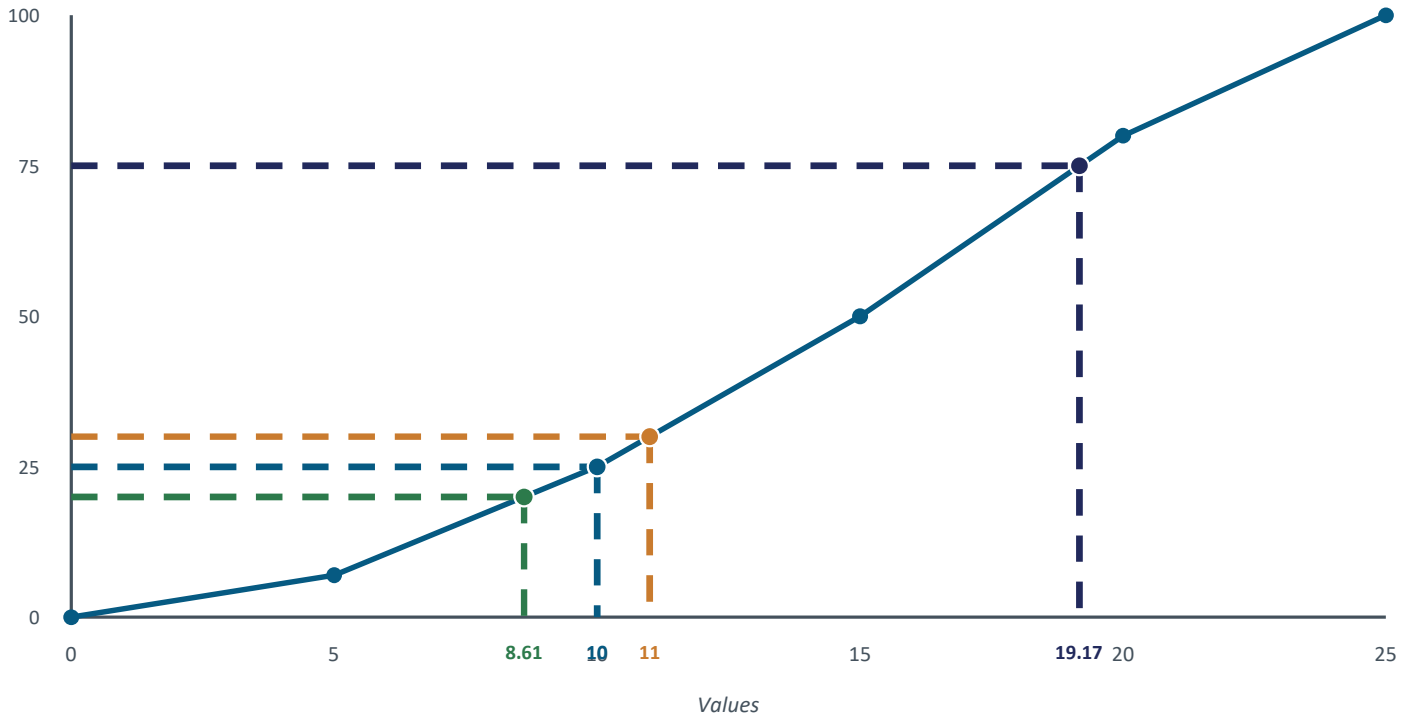
$P_{20} = 5 + ((20-7)/18) \times 5$ **$P_{20} = 5 + 3.61 = 8.61$**

Quartiles, Decile & Percentile: Continuous Series

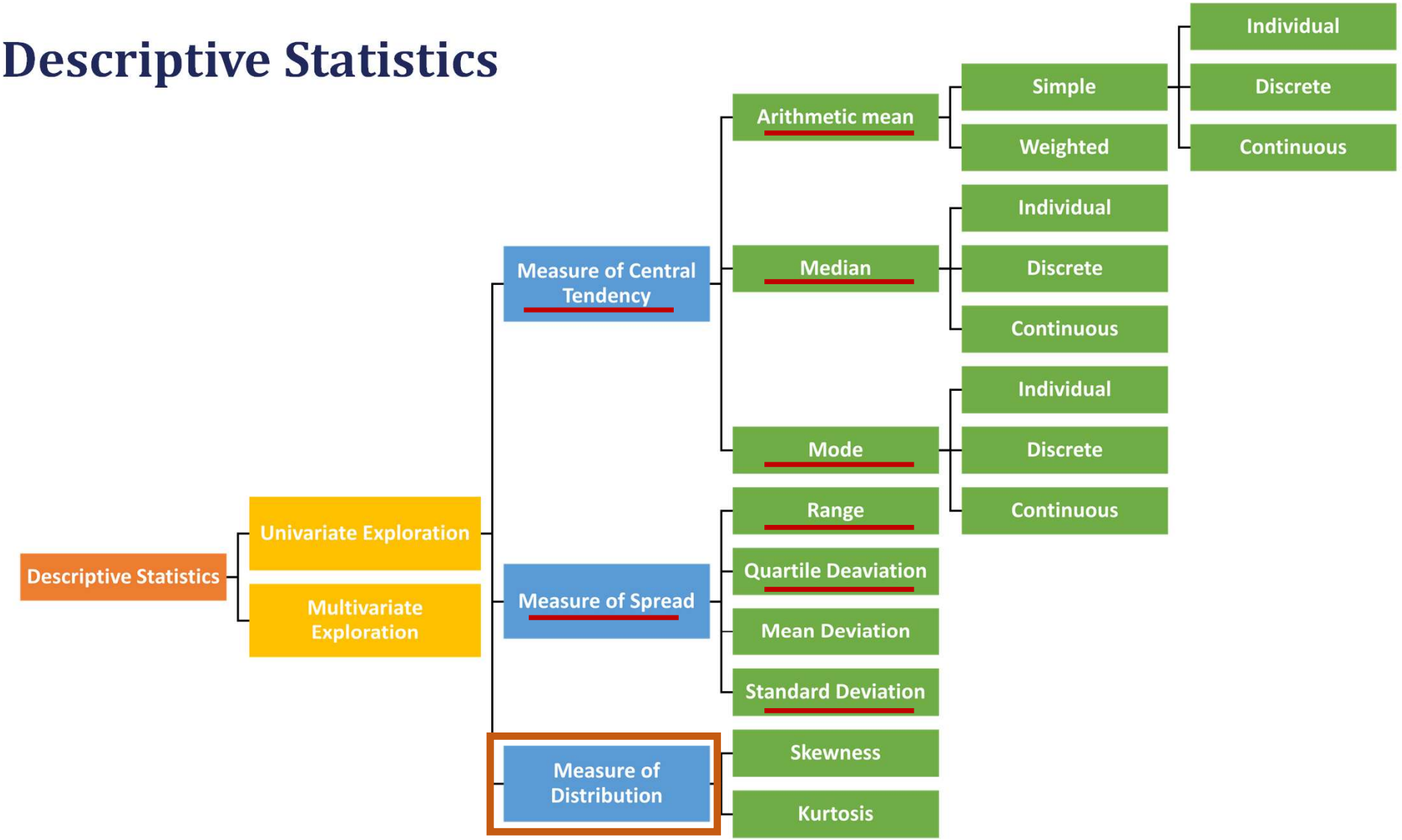
Locating Q_1, Q_3, D_3, P_{20} on the cumulative-frequency curve (ogive)

Class	f	cf
0–5	7	7
5–10	18	25
10–15	25	50
15–20	30	80
20–25	20	100
N = 100		

- Q_1 (25th item)
- Q_3 (75th item)
- D_3 (30th item)
- P_{20} (20th item)



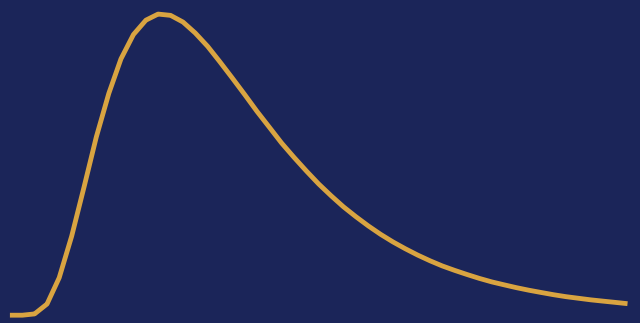
Descriptive Statistics



SKEWNESS

Measuring Asymmetry in a Frequency Distribution

Statistical Methods · Skewness, Moments & Kurtosis



What Is Skewness?

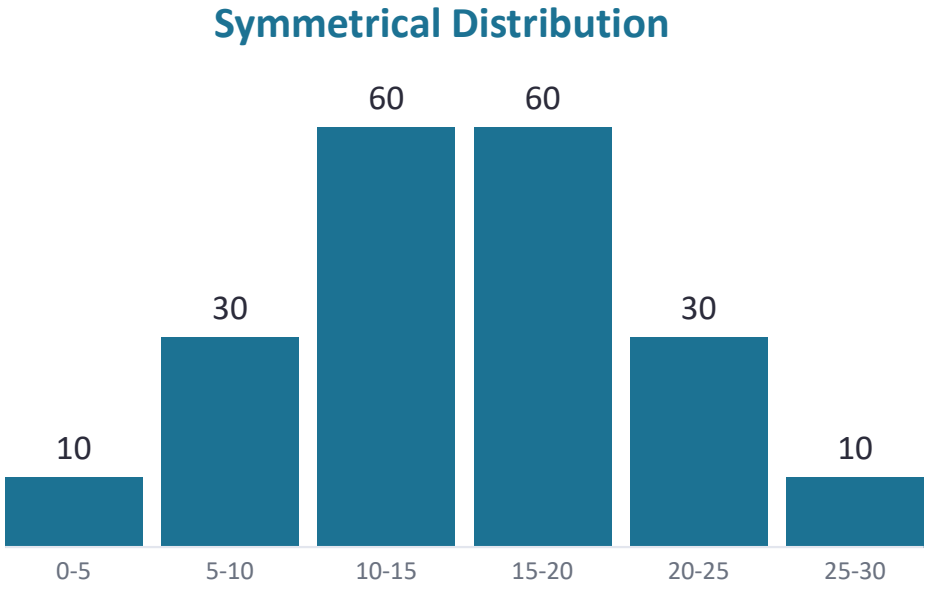
- Two distributions can share the same mean and standard deviation, yet look completely different in shape.
- Skewness measures that difference: the lack of symmetry in a frequency distribution.
- In a symmetrical distribution, Mean = Median = Mode.
- The further the mean moves away from the mode, the greater the skewness.

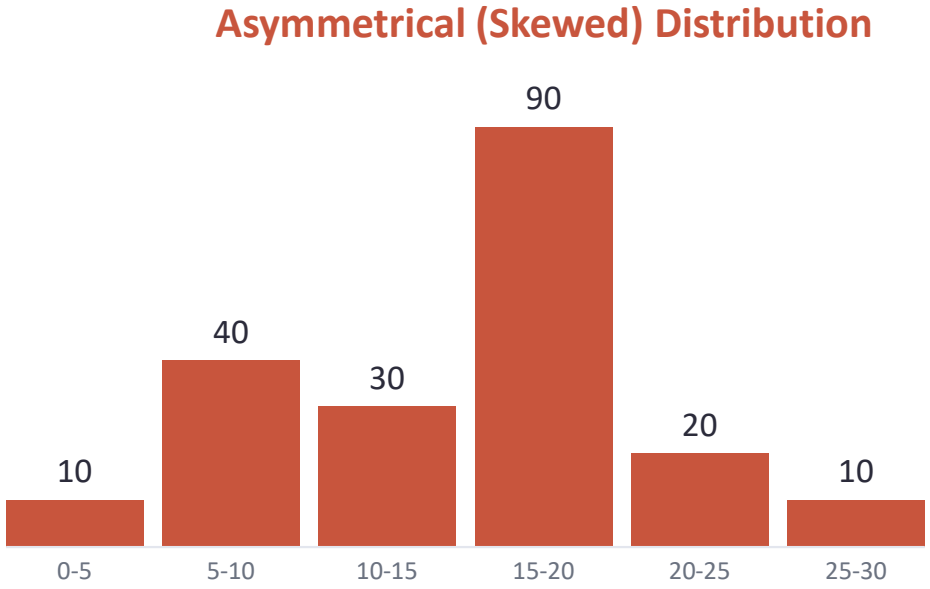
> *A distribution is skewed when the mean and median fall at different points, and the balance is shifted to one side.*
>
> — Garret

Skewness tells us the DIRECTION and EXTENT of asymmetry in a distribution.

Same Mean & Standard Deviation — Different Shape

Both distributions below share $\bar{X} = 15$ and $\sigma = 6$

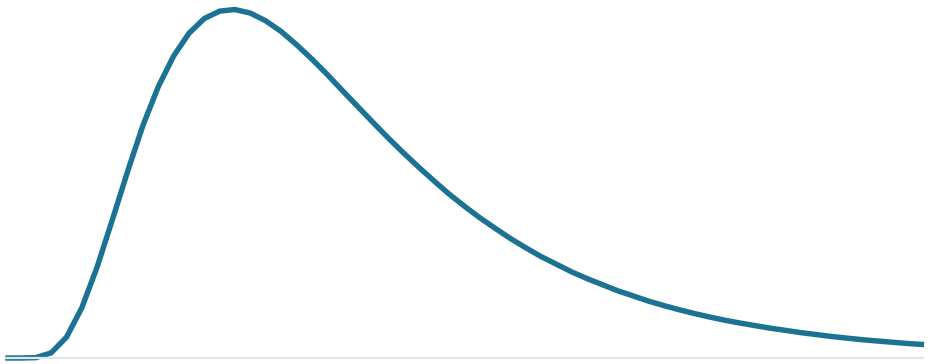




Mean and standard deviation alone cannot distinguish these two shapes — skewness can.

Positively vs Negatively Skewed

Positively Skewed (tail to the right)



Negatively Skewed (tail to the left)



Distribution	Order of Mean, Median, Mode	Relative Value
Positively Skewed	Mode < Median < Mean	Mean is highest, Mode is lowest
Negatively Skewed	Mean < Median < Mode	Mode is highest, Mean is lowest

In a symmetrical distribution all three coincide: $\bar{X}$ = Median = Mode.

Tests of Skewness

A distribution is skewed if any of the following hold true:

1. Mean, median and mode do not coincide
2. Plotted data does not form a normal bell-shaped curve
3. The two halves, when cut vertically through the centre, are not equal
4. Sum of positive deviations from the median ≠ sum of negative deviations
5. Quartiles are not equidistant from the median
6. Frequencies are not equally spread at equal distances from the mode

Dispersion vs. Skewness

Two related but distinct ideas

Dispersion	Skewness
Concerned with the amount of variation	Concerned with the direction of variation (departure from symmetry)
Tells us how spread out the values are	Tells us which side the distribution leans toward
A fundamental characteristic — always computed	Depends on the amount of dispersion; rarely calculated in business/economic series

Measures of Skewness

Measures of skewness show the direction and extent of asymmetry, and let us compare distributions.

Absolute Measure

Skewness measured as the difference between mean and mode:

$$\text{Absolute } S_k = \overline{X} - \text{Mode}$$

- Mean > Mode → positive skewness
- Mode > Mean → negative skewness

Limitation: expressed in the unit of the data, so it cannot compare series in different units or scales.

Relative Measures

Absolute differences expressed relative to spread — usable for direct comparison: (more distributions)

1. Karl Pearson's coefficient of skewness
2. Bowley's coefficient of skewness
3. Kelly's coefficient of skewness
4. Measure based on the third moment

These are mainly used to compare two or more distributions — as a description of a single distribution, they are necessarily vague.

Karl Pearson's Coefficient of Skewness

Suggested by Karl Pearson, a British biometrician and statistician, this measure is based on the difference between mean and mode. Dividing that difference by the standard deviation converts it into a relative measure.

FORMULA (i)

$$Sk_P = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} = \frac{\bar{X} - \text{Mode}}{\sigma}$$

Sk_p = Karl Pearson's coefficient of skewness

Symmetrical

Sk = 0

Mean, median and mode coincide exactly.

Positively Skewed

Sk > 0

Coefficient carries a plus sign — long tail to the right.

Negatively Skewed

Sk < 0

Coefficient carries a minus sign — long tail to the left.

WHAT THE NUMBER TELLS YOU

Direction — the sign (+ / −) shows whether the distribution leans right or left.

Extent — the numerical value (e.g. 0.8 or 0.2) shows how pronounced the skew is.

Karl Pearson's Coefficient of Skewness: When Mode Is Ill-Defined

The mean–mode formula breaks down when the mode cannot be clearly identified. In a moderately skewed distribution, the empirical relationship between the averages provides a workaround.

EMPIRICAL RELATIONSHIP

$$\text{Mode} = 3\ \text{Median} - 2\ \text{Mean}$$

Substitute this expression for Mode into formula (i).

RESULT — FORMULA (ii)

$$\text{Sk}_p = \frac{3(\bar{X} - \text{Median})}{\sigma}$$

DERIVATION

$$[\bar{X} - \text{Mode}] / \sigma = [\bar{X} - (3\ \text{Med.} - 2\bar{X})] / \sigma = (\bar{X} - 3\ \text{Med.} + 2\ \bar{X}) / \sigma = 3(\bar{X} - \text{Med.}) / \sigma$$

Theoretically this coefficient can vary between ±3; in practice, though, it rarely exceeds +1.

Pearson's Coefficient of Skewness: Example1

Calculate Pearson's Coefficient of Skewness. *Assumed mean* **A = 27.5**

X	12.5	17.5	22.5	27.5	32.5	37.5	42.5	47.5
f	28	42	54	108	129	61	45	33

- **i = 5 & A = 27.5**
- Mean ($\bar{X}$) via step-deviation $\bar{X} = A + \Sigma fd / N$

  where $d = (X-A)/i$
- Modal class value: Maximum frequency class
- Standard deviation (σ) $= \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i$
- Karl Pearson's Coeff. of Sk= $[\bar{X} - \text{Mode}] / \sigma$

X	f	d = (X-A)/i	fd	fd^2
12.5	28	-3	-84	252
17.5	42	-2	-84	168
22.5	54	-1	-54	54
27.5	108	0	0	0
32.5	**129**	**1**	**129**	**129**
37.5	61	2	122	244
42.5	45	3	135	405
47.5	33	4	132	528
N = 500			**Σfd = 296**	**Σfd^2 = 1780**

Pearson's Coefficient of Skewness: Example1

A = 27.5

X	f	d = (X-A)/i	fd	fd^2
12.5	28	-3	-84	252
17.5	42	-2	-84	168
22.5	54	-1	-54	54
27.5	108	0	0	0
32.5	**129**	**1**	**129**	**129**
37.5	61	2	122	244
42.5	45	3	135	405
47.5	33	4	132	528
N = 500			**Σfd = 296**	**$Σfd^2$ = 1780**

MEAN ($\bar{X}$)

$\bar{X} = A + Σfd / N$

$27.5 + (296/500) \times 5$

30.46

MODE

Class with max frequency (129)

32.5

STD. DEV. (σ)

$\sqrt{(Σfd^2/N - (Σfd/N)^2)} \times i$

$\sqrt{(1780/500 - (296/500)^2)} \times 5$

8.96

COEFFICIENT OF SKEWNESS

$$Sk = \frac{30.46 - 32.5}{8.96}$$

–0.228

Negatively skewed

Pearson's Coefficient of Skewness: Example2

Calculate Pearson's Coefficient of Skewness.
A = 115

Profits	70–80	80–90	90–100	100–110	110–120	120–130	130–140	140–150
Companies	12	18	35	42	50	45	30	8

- **i = 10 & A = 115**
- Mean ($\bar{X}$) via step-deviation $\bar{X} = (A + \Sigma fd / N)*i$
  where d = (m-A)/i
- Modal class value: Mode = $L + \Delta_1 / (\Delta_1 + \Delta_2) \times i$
- Standard deviation (σ) = $\sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times i$
- Karl Pearson's Coeff. of Sk= $[\bar{X} - \text{Mode}] / \sigma$

Profits	m	f	d	fd	fd^2
70–80	75	12	-4	-48	192
80–90	85	18	-3	-54	162
90–100	95	35	-2	-70	140
100–110	105	42	-1	-42	42
110–120	**115**	**50**	**0**	**0**	**0**
120–130	125	45	1	45	45
130–140	135	30	2	60	120
140–150	145	8	3	24	72
N = 240				**Σfd = –85**	**Σfd^2 = 773**

Pearson's Coefficient of Skewness: Example2

Profits	m	f	d	fd	fd^2
70–80	75	12	-4	-48	192
80–90	85	18	-3	-54	162
90–100	95	35	-2	-70	140
100–110	105	42	-1	-42	42
110–120	**115**	**50**	**0**	**0**	**0**
120–130	125	45	1	45	45
130–140	135	30	2	60	120
140–150	145	8	3	24	72
N = 240				**Σfd = −85**	**Σfd² = 773**

MODE — MODAL CLASS 110–120

Mode = $L + [\Delta 1 / (\Delta 1+\Delta 2)] \times i$

L=110,
$\Delta 1 = | f_1 - f_o | = |50-42| = 8$,
$\Delta 2 = | f_1 - f_2 | = |50-45| = 5$,
i=10

$= 110 + (8/13) \times 10 =$

116.15

MEAN & STANDARD DEVIATION

$\bar{X} = (A + \Sigma fd / N)*i$

$\bar{X} = 115 + (-85/240) \times 10$

111.46

$\sigma = \sqrt{(\Sigma fd^2/N - (\Sigma fd/N)^2)} \times I$

$\sigma = \sqrt{(773/240 - (-85/240)^2)} \times 10 =$

17.595

COEFFICIENT OF SKEWNESS

$$Sk = \frac{111.46 - 116.15}{17.595}$$

−0.267

Negatively skewed

Bowley's Coefficient of Skewness

An alternative measure proposed by the late Professor Bowley, based entirely on quartiles rather than the mean, mode and standard deviation. This makes it useful when the mode is unreliable or distributions have open-ended classes.

THE SYMMETRICAL CASE

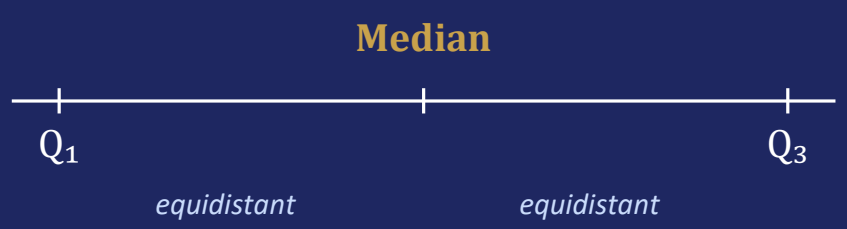
In a symmetrical distribution, the first and third quartiles are equidistant from the median:

$$Q_3 - \text{Med.} = \text{Med.} - Q_1$$

or equivalently:

$$Q_3 + Q_1 - 2\,\text{Med.} = 0$$

SYMMETRICAL DISTRIBUTION



WHY QUARTILES?

Quartiles depend only on the middle 50% of ordered data, so this measure sidesteps the need for a well-defined mode entirely.

Bowley's Coefficient of Skewness

For an asymmetrical distribution, the gap above and below the median is compared directly.

FORMULA

$$Sk_B = \frac{(Q_3 - \text{Med.}) - (\text{Med.} - Q_1)}{(Q_3 - \text{Med.}) + (\text{Med.} - Q_1)}$$

or, equivalently

$$\frac{Q_3 + Q_1 - 2\,\text{Med.}}{Q_3 - Q_1}$$

Sk_B = Bowley's coefficient of skewness

Positive Skew

Top 25% of values lie farther from the median than the bottom 25%; Q_3 is farther from Med. than Q_1.

Negative Skew

Bottom 25% lie farther from the median than the top 25% — the reverse pattern.

Because of its computational basis, Bowley's measure is bounded between −1 and +1.

Bowley's Coefficient of Skewness: Example

Worked example: weight distribution of 589 students

Weight (lbs)	f
Below 100	1
100 – 109	14
110 – 119	66
120 – 129	122
130 – 139	145
140 – 149	121

Weight (lbs)	f
150 – 159	65
160 – 169	34
170 – 179	12
180 – 189	5
190 – 199	2
200 & over	2

Bowley's Coefficient of Skewness $= (Q_3 + Q_1 - 2 \times \text{Median}) \div (Q_3 - Q_1)$

Bowley's Coefficient of Skewness: Example

Step1: Convert to exclusive class and find N

Weight (lbs)	f	cf
Below 99.5	1	1
99.5 – 109.5	14	15
109.5 – 119.5	66	81
119.5 – 129.5	122	203
129.5 – 139.5	145	348
139.5 – 149.5	121	469

Weight (lbs)	f	cf
149.5 – 159.5	65	534
159.5 – 169.5	34	568
169.5 – 179.5	12	580
179.5 – 189.5	5	585
189.5 – 199.5	2	587
199.5 & over	2	589

N = Σf = 589 students

Bowley's Coefficient of Skewness: Example

Step 2 — Find the Median

Median position = N/2 = 589/2 = **294.5th item**

The 294.5th student falls inside the class 130–139 (real limits 129.5–139.5).

129.5	294.5	203	145	10
L (lower limit)	N/2	cf (before class)	f (class freq.)	i (class width)

Median = L + [(N/2 − cf) / f] × i = 129.5 + [(294.5 − 203) / 145] × 10

= 129.5 + 6.31

Median = 135.81 lbs

Weight (lbs)	f	cf
Below 99.5	1	1
99.5 – 109.5	14	15
109.5 – 119.5	66	81
119.5 – 129.5	122	203
129.5 – 139.5	145	348
139.5 – 149.5	121	469
149.5 – 159.5	65	534
159.5 – 169.5	34	568
169.5 – 179.5	12	580
179.5 – 189.5	5	585
189.5 – 199.5	2	587
199.5 & over	2	589

Bowley's Coefficient of Skewness: Example

Step 3 — Find Q_1

Q_1 — Lower Quartile

Position = N/4 = 589/4 = **147.25th item**

Falls in class 120–129 (real limits 119.5–129.5)

119.5	147.25	81	122	10
L	N/4	cf	f	i

$Q_1 = L + [(N/4 - cf)/f] \times i$

$= 119.5 + [(147.25-81)/122] \times 10 = 119.5 + 5.43$

$Q_1 = 124.93$

Weight (lbs)	f	cf
Below 99.5	1	1
99.5 – 109.5	14	15
109.5 – 119.5	66	81
119.5 – 129.5	122	203
129.5 – 139.5	145	348
139.5 – 149.5	121	469
149.5 – 159.5	65	534
159.5 – 169.5	34	568
169.5 – 179.5	12	580
179.5 – 189.5	5	585
189.5 – 199.5	2	587
199.5 & over	2	589

Bowley's Coefficient of Skewness: Example

Step 4 — Find Q_3

Q_3 — Upper Quartile

Position = 3N/4 = 3×589/4 = **441.75th item**

Falls in class 140–149 (real limits 139.5–149.5)

139.5	441.75	348	121	10
L	3N/4	cf	f	i

$Q_3 = L + [(3N/4 - cf)/f] \times i$

$= 139.5 + [(441.75-348)/121]\times 10 = 139.5 + 7.75$

$Q_3 = 147.25$

Weight (lbs)	f	cf
Below 99.5	1	1
99.5 – 109.5	14	15
109.5 – 119.5	66	81
119.5 – 129.5	122	203
129.5 – 139.5	145	348
139.5 – 149.5	121	469
149.5 – 159.5	65	534
159.5 – 169.5	34	568
169.5 – 179.5	12	580
179.5 – 189.5	5	585
189.5 – 199.5	2	587
199.5 & over	2	589

Bowley's Coefficient of Skewness: Example

Put Q_1, Q_3, and the Median into Bowley's formula

135.81	124.93	147.25
Median	Q_1	Q_3

Coeff. of Sk = $(Q_3 + Q_1 - 2 \times \text{Median}) / (Q_3 - Q_1)$

= (147.25 + 124.93 − 271.62) / (147.25 − 124.93)

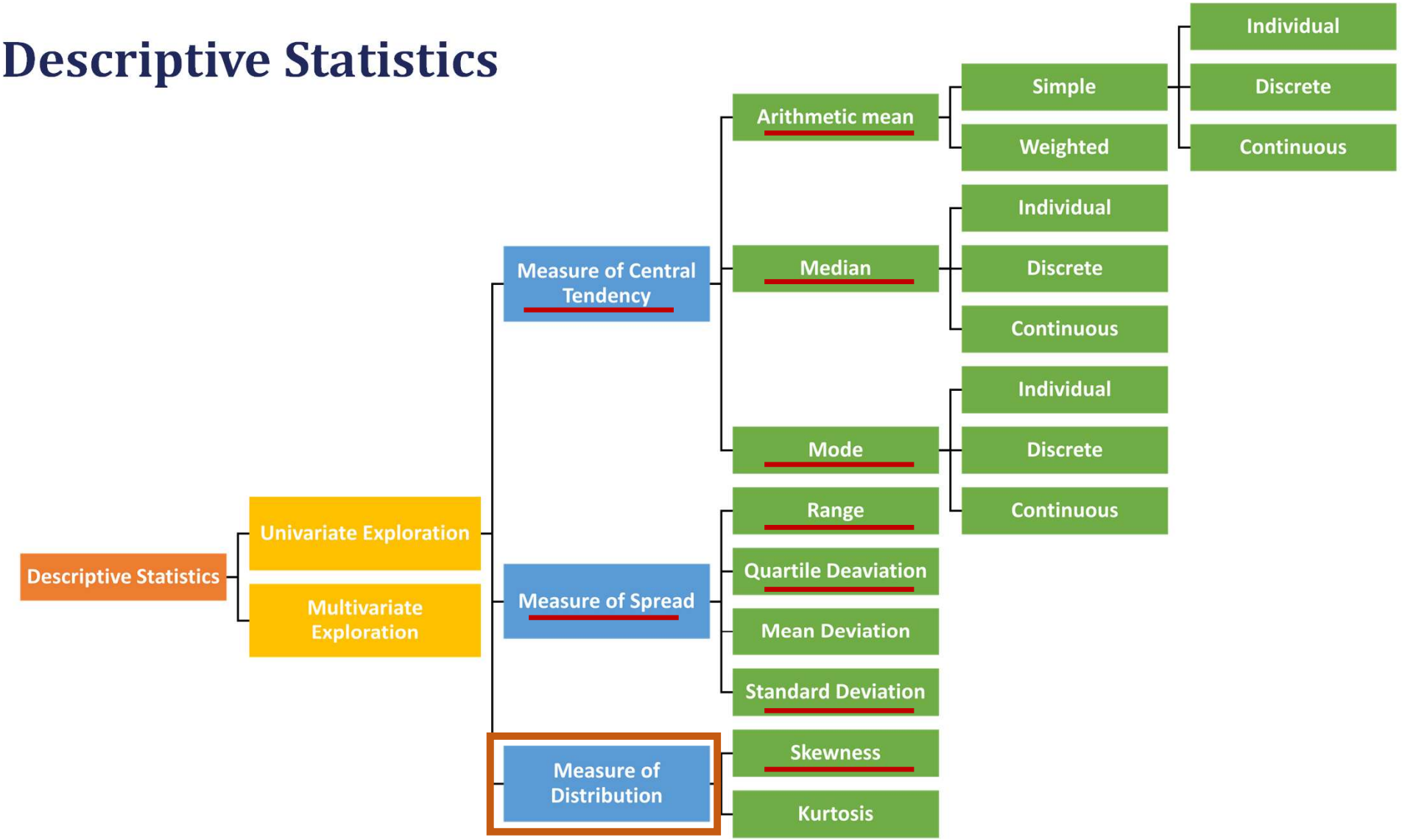
= 0.56 / 22.32 = **0.025**

What 0.025 means: The value is very close to zero and slightly positive. This tells us the weight distribution of students is **almost symmetrical**, with only a very mild pull toward higher weights (a few heavier students in the tail).

Pearson vs Bowley

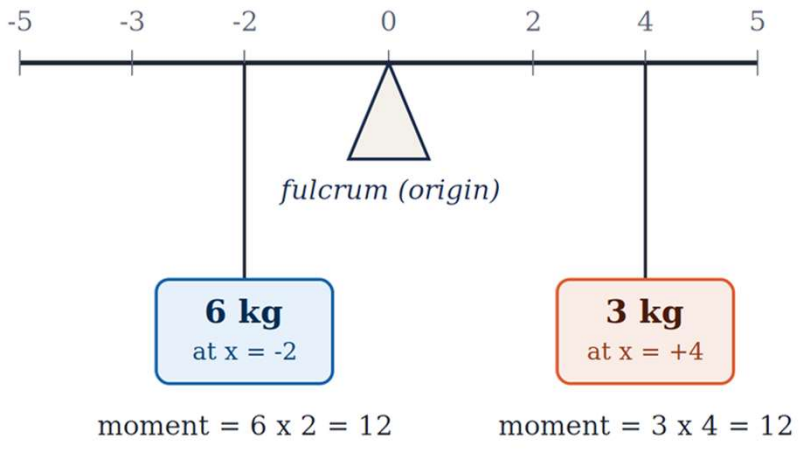
	Karl Pearson's Sk_p	**Bowley's Sk_β**
Based on	Mean, mode, standard deviation	Quartiles (Q_1, Median, Q_3)
Theoretical range	No limit (± unbounded)	Bounded: −1 to +1
Typical practical range	Rarely exceeds ±1	Rarely exceeds ±1
Needs a defined mode?	Yes (or the median-based variant)	No — mode not required
Best suited when	Mode is clearly identifiable	Mode is ill-defined or data is open-ended

Descriptive Statistics



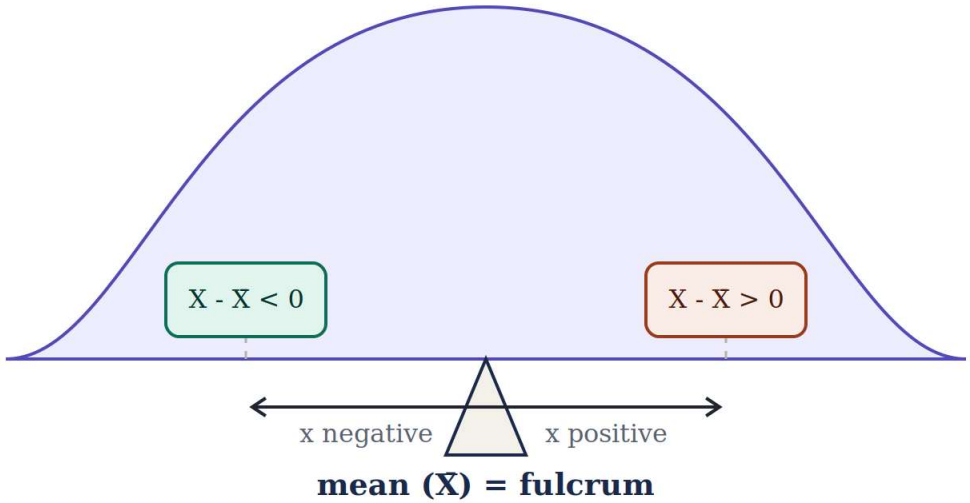
Moments — The Balance Beam Idea

Moment = **Force** × **Distance**. A beam balances when moments on both sides of the fulcrum are equal



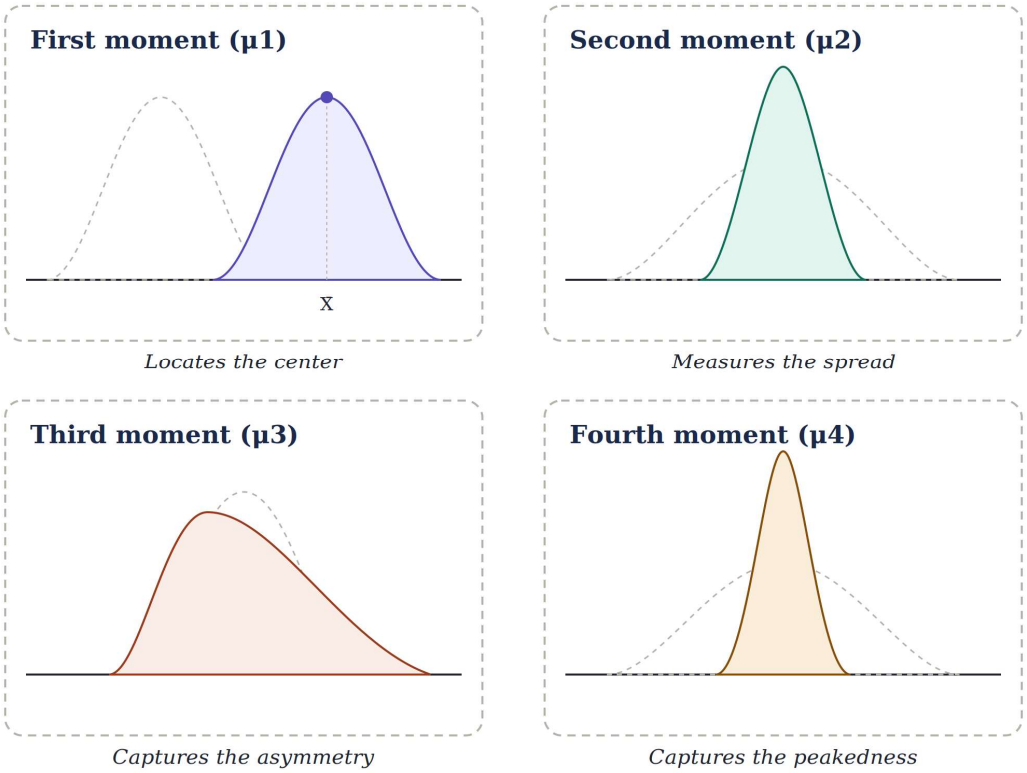
Equal and opposite moments -> beam balances

Statistics Borrows the Same Idea



Deviations cancel at r = 1, but not at r = 2, 3, 4

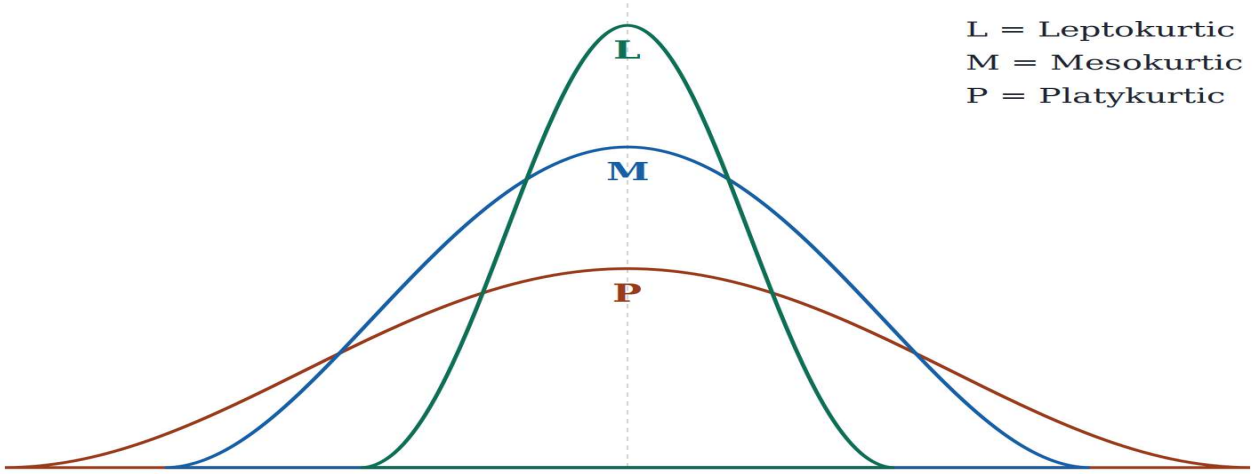
The Four Moments at a Glance



Moment	What it measures
First moment about origin (μ1)	Mean
Second moment about mean (μ2)	Variance
Third moment about mean (μ3)	Skewness
Fourth moment about mean (μ4)	Kurtosis

Kurtosis

Kurtosis (Greek: "bulginess") measures the peakedness or flatness of a distribution near the mode, relative to the normal curve.

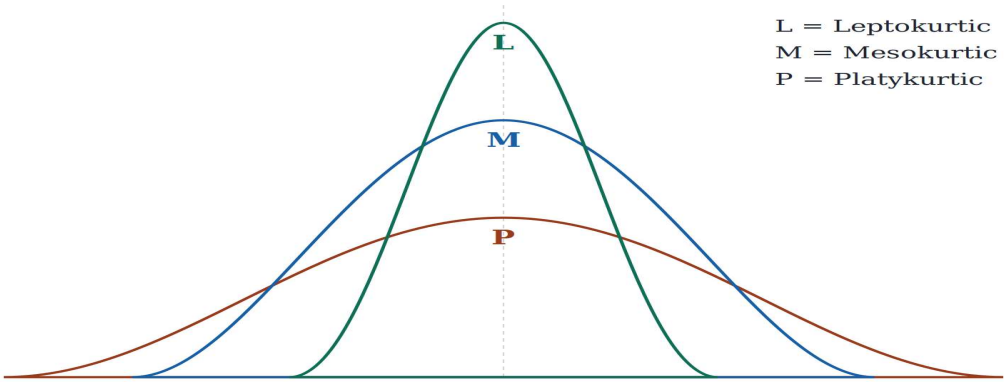


Curve	Shape	Meaning
L — Leptokurtic	Narrow, tall peak	More peaked than normal
M — Mesokurtic	Normal curve	Same peakedness as normal
P — Platykurtic	Broad, flat top	Less peaked than normal

Measuring Kurtosis — β2

```
β2 = μ4 / μ2²          (μ4 = 4th moment, μ2 = 2nd moment)
```

Value of β2	Interpretation	Shape	Meaning
β2 = 3	Mesokurtic	Narrow, tall peak	More peaked than normal
β2 > 3	Leptokurtic	Normal curve	Same peakedness as normal
β2 < 3	Platykurtic	Broad, flat top	Less peaked than normal



Kurtosis for individual series

Dataset: x = [2, 4, 6, 8, 10] **Mean:** $\bar{x} = \Sigma x / n = (2+4+6+8+10) / 5 = 30 / 5 = 6$

x	$x - \bar{x}$	$(x - \bar{x})^2$	$(x - \bar{x})^4$
2	2 − 6 = −4	16	256
4	4 − 6 = −2	4	16
6	6 − 6 = 0	0	0
8	8 − 6 = 2	4	16
10	10 − 6 = 4	16	256
Sum	**—**	**Σ = 40**	**Σ = 544**

Central Moments

$\mu_2 = \Sigma(x - \bar{x})^2 / n = 40 / 5 = 8$

$\mu_4 = \Sigma(x - \bar{x})^4 / n = 544 / 5 = 108.8$

Kurtosis

$\beta_2 = \mu_4 / \mu_2^2 = 108.8 / 8^2 = 108.8 / 64 = 1.7$

Interpretation: $\beta_2 = 1.7$, which is less than 3, so the distribution is ***Platykurtic***. It has lighter tails and is less outlier-prone than a normal distribution (which has $\beta_2 = 3$).

Kurtosis for a Distributed Series

x	f	f·x	$x-\bar{x}$	$(x-\bar{x})^2$	$f(x-\bar{x})^2$	$(x-\bar{x})^4$	$f(x-\bar{x})^4$
2	1	2	-2	4	4	16	16
3	4	12	-1	1	4	1	4
4	6	24	0	0	0	0	0
5	4	20	1	1	4	1	4
6	1	6	2	4	4	16	16
Total	**N = 16**	**Σfx = 64**	**—**	**—**	**Σ = 16**	**—**	**Σ = 40**

Step-by-Step Calculation

1. Mean: $\bar{x} = \Sigma fx / N = 64 / 16 = 4$
2. 2nd Central Moment Variance: $\mu_2 (\sigma^2) = \Sigma f(x-\bar{x})^2 / N = 16 / 16 = 1$
3. 4th Central Moment: $\mu_4 = \Sigma f(x-\bar{x})^4 / N = 40 / 16 = 2.5$
4. Kurtosis: $\beta_2 = \mu_4 / \mu_2{}^2 = 2.5 / (1)^2 = 2.5$

Interpretation

$\beta_2 = 2.5 < 3$

Excess kurtosis is negative.

The distribution is ***Platykurtic*** — it has lighter tails and is less outlier-prone than a normal distribution.

Solved Example: Kurtosis for a Continuous Frequency Distribution

Class Interval	f	Midpoint (x)	f·x	x−x̄	(x−x̄)²	f(x−x̄)²	(x−x̄)⁴	f(x−x̄)⁴
0–10	1	5	5	-20	400	400	160,000	160,000
10–20	3	15	45	-10	100	300	10,000	30,000
20–30	8	25	200	0	0	0	0	0
30–40	3	35	105	10	100	300	10,000	30,000
40–50	1	45	45	20	400	400	160,000	160,000
Total	**N = 16**	**—**	**Σfx = 400**	**—**	**—**	**Σ = 1,400**	**—**	**Σ = 380,000**

Step-by-Step Calculation

1. Mean: $\bar{x} = \Sigma fx / N = 400 / 16 = 25$
2. 2nd Central Moment: $\mu_2 = \Sigma f(x-\bar{x})^2 / N = 1400 / 16 = 87.5$
3. 4th Central Moment: $\mu_4 = \Sigma f(x-\bar{x})^4 / N = 380,000 / 16 = 23,750$
4. Kurtosis: $\beta_2 = \mu_4 / \mu_2^2 = 23,750 / (87.5)^2 = 23,750 / 7,656.25 \approx 3.102$

Interpretation $\beta_2 \approx 3.102 > 3$

Excess kurtosis $\gamma_2 = \beta_2 - 3 \approx 0.102$ (positive)

The distribution is ***Leptokurtic*** — it has a slightly sharper peak and heavier tails than a normal distribution.

Worked Example — Illustration 13

First four central moments: $\mu_1 = 0$, $\mu_2 = 2.5$, $\mu_3 = 0.7$, $\mu_4 = 18.75$

SKEWNESS

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

$$= \frac{(0.7)^2}{(2.5)^3}$$

$$= \frac{0.49}{15.625}$$

$$= 0.031$$

KURTOSIS

$$\beta_2 = \frac{\mu_4}{\mu_2^2}$$

$$= \frac{18.75}{(2.5)^2}$$

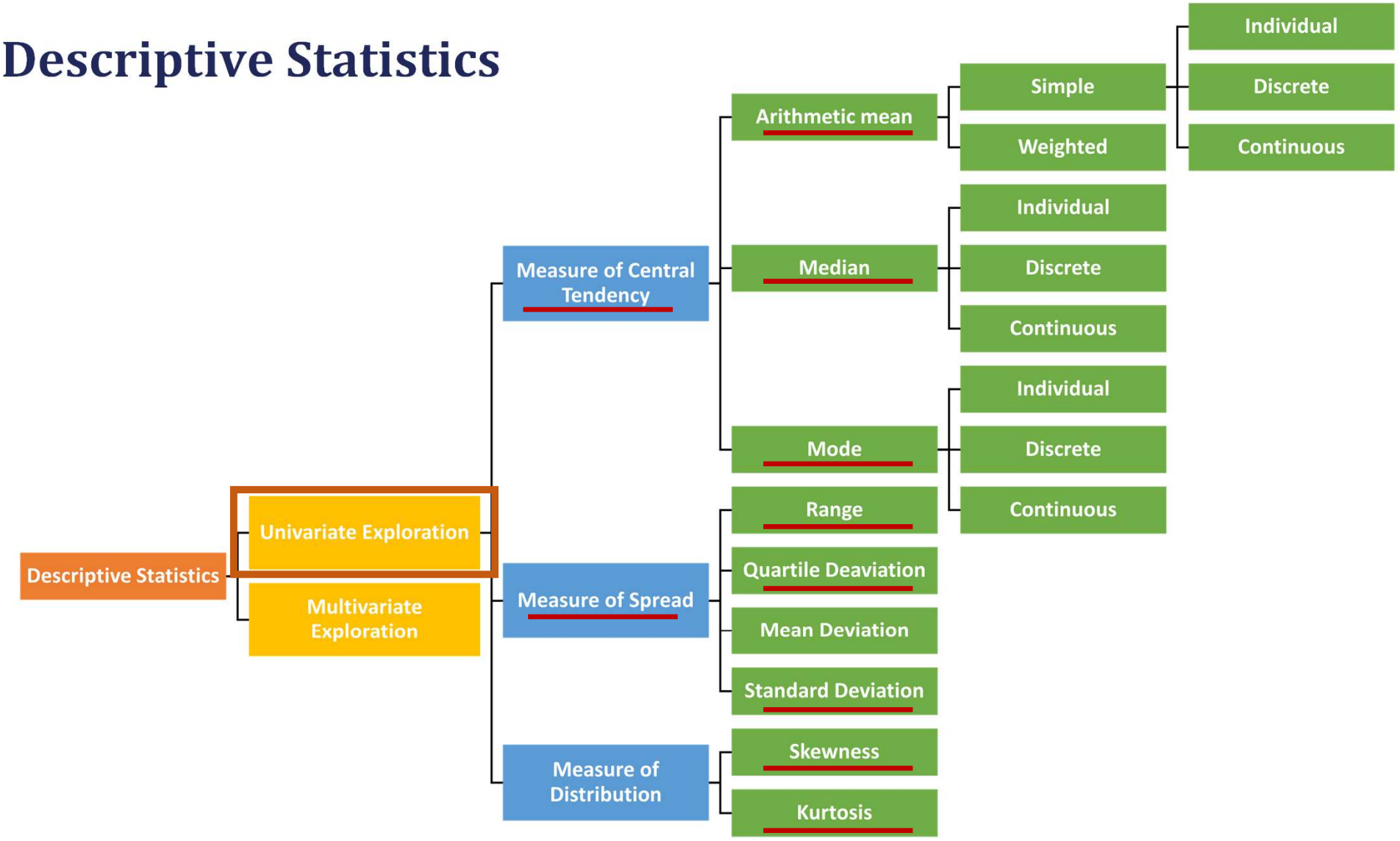
$$= \frac{18.75}{6.25}$$

$$= 3$$

Conclusion: $\beta_1 = 0.031$ is close to 0, so the distribution is only slightly skewed — nearly symmetrical.

$\beta_2 = 3$ exactly, so the distribution is mesokurtic — its peak has the same shape as the normal curve.

Descriptive Statistics



Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

Multivariate Exploration /Correlation Analysis

Measuring the Degree of Relationship Between Variables

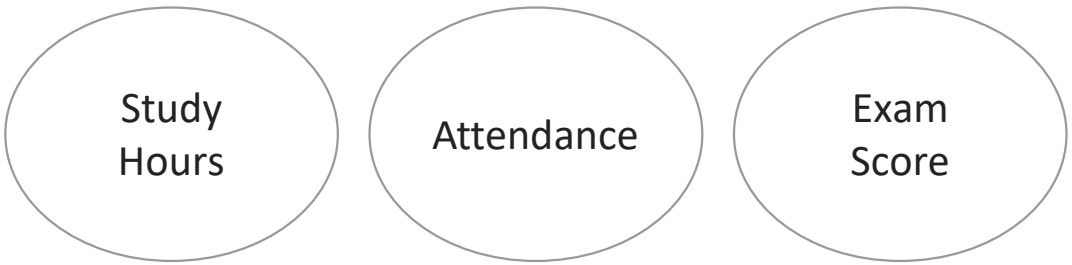
Multivariate Exploration

Definition

Analyzes more than two variables simultaneously to understand relationships, patterns, and interactions among them.

Example

In a student performance dataset with Study Hours, Attendance, and Exam Score, multivariate exploration helps analyze how Study Hours and Attendance together affect Exam Score — something that cannot be fully understood by looking at each variable separately.



Multivariate exploration studies how all three interact together — not just pairwise.

Central Data Point

Definition

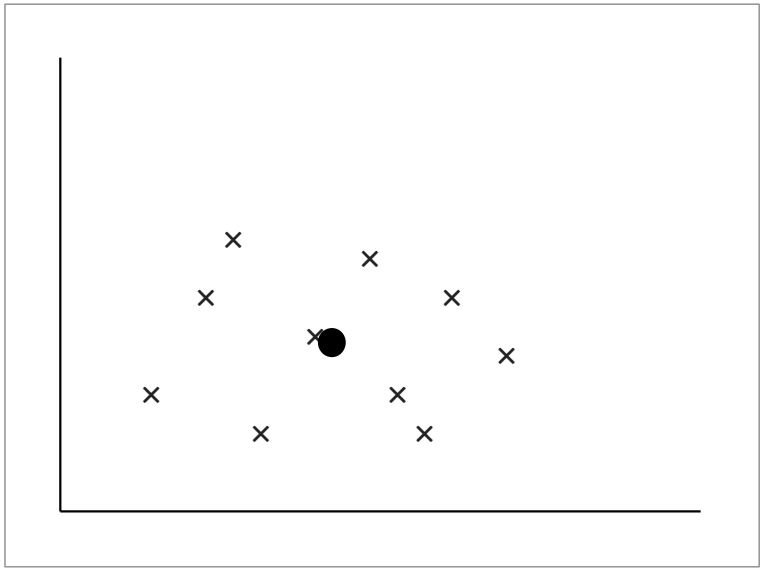
A single point that represents the overall center or typical location of a dataset, considering all variables together — the "average point" in multi-dimensional space.

Example

If a dataset has variables (Height, Weight, Age), the central data point is:**(Mean Height, Mean Weight, Mean Age)**

It helps in:

- Understanding the overall tendency of multivariate data
- Comparing how far other observations deviate from the center
- Detecting outliers and clusters



Central point(mean of all variables)

What Is Correlation?

A statistical descriptive measure of the strength of a linear relationship between two variables.

- Age of husband & age of wife
- Price of a commodity & amount demanded
- Rainfall & production of rice
- TV licences & number of cinema-goers

KEY DEFINITION

"Correlation is an analysis of the covariation between two or more variables."

— A. M. Tuttle

"Correlation is a measure of degree of relatedness of variables."

— Ken Black

Correlation and Causation

Correlation analysis only measures the degree of relationship between variables — it says nothing about cause and effect. **Causation always implies correlation, but correlation does NOT imply causation.**

A significant correlation may arise from any one, or a combination, of the following reasons:

1 Pure Chance

- ✓ Especially in small samples
- ✓ A high correlation may appear in the sample with no real relationship in the universe

Example: Income vs Weight below shows a perfect positive relationship — purely coincidental.

Income (₹)	20K	25K	30K	35K	40K
Weight (lbs)	120	140	160	180	200

2 Common Third Variable

Both correlated variables may be influenced by one or more other variables, or different causes with the same effect — neither is the cause of the other.

Example: Rice & tea yields both depend on rainfall.

3 Mutual Influence

Both variables may influence each other continuously or intermittently, so neither can be clearly designated as cause or effect — common in economic variables.

Example: Price → Demand (price rise cuts demand) AND Demand → Price

Spurious Correlation: Production of pigs vs. pig iron shows extremely high correlation with no causal link. It is the ***interpretation*** that is spurious — not the mathematical degree of correlation itself.

Types of Correlation

Correlation is described or classified in several different ways. Three of the most important ways of classifying correlation are as follows:

1

Positive or Negative Correlation

Based on the direction in which the variables move — together (positive) or in opposite directions (negative).

2

Simple, Partial and Multiple Correlation

Based on the number of variables studied — two (simple), or three or more (partial/multiple).

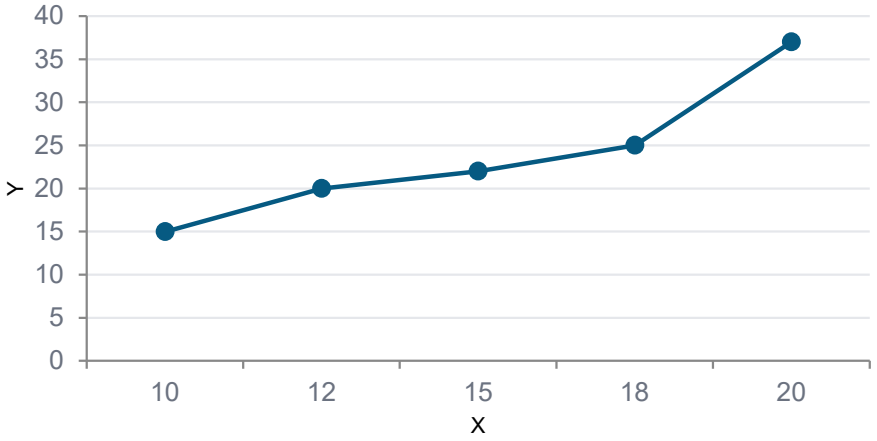
3

Linear and Non-Linear (Curvilinear) Correlation

Based on the constancy of the ratio of change between the variables — constant (linear) or not (curvilinear).

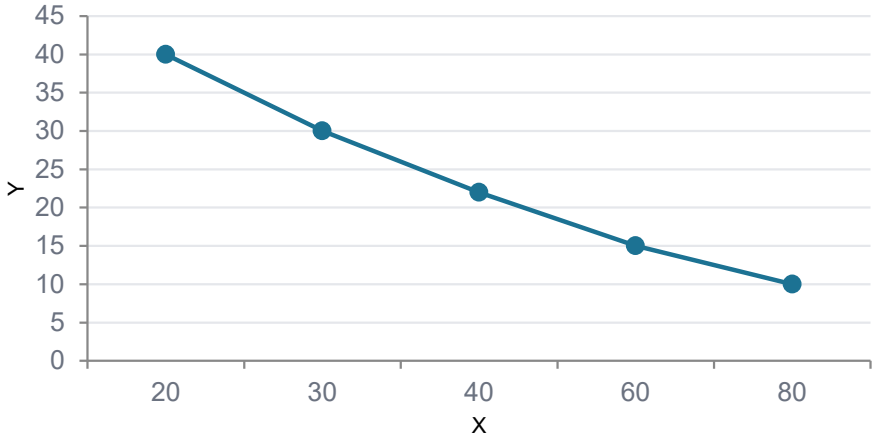
Types of Correlation: Positive vs Negative

Positive Correlation



As X increases, Y also increases (same direction).

Negative Correlation



As X increases, Y decreases (opposite directions).

Rule: Both variables move in the same direction → Positive. Both move in opposite directions → Negative.

Types of Correlation: By Variables & Shape

By Number of Variables

S **Simple**

Only two variables studied (e.g., rainfall and yield of rice).

M **Multiple**

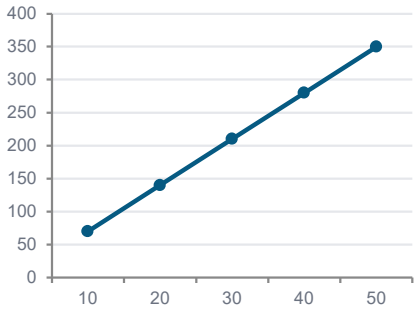
Three or more variables studied simultaneously (e.g., yield of rice with rainfall AND fertilizer).

P **Partial**

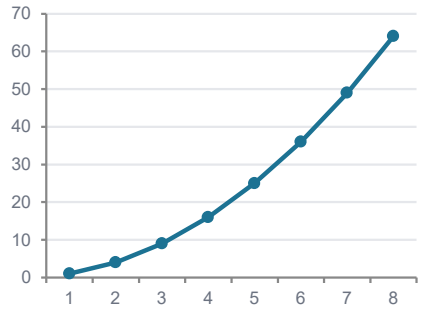
More than two variables recognised, but only two considered, with the effect of others held constant.

By Shape of Relationship

Linear



Curvilinear



Linear: constant ratio of change between variables — points fall on a straight line.

Curvilinear: the ratio of change is NOT constant — points fall on a curve.

Methods of Studying Correlation

1. Scatter Diagram Method
2. Graphic Method
3. Karl Pearson's Coefficient of Correlation
4. Concurrent Deviation Method
5. Method of Least Squares

The first two are diagram/graph-based; the rest are mathematical methods.

Methods of Studying Correlation

1. ~~Scatter Diagram Method~~
2. ~~Graphic Method~~
3. Karl Pearson's Coefficient of Correlation
4. ~~Concurrent Deviation Method~~
5. ~~Method of Least Squares~~

The first two are diagram/graph-based; the rest are mathematical methods.

What is Pearson's Coefficient of Correlation?

- ✓ Karl Pearson's method is the most widely used in practice.
- ✓ It is denoted by the symbol r — used universally for describing the degree of correlation between two variables.
- ✓ Also known as the Product Moment Correlation Coefficient, named after Karl Pearson (1857–1936), an English statistician.

Formula (i) $\mathbf{r = \Sigma xy / (N \sigma x \sigma y)}$

$x = (X - \bar{X})$; $y = (Y - \bar{Y})$

σx, σy = std. deviations of X, Y

N = number of pairs of observations

Key Property: **r always lies between +1 and −1**

+1 → perfect positive correlation

−1 → perfect negative correlation

0 → no relationship

Sign shows direction; magnitude shows strength

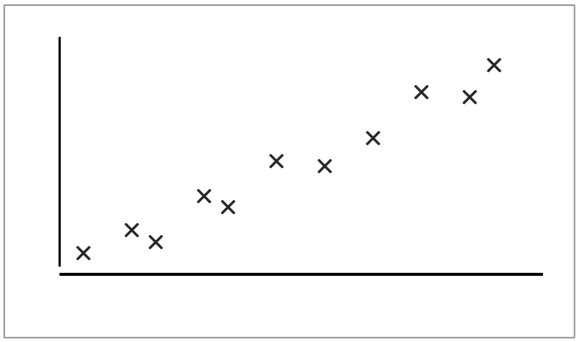
(e.g. −0.26 = low negative correlation)

Pearson's Coefficient of Correlation?

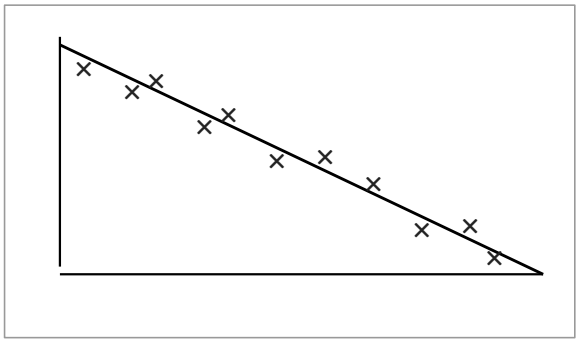
Positive Correlation: Both variables move in the same direction (e.g., as study hours increase, marks increase).

Negative Correlation: Variables move in opposite directions (e.g., as stress increases, performance decreases).

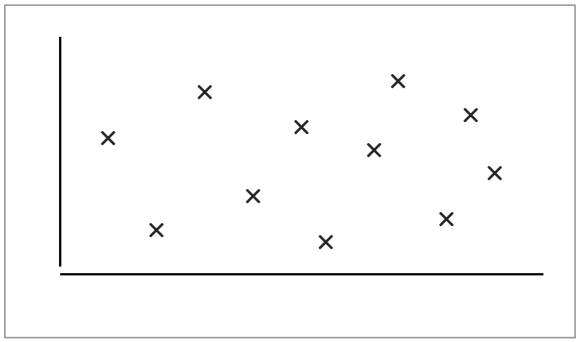
Zero (No) Correlation: No relationship between the variables (e.g., shoe size and exam marks).



Positive Correlation



Negative Correlation

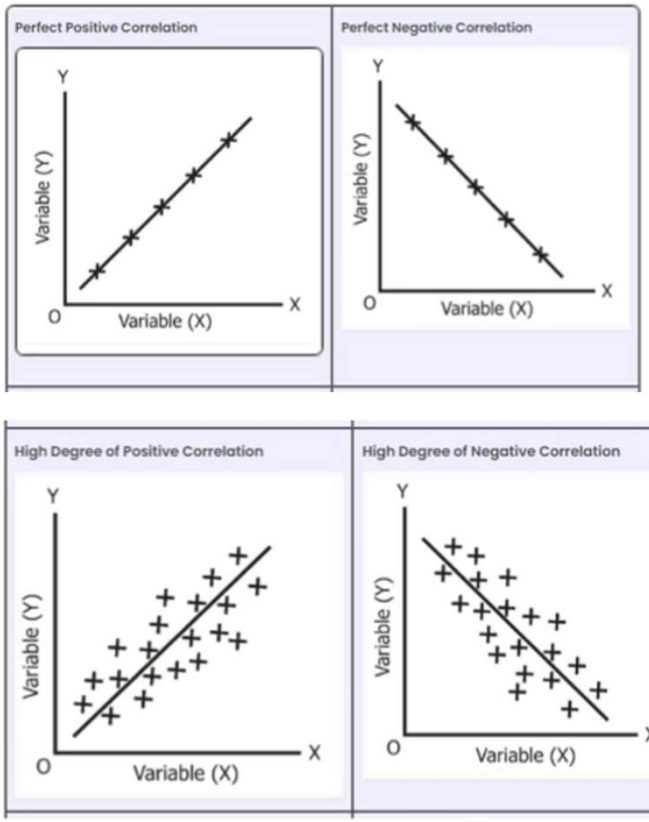


No Correlation

Degrees of Correlation

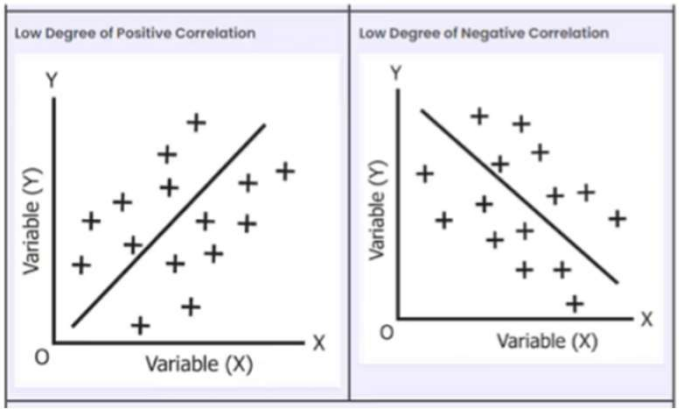
Type	Coefficient Range	Description
Perfect Positive	r = +1	All points fall exactly on an upward line.
Perfect Negative	r = −1	All points fall exactly on a downward line.

Type	Coefficient Range	Description
High Degree Positive	0.75 to +0.99	Points closely follow an upward trend.
High Degree Negative	−0.75 to −0.99	Points closely follow a downward trend.

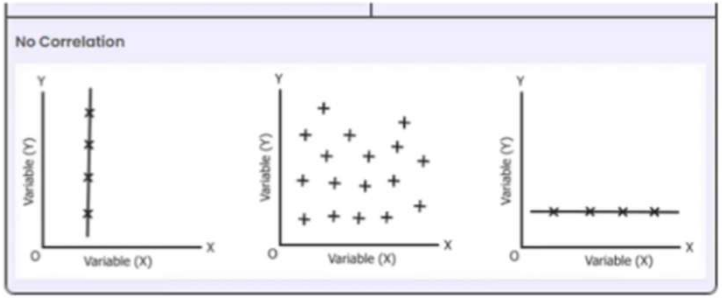


Degrees of Correlation

Type	Coefficient Range	Description
Low Degree Positive	0 to +0.5 (approx.)	Points loosely follow an upward trend.
Low Degree Negative	0 to −0.5 (approx.)	Points loosely follow a downward trend.



Type	Coefficient Range	Description
No Correlation	r ≈ 0	Points show no linear pattern (scattered, vertical, or horizontal).



Pearson's Coefficient of Correlation

1

METHOD 1

Deviations Taken from Actual Mean

2

METHOD 2

Direct Method — Without Taking Deviations

3

METHOD 3

Deviations Taken from an Assumed Mean

Pearson's Coefficient of Correlation: Method 1 (Deviations Taken from Actual Mean)

Simplified formula $r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)}$ **OR** $r = \Sigma xy / (N \sigma x \sigma y)$

where $x = (X - \bar{X})$ and $y = (Y - \bar{Y})$.

This avoids separately calculating σx and σy — greatly simplifying the task.

Steps to Compute r

1. Take deviations of the X series from mean of X → denote as x
2. Square these deviations; obtain the total Σx^2
3. Take deviations of the Y series from mean of Y → denote as y
4. Square these deviations; obtain the total Σy^2
5. Multiply deviations of X and Y series; obtain the total Σxy
6. Substitute Σxy, Σx^2 and Σy^2 into the formula

Pearson's Coefficient of Correlation: Example1

Calculate Karl Pearson's coefficient of correlation from the following data and interpret its value.

Roll Number of Students	1	2	3	4	5
Marks in Accountancy (X)	48	35	17	23	47
Marks in Statistics (Y)	45	20	40	25	45

Let marks in Accountancy be denoted by X and marks in Statistics by Y.

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)}$$

Pearson's Coefficient of Correlation: Example1

Roll No.	X	Y	x=(X-34)	x^2	y=(Y-35)	y^2	xy
1	48	45	+14	196	+10	100	+140
2	35	20	+1	1	-15	225	-15
3	17	40	-17	289	+5	25	-85
4	23	25	-11	121	-10	100	+110
5	47	45	+13	169	+10	100	+130
Total	**ΣX=170**	**ΣY=175**	**Σx=0**	**Σx^2=776**	**Σy=0**	**Σy^2=550**	**Σxy=280**
	$\bar{X}$ =ΣX/N =170/5=34	$\bar{Y}$ = ΣY/N =175/5 =35					

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)}$$
$$= 280 / \sqrt{(776 \times 550)}$$
$$= 280 / 653.299 = 0.429$$

Interpretation: r = +0.429 indicates a moderate degree of positive correlation between marks in Accountancy and Statistics.

Pearson's Coefficient of Correlation: Example2

Making use of the data summarized below, calculate the coefficient of correlation, r_{12}.

Case	A	B	C	D	E	F	G	H
X1	10	6	9	10	12	13	11	9
X2	9	4	6	9	11	13	8	4

$$r12 = \Sigma x1x2 \,/\, \sqrt{(\Sigma x1^2 \times \Sigma x2^2)}$$

Pearson's Coefficient of Correlation: Example2

Case	X1	X2	$x1 = X1 - \overline{X_1}$	$x2 = X2 - \overline{X_2}$	$x1^2$	$x2^2$	x1x2
A	10	9	0	1	0	1	0
B	6	4	-4	-4	16	16	16
C	9	6	-1	-2	1	4	2
D	10	9	0	1	0	1	0
E	12	11	2	3	4	9	6
F	13	13	3	5	9	25	15
G	11	8	1	0	1	0	0
H	9	4	-1	-4	1	16	4
Σ (Total)	**80**	**64**	**0**	**0**	**32**	**72**	**43**
	$\overline{X_1} = \Sigma X_1/N$ $= 80/8 = 10$	$\overline{X_2} = \Sigma X_2/N$ $= 64/8 = 08$					

Solution $r12 = \Sigma x1x2 / \sqrt{(\Sigma x1^2 \times \Sigma x2^2)}$

$= 43 / \sqrt{(32 \times 72)} = 43 / \sqrt{2,304} = 43 / 48$

r12 = +0.896

Interpretation: r12 = +0.896 shows a high degree of positive correlation between X1 and X2.

Pearson's Coefficient of Correlation: Example3

The following table gives indices of industrial production and the number of registered unemployed (in lakhs). Calculate the value of the coefficient of correlation.

Year	2008	2009	2010	2011	2012	2013	2014	2015
Index of Production	100	102	104	107	105	112	103	99
Number Unemployed	15	12	13	11	12	12	19	26

Pearson's Coefficient of Correlation: Example3

Year	X	x=(X-104)	x^2	Y	y=(Y-15)	y^2	xy
2008	100	-4	16	15	0	0	0
2009	102	-2	4	12	-3	9	+6
2010	104	0	0	13	-2	4	0
2011	107	+3	9	11	-4	16	-12
2012	105	+1	1	12	-3	9	-3
2013	112	+8	64	12	-3	9	-24
2014	103	-1	1	19	+4	16	-4
2015	99	-5	25	26	+11	121	-55
Total	**ΣX=832**	**Σx=0**	**Σx^2=120**	**ΣY=120**	**Σy=0**	**Σy^2=184**	**Σxy=-92**
	$\bar{x}$ = 832/8 = 104	$\bar{Y}$ = 120/8 = 15					

$$r = \Sigma xy / \sqrt{(\Sigma x^2 \times \Sigma y^2)} = -92 / \sqrt{(120 \times 184)} = -92 / 148.593 = -0.619$$

Pearson's Coefficient of Correlation: Method 2 (Direct Method — Without Taking Deviations)

$$r = [\, N\Sigma XY - (\Sigma X)(\Sigma Y) \,] \;/\; \sqrt{[\, N\Sigma X^2 - (\Sigma X)^2 \,]} \times \sqrt{[\, N\Sigma Y^2 - (\Sigma Y)^2 \,]}$$

Correlation coefficient can also be calculated without taking deviations of items either from actual mean or assumed mean — i.e. using the actual X and Y values directly. This gives the same answer as the deviation-based formulas.

Pearson's Coefficient of Correlation: Method 2: Example1

Calculate coefficient of correlation from the data given below by the direct method, i.e., without taking deviations of items from actual or assumed mean.

	1	2	3	4	5	6	7	8	9
X	9	8	7	6	5	4	3	2	1
Y	15	16	14	13	11	12	10	8	9

Pearson's Coefficient of Correlation: Method 2: Example1

X	X^2	Y	Y^2	XY
9	81	15	225	135
8	64	16	256	128
7	49	14	196	98
6	36	13	169	78
5	25	11	121	55
4	16	12	144	48
3	9	10	100	30
2	4	8	64	16
1	1	9	81	9
$\Sigma X=45$	**$\Sigma X^2=285$**	**$\Sigma Y=108$**	**$\Sigma Y^2=1356$**	**$\Sigma XY=597$**

N = 9, $\Sigma XY = 597$, $\Sigma X = 45$, $\Sigma Y = 108$, $\Sigma X^2 = 285$, $\Sigma Y^2 = 1{,}356$

$$r = [\, N\,\Sigma XY - (\Sigma X)(\Sigma Y)\,] \;/\; \sqrt{[\, N\,\Sigma X^2 - (\Sigma X)^2\,]} \times \sqrt{[\, N\,\Sigma Y^2 - (\Sigma Y)^2\,]}$$

$r = (9\times597 - 45\times108) / [\sqrt{(9\times285 - 45^2)} \times \sqrt{(9\times1356 - 108^2)}] = 513/540 = 0.95$

Pearson's Coefficient of Correlation: Method 3 (Deviations Taken from an Assumed Mean)

$$r = [\, N \Sigma dx\, dy - (\Sigma dx)(\Sigma dy) \,] \;/\; \sqrt{[\, N \Sigma dx^2 - (\Sigma dx)^2 \,]} \times \sqrt{[\, N \Sigma dy^2 - (\Sigma dy)^2 \,]}$$

When actual means are in fractions, calculation becomes lengthy. Deviations are instead taken from an assumed mean A.

dx	deviations of X series from assumed mean, i.e. (X − A)
dy	deviations of Y series from assumed mean, i.e. (Y − A)
$\Sigma dx\, dy$	sum of the product of deviations of X and Y series
$\Sigma dx^2, \Sigma dy^2$	sum of squares of deviations of X and Y series
$\Sigma dx, \Sigma dy$	sum of the deviations of X and Y series

Pearson's Coefficient of Correlation: Method 3: Example 1

Calculate the coefficient of correlation between X and Y from the following data and calculate the probable error. Assume 69 and 112 as the mean values for X and Y respectively.

	1	2	3	4	5	6	7	8
X	78	89	99	60	59	79	68	61
Y	125	137	156	112	107	136	123	108

Pearson's Coefficient of Correlation: Method 3: Example 1 (Assume 69 and 112)

X	Y	dx=(X-69)	dx^2	dy=(Y-112)	dy^2	dxdy
78	125	+9	81	+13	169	+117
89	137	+20	400	+25	625	+500
99	156	+30	900	+44	1936	+1320
60	112	-9	81	0	0	0
59	107	-10	100	-5	25	+50
79	136	+10	100	+24	576	+240
68	123	-1	1	+11	121	-11
61	108	-8	64	-4	16	+32
ΣX=593	**ΣY=1004**	**Σdx=41**	**Σdx^2=1727**	**Σdy=108**	**Σdy^2=3468**	**Σdxdy=2248**

$$r = [\, N \Sigma dx\, dy - (\Sigma dx)(\Sigma dy) \,] \;/\; \sqrt{[\, N \Sigma dx^2 - (\Sigma dx)^2 \,]} \times \sqrt{[\, N \Sigma dy^2 - (\Sigma dy)^2 \,]}$$

$r = [(8)\,(41)(108) - (41)(108)] / [\sqrt{(8\times1727-41^2)} \times \sqrt{(8\times3468-108^2)}] = 13556/13969 = 0.97$

$r = 0.6745 \times (1-r^2)/\sqrt{N} = 0.6745 \times (1-0.9409)/2.828 = 0.014$

Pearson's Coefficient of Correlation: Method 3: Example 4

Given:

Total of the product of deviations of X and Y series = 3,044
Number of pairs of observations = 10
Total of the deviations of X series = -170
Total of the deviations of Y series = -20
Total of the squares of deviations of X series = 8,288
Total of the squares of deviations of Y series = 2,264

Find the coefficient of correlation when the assumed means of X series and Y series are 82 and 68 respectively.

$\Sigma dxdy=3044$, $\Sigma dx=-170$, $\Sigma dy=-20$, $\Sigma dx^2=8288$, $\Sigma dy^2=2264$, $N=10$

$$r = [(10\times3044) - (-170)(-20)] / [\sqrt{(10\times8288-(-170)^2)} \times \sqrt{(10\times2264-(-20)^2)}]$$

$$= 27040 / (\sqrt{53980} \times \sqrt{22240}) = 27040/34648.5 = 0.7804$$

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.

Data Visualization

$$r_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{N \cdot \sigma_x \cdot \sigma_y}$$

σ_x , σ_y are the standard deviations of x and y.

- Only captures linear relationships — misses quadratic or other nonlinear patterns
- Sensitive to outliers, which can skew the measure
- A single number can never reveal the full shape of a distribution

Visualization should be the first step in understanding correlation — it reveals nonlinear relationships and outliers a coefficient alone cannot.

Data Visualization

Four datasets · same mean, variance, and r = 0.816 · four very different stories

Dataset I

A clean linear relationship — the case correlation is built for.

Dataset II

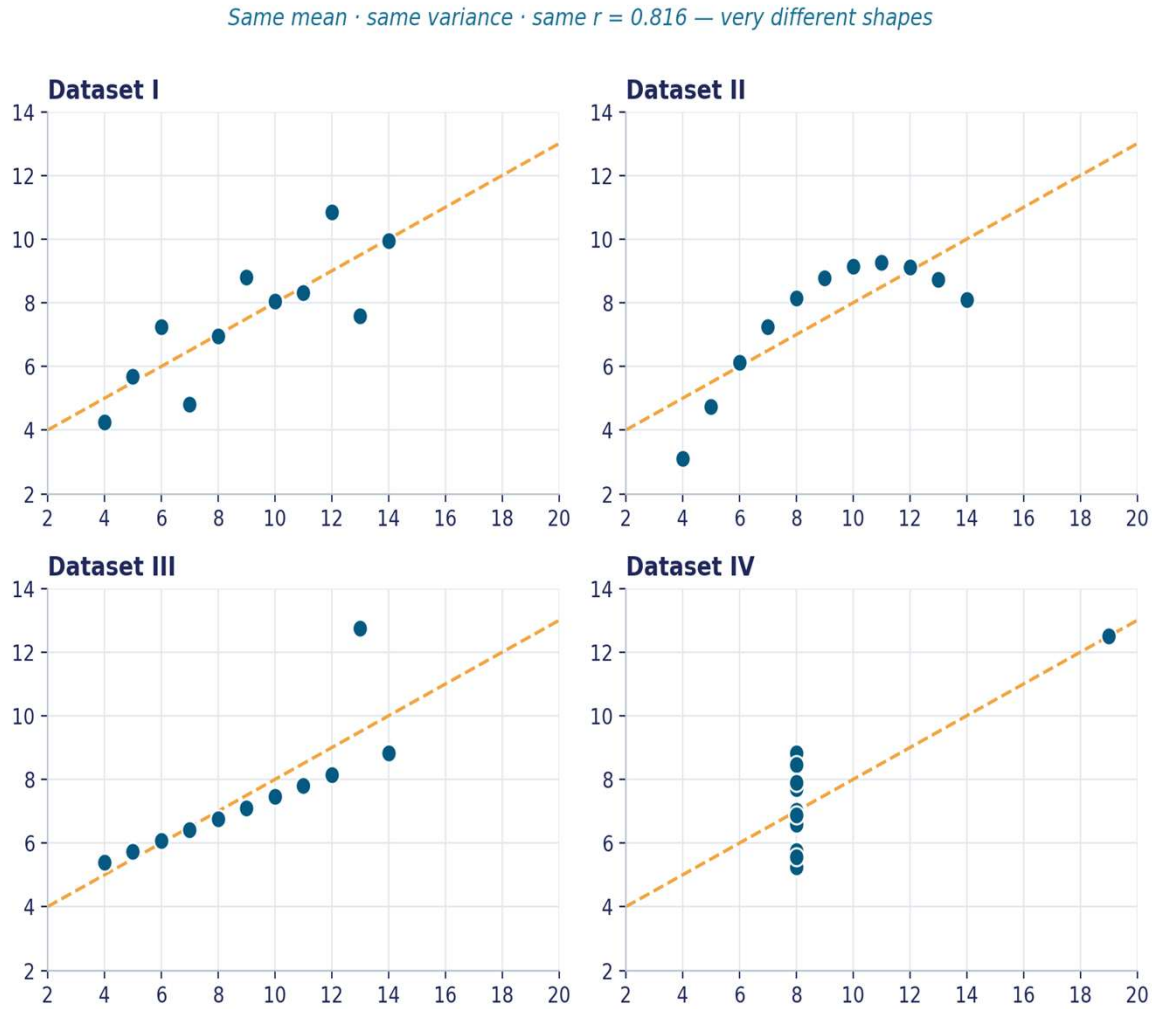
A clear curve. Same r, but the true relationship is nonlinear.

Dataset III

A perfect line broken by one outlier that drags the fit.

Dataset IV

No relationship at all — one extreme x-value creates the illusion of one.



Data Visualization: Why visualize

Human vision is trained to spot patterns and anomalies — but only if the data is presented effectively.

1 Comprehension of Dense Information

A single chart can hold thousands of data points, revealing the big picture and long-term trends that are hard to interpret from numbers alone.

2 Relationships

Cartesian coordinates expose relationships between attributes; size, color, and shape extend a 2D chart to encode even more dimensions.

3 Pattern & Anomaly Detection

Vision is one of our most powerful senses — well-chosen visuals let people discover hidden patterns and outliers almost instantly.

Data Visualization

01

Univariate Visualization

Histograms, box-whisker (quartile) charts, and distribution curves.

02

Multivariate Visualization

Scatterplots, scatter matrices, and bubble charts for 2-4 attributes.

UNIVARIATE VISUALIZATION

Histogram, Quartile & Distribution Chart

Univariate visualization represents a single variable to understand its distribution, spread, and patterns.

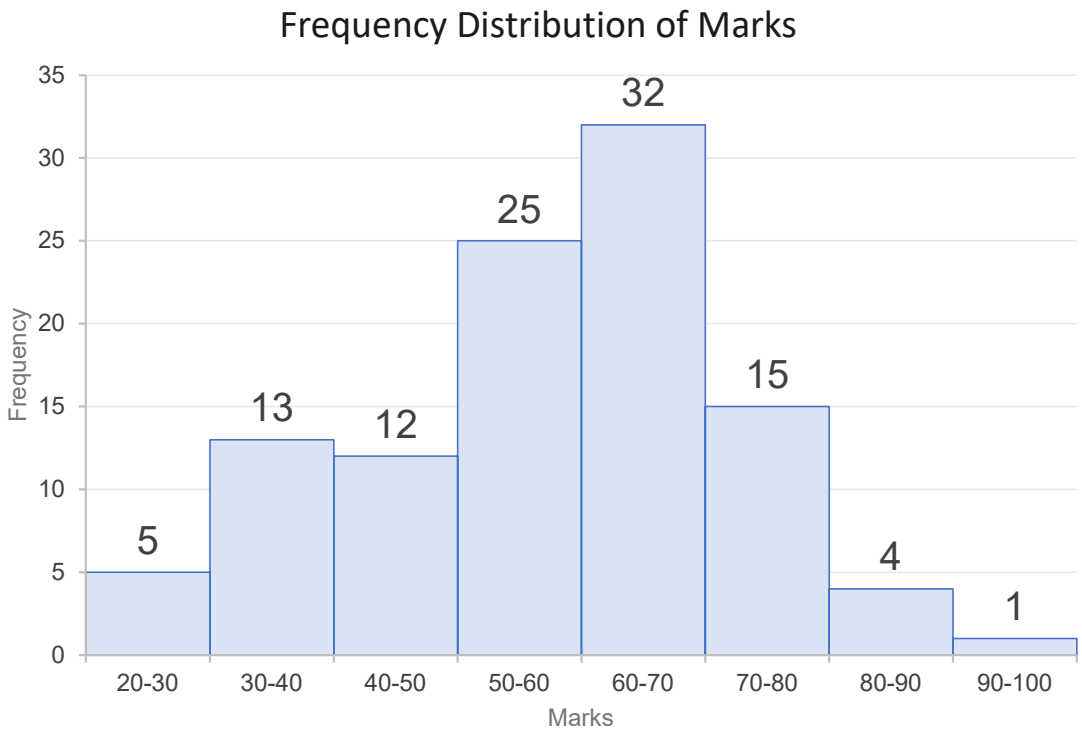
Histogram

- Shows the frequency distribution of a variable by grouping values into intervals (bins).
- Horizontal axis represents data values (class intervals); vertical axis represents frequency.
- Bars are drawn adjacent to each other, without gaps between them.
- Drawn on the basis of the number of class intervals.
- Bar shape/width depends on whether class intervals are equal or unequal.

Histogram — Example

Marks (Classes)

Marks	No. of Students
20–30	5
30–40	13
40–50	12
50–60	25
60–70	32
70–80	15
80–90	4
90–100	1



Choosing Bins & Applications

Choosing the Number of Bins

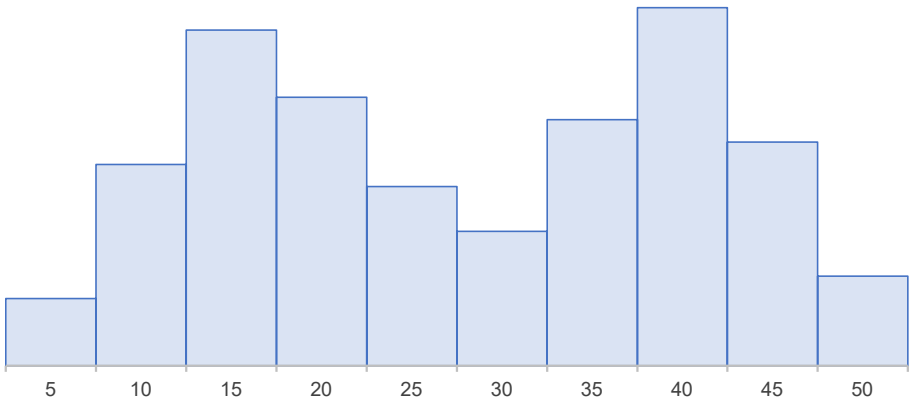
- Used for continuous numerical data.
- Too small → more detail, but noisy.
- Too large → loss of detail.

Rule of Thumb

Bins ≈ $\sqrt{n}$ or $\sqrt[3]{n}$ (n = frequency count)

e.g. if n = 100, then $\sqrt{100} = 10$ — 10 bins are sufficient.

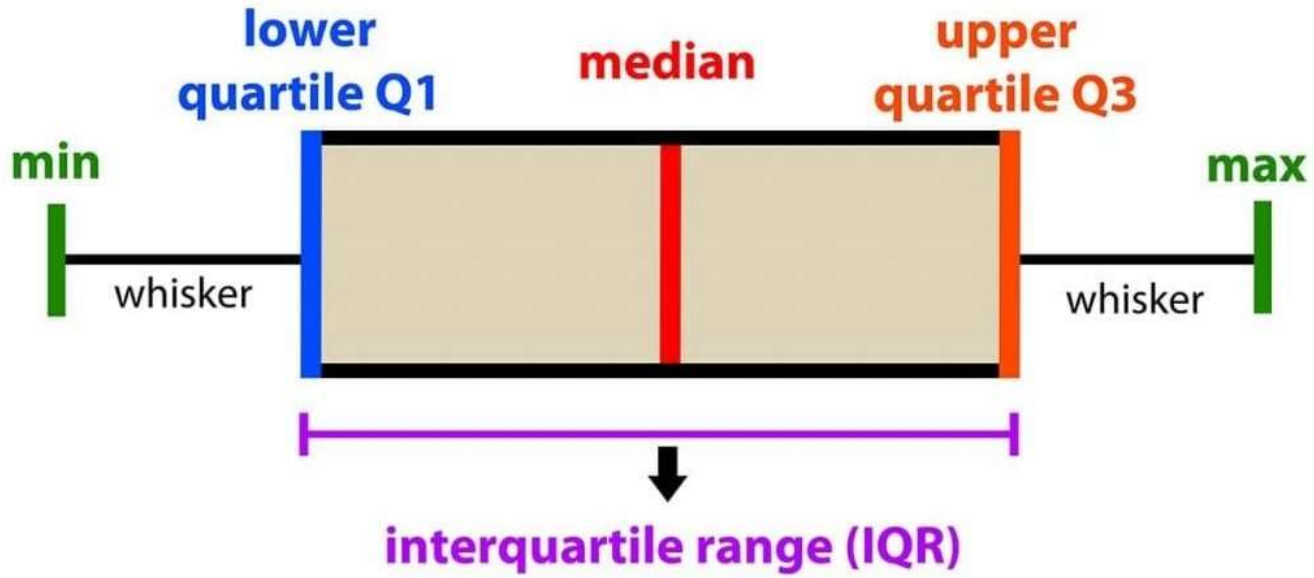
Application: Detecting Distribution Shape



- Detects normal or skewed distribution.
- Identifies multiple peaks (multimodal patterns).
- Enables comparison of grouped patterns.

What Is a Box Plot?

- In descriptive statistics, a box plot (also known as a box and whisker plot) is a type of chart often used in explanatory data analysis.
- Box plots visually show the distribution of numerical data and skewness by displaying the data quartiles (or percentiles) and averages.
- Box plots show the five-number summary of a set of data: including the minimum score, first (lower) quartile, median, third (upper) quartile, and maximum score.



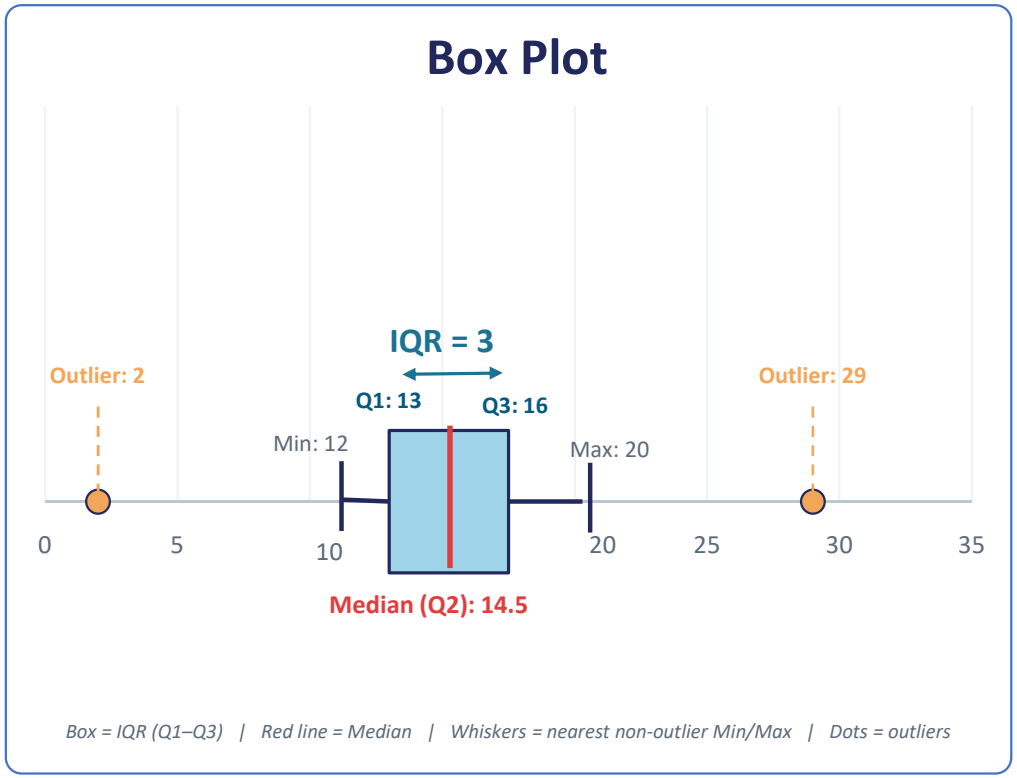
Box-and-Whisker Plot Analysis

Dataset (10 values): 2, 12, 13, 13, 14, 15, 15, 16, 20, 29

Min / Max	**2 / 29**
Q1 (Lower Quartile)	**13**
Q2 (Median)	**14 or 14.5**
Q3 (Upper Quartile)	**16**
IQR = Q3 − Q1	**16 − 13 = 3**
Lower Limit = Q1 − 1.5×IQR	**13 − 4.5 = 8.5**
Upper Limit = Q3 + 1.5×IQR	**16 + 4.5 = 20.5**

Outliers: 2 and 29

Values outside 8.5–20.5 are plotted individually



Why Are Box Plots Useful?

1. **A concise data snapshot**

   One glance shows the median, spread (IQR), and any outliers — no need to scan raw numbers.

2. **Four roughly equal sections**

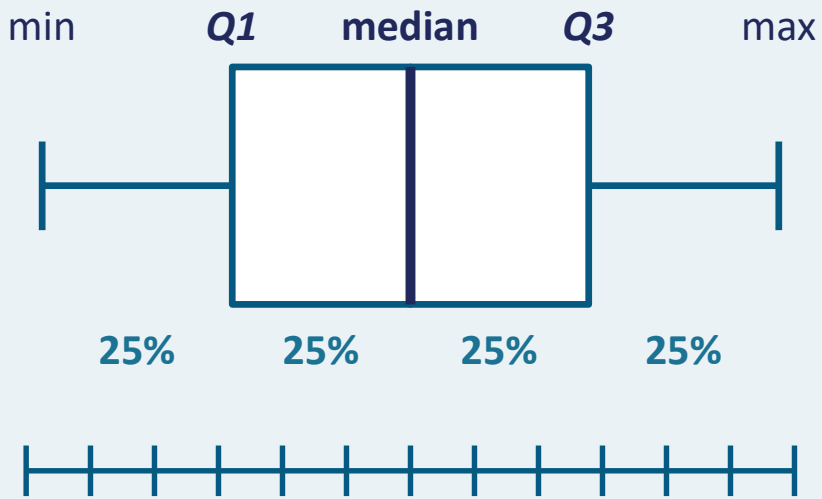
   The whiskers and box split the dataset into four sections, each holding about 25% of the values.

3. **The median splits the box in half**

   Half of all values sit above the median line; half sit below it.

4. **Shape reveals distribution**

   A symmetric box suggests a normal spread; a lopsided box signals skewed data.



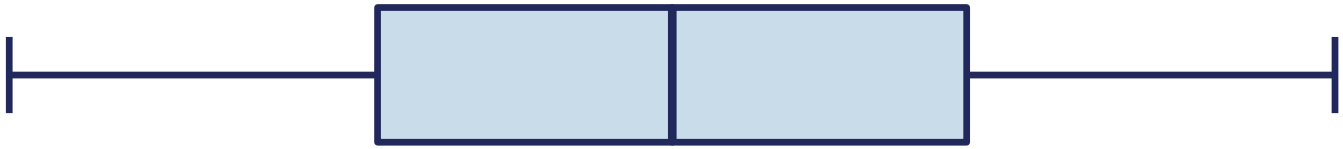
Each mark represents one data point along the sorted dataset

Skewness of a Data Set

The box plot's shape shows whether a dataset is symmetric or skewed — look at where the median sits inside the box, and which whisker is longer.

Normal Distribution

Median centered; both whiskers and both box halves are roughly equal.



Positive Skew

Median sits closer to Q1; the right whisker stretches out toward high-value outliers.



Negative Skew

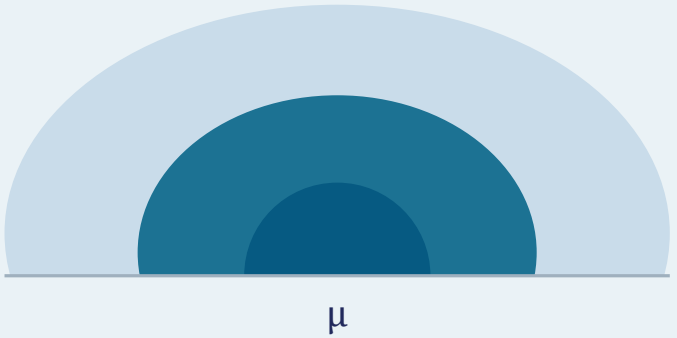
Median sits closer to Q3; the left whisker stretches out toward low-value outliers.



What Is a Distribution Chart?

- Also called a Normal Distribution or Density Curve
- Represents the probability distribution of continuous numerical data using a smooth curve instead of raw data points
- Often assumes data follows a normal (Gaussian) distribution, characterized by mean (μ) and standard deviation (σ)

- **Bell-shaped curve**
  Symmetrical around the mean
- **Mean = center**
  The peak marks the average value
- **Std. deviation (σ) = spread**
  Controls how wide or narrow the curve is



The Empirical Rule (68–95–99.7)

68.2%

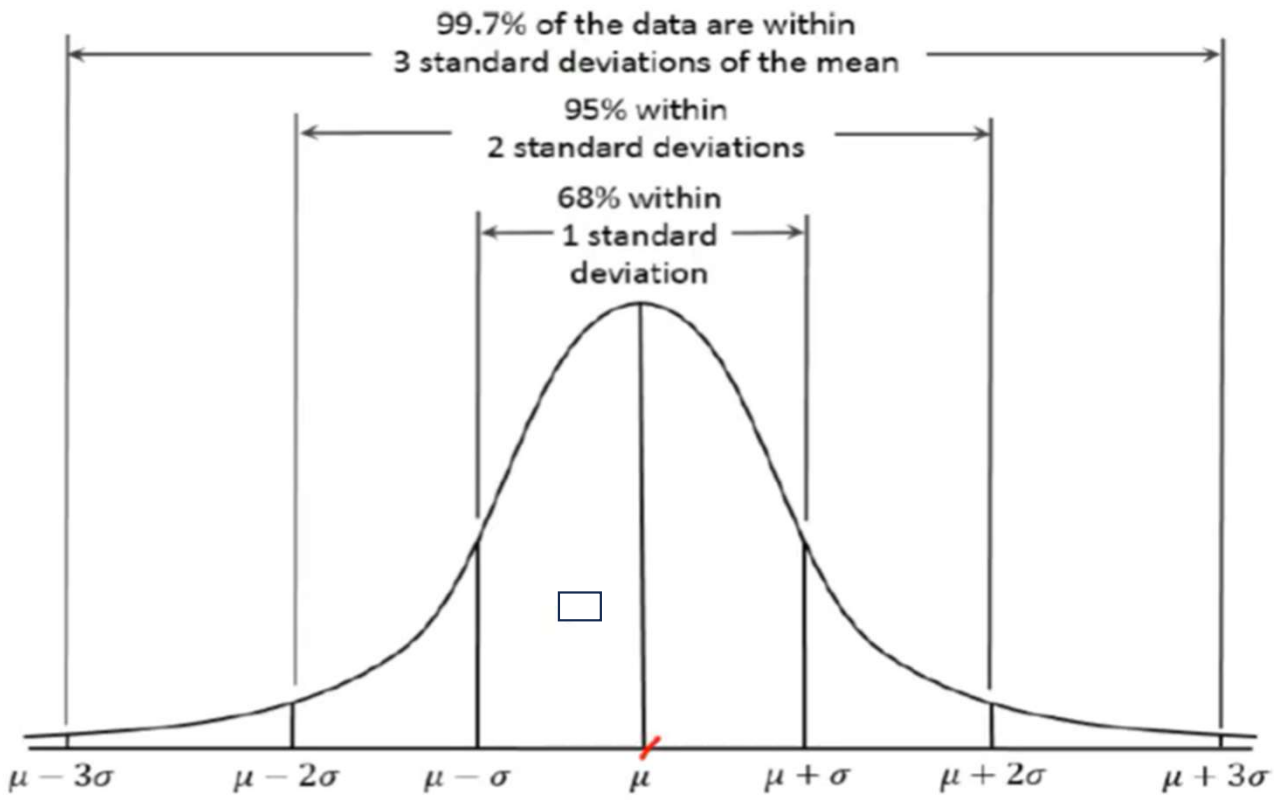
of data within ±1σ

95.4%

of data within ±2σ

99.7%

of data within ±3σ



The Normal Distribution Equation

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}$$

Variables

- x = value of the variable
- μ = mean (central value)
- σ = standard deviation
- σ^2 = variance
- e = Euler's constant (≈ 2.718)
- π = 3.1416

Key Properties

- Symmetrical around the mean (μ)
- Mean = Median = Mode
- Shape depends on standard deviation (σ)
- Small σ → narrow curve | Large σ → wide curve

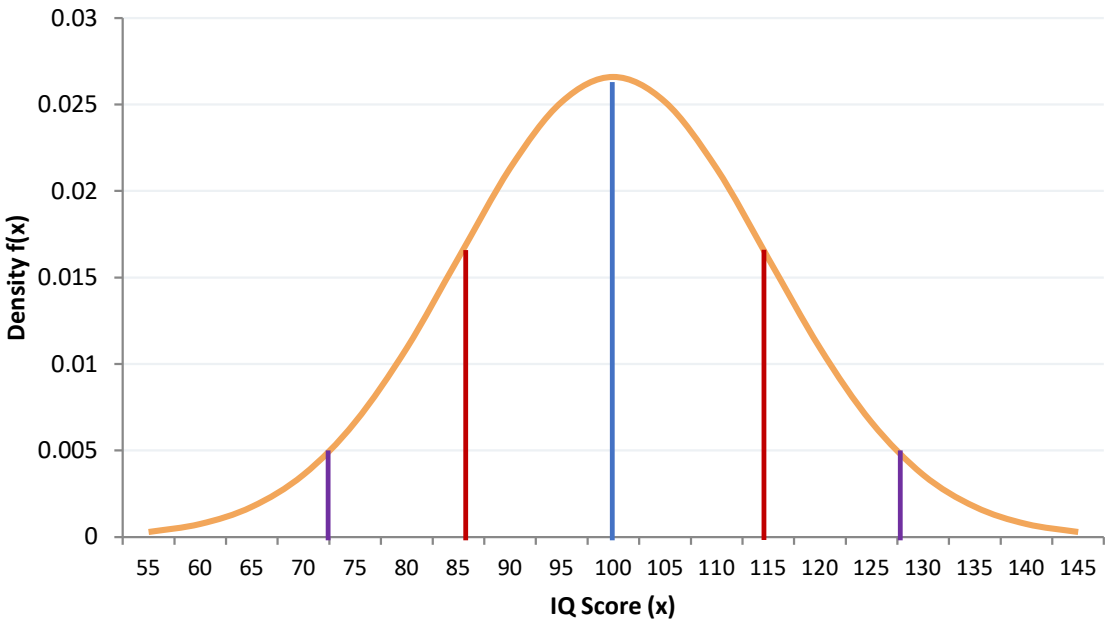
The Normal Distribution Equation

$$f(x) = {}^{1}/\sqrt{(2\pi\sigma^2)} \cdot e^{[(x-\mu)^2/2\sigma^2]} \qquad \mu = 100 \text{ (mean)} \qquad \sigma = 15 \text{ (std. dev.)}$$

Density at Each σ Band

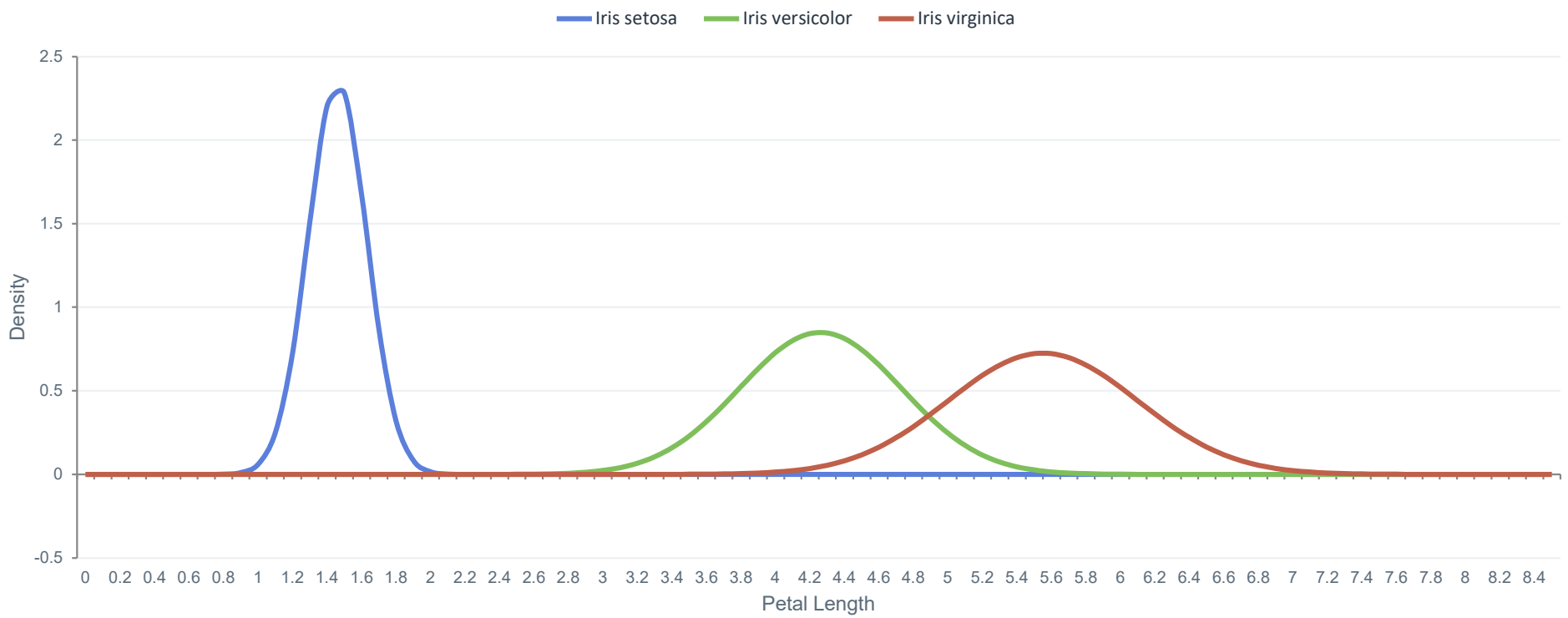
x	Dist.	f(x)
100	**μ**	**0.02660**
85 / 115	**±1σ**	0.01613
70 / 130	**±2σ**	0.00360
55 / 145	**±3σ**	0.00030

Normal Curve: N(100, 15²)



Comparing Multiple Distributions

Density curves make it easy to compare groups side by side. Here, petal length is plotted for three Iris species — each with its own mean and spread.



MULTIVARIATE VISUALIZATION

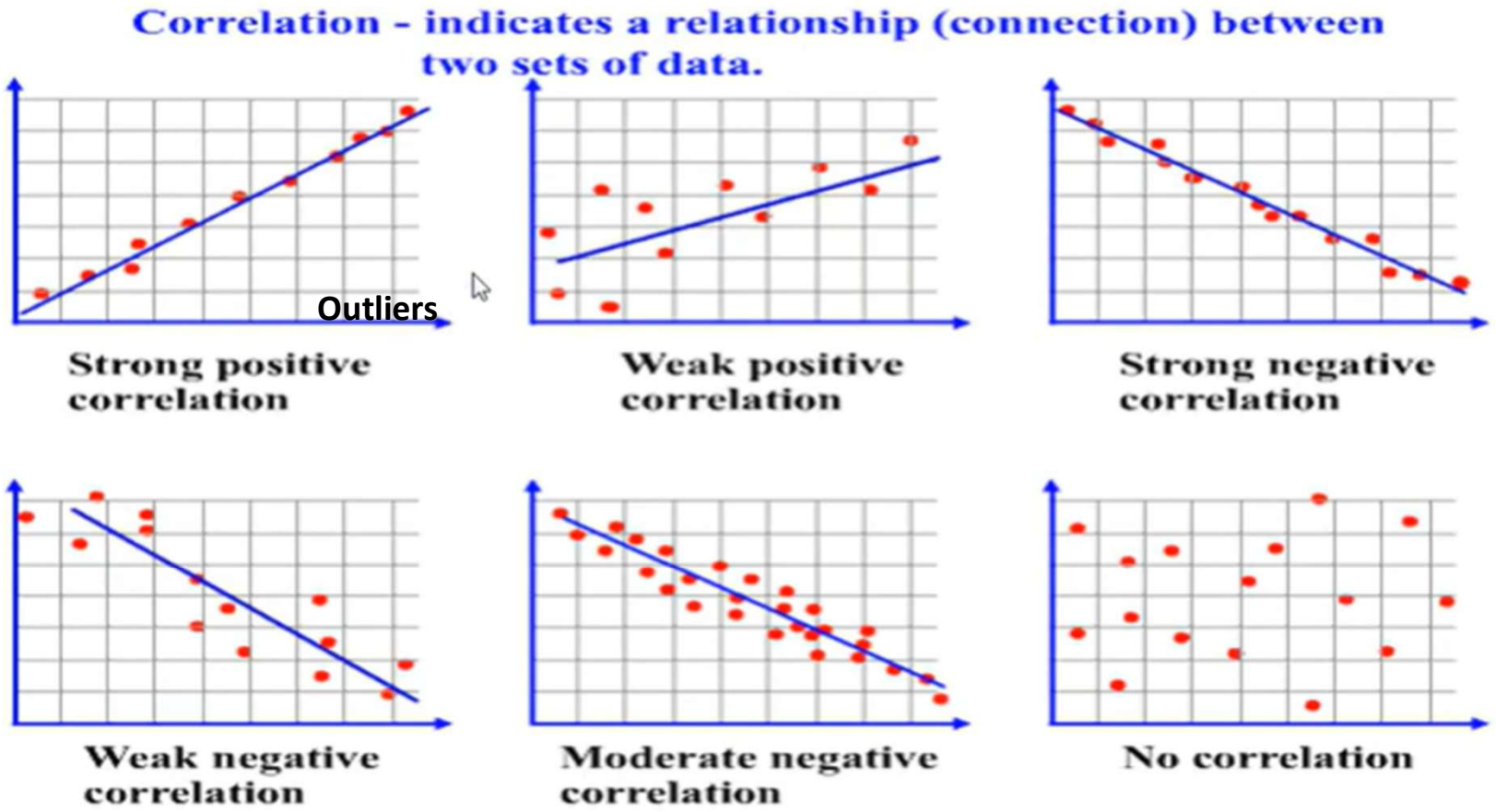
Scatter Plot • Scatter Multiple • Scatter Matrix
Bubble Chart

Multivariate visualization represents a multiple variable to understand its distribution, spread, and patterns.

Scatter Plot

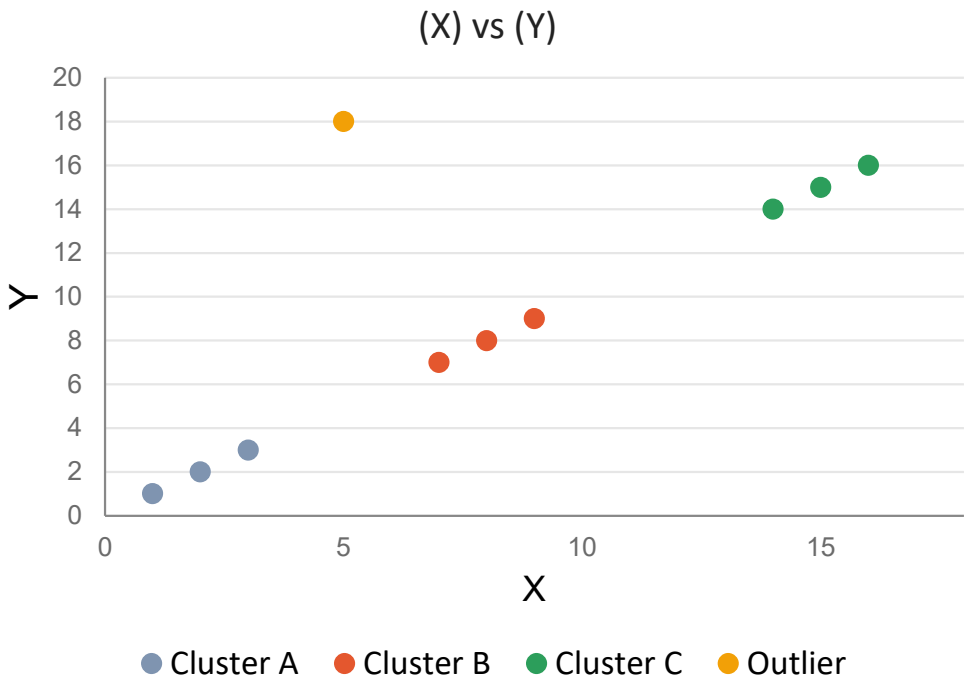
- A scatter plot is a simple and powerful graphical method used to display the relationship between two continuous variables.
- In a scatter plot, data points are plotted on Cartesian coordinates, where one variable is represented on the x-axis and another variable on the y-axis.
- **Correlation (Relationship)**
  - If points form a straight-line pattern → **Linear correlation**
  - If points are scattered → **No correlation**
- **Patterns:** Increasing pattern (positive correlation), Decreasing pattern (negative correlation)
- **Clusters:** Groups of similar data points can be identified. **Useful in classification problems**
- **Outliers:** Points far away from others. Helps in detecting abnormal values.

Scatter Plot



Reading a Scatterplot: Pattern, Clusters & Outlier

Point	X	Y	Type
1	1	1	Cluster A
2	2	2	Cluster A
3	3	3	Cluster A
4	7	7	Cluster B
5	8	8	Cluster B
6	9	9	Cluster B
7	14	14	Cluster C
8	15	15	Cluster C
9	16	16	Cluster C
10	5	18	Outlier



Pattern

X and Y rise together — a consistent positive trend.

Clusters

Points 1–9 form 3 tight cluster along the trend.

Outlier

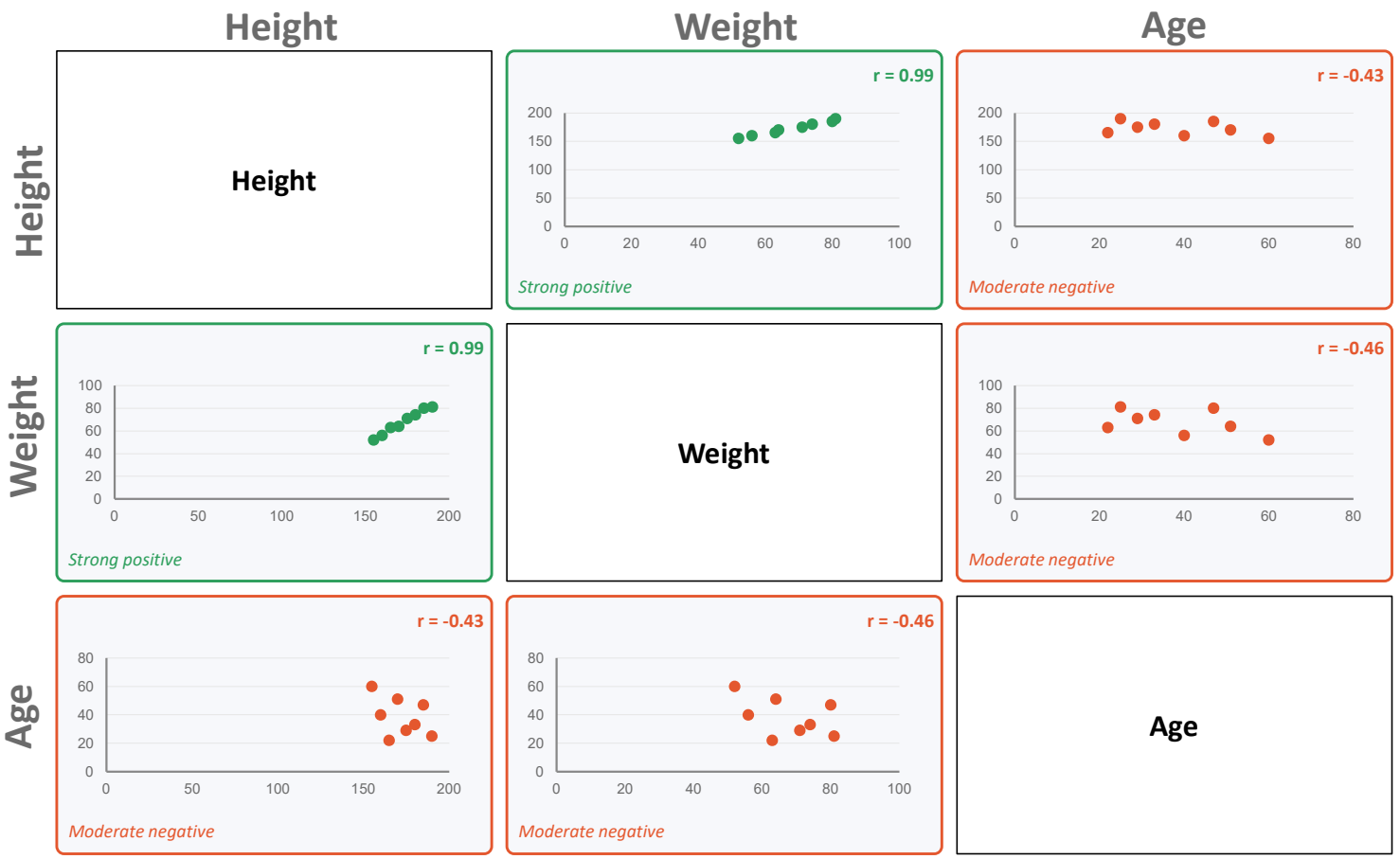
Point 10 sits far from every cluster and the trend.

Scatter Matrix

- A scatter matrix is used to visualize all pairwise relationships between multiple variables by arranging scatter plots in a matrix form.
- If there are n variables → n × n plots
- Each cell = scatter plot between two variables
- Helps analyze correlation, patterns, clusters

Scatter Matrix (Pairs Plot)

Point	Height	Weight	Age
1	160	56	40
2	165	63	22
3	170	64	51
4	175	71	29
5	180	74	33
6	185	80	47
7	190	81	25
8	155	52	60



	Scatter Plot	Scatter Matrix
Variables	2 variables	All pairwise combinations
Layout	Single plot	Matrix / grid of plots
Purpose	Relationship between two variables	Comprehensive multivariate comparison
Type	Bivariate	Multivariate
Example	Study hours vs. marks	All Iris dataset attributes

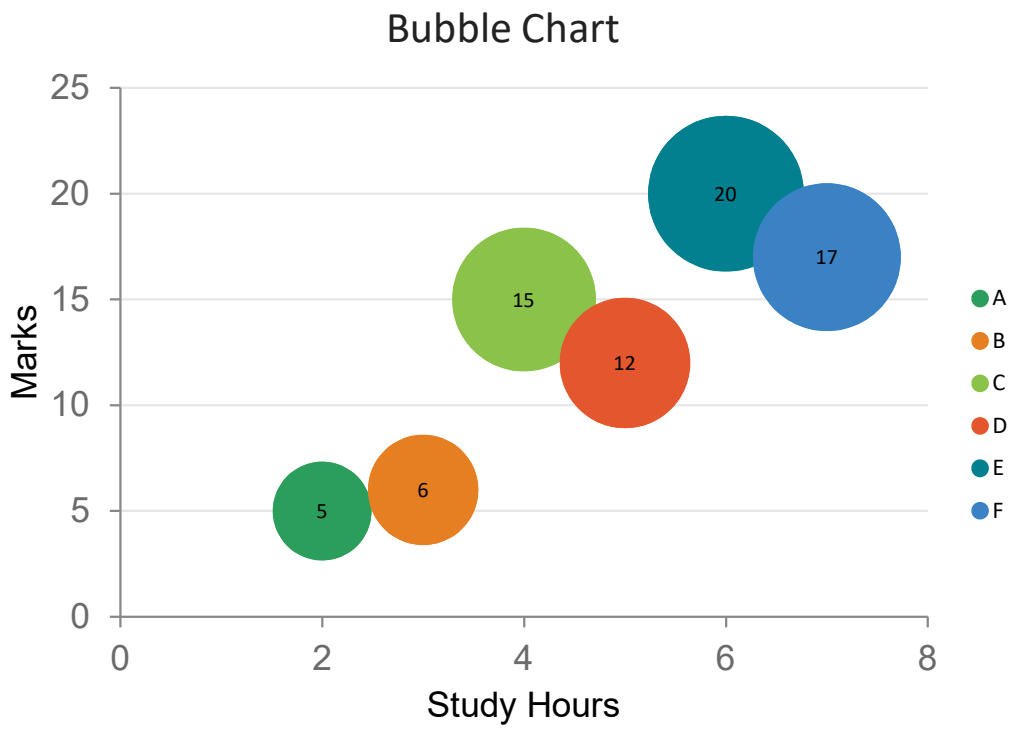
Bubble Chart

- A bubble chart is an advanced scatter plot where x-axis and y-axis represent two variables, while bubble size represents a third variable.
- Color is often used to represent class label.
- X-axis → First attribute, Y-axis → Second attribute, Bubble size → Third attribute, Color → Class/category
- Visualize relationships among 3–4 variables
- Compare magnitude
- Detect clusters and patterns
- Analyze multidimensional data

Bubble Chart Example

X = Study Hours, Y = Marks, Bubble size = Attendance, Color = Student

Student	Study Hours	Marks	Attendance
A	2	5	40
B	3	6	50
C	4	15	85
D	5	12	70
E	6	20	100
F	7	17	90



Bubble size reflects Attendance
bigger bubble = higher attendance.

Syllabus Content

Unit 2 · Statistical Foundations & Data Visualization

2.1 **Types of Data & Properties of Data**

Foundational classification of data and its key properties.

2.2 **Statistical Data**

Classification, Tabulation, and Diagrammatic & Graphic representation of data.

2.3 **Univariate Exploration**

Measures of Central Tendency & Spread; Symmetry and Skewness (Karl Pearson, Bowley's Coefficient); Kurtosis.

2.4 **Multivariate Exploration**

Central Data Point, Correlation & its forms, Karl Pearson Correlation Coefficient for bivariate distribution.

2.5 **Data Visualization**

Univariate: Histogram, Quartile, Distribution Chart

Multivariate: Scatter Plot, Scatter Matrix, Bubble Chart.